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Master's Thesis

Improving the Reconstruction of Top-Quark Pairs in Dileptonic Final States in ATLAS Using Machine Learning

prepared by

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Abstract

Recent advances in particle physics research have focused on top-quark physics. These advances include stringent tests of spin correlations and quantum effects in top-quark pairs ($t\bar{t}$), as well as the observation of a possible quasi-bound-state resonance in the $t\bar{t}$ invariant mass spectrum. The latter presents an exciting opportunity to advance the current understanding of non-relativistic Quantum Chromodynamics. Both phenomena are predominantly investigated in the dilepton decay channel of top-quark pairs near the $t\bar{t}$ production threshold.

The dilepton final state consists of two neutrinos, two charged leptons, and two b-quark jets. Accurate analysis of this system requires precise reconstruction of the top quarks, a task complicated by the presence of two neutrinos. Previous reconstruction strategies have primarily focused on inferring neutrino momenta, while the challenge of correctly assigning b-jets to their parent top quarks has received less attention. Nevertheless, many sensitive variables used in $t\bar{t}$ precision measurements depend critically on the correct assignment of jets. Therefore, this work aims to close this gap in the research.

Building on the success of machine learning architectures in addressing assignment challenges in hadronic decay channels and neutrino regression in the dilepton channel, this study demonstrates that both tasks can be accomplished by a single model. Various models and configurations are implemented and evaluated. The discussion covers machine learning aspects of the different architectures, followed by an in-depth analysis of the method's strengths and advantages. Using a simplified analysis setup, the model's ability to improve the measurement sensitivity is demonstrated. Comparison of distributions from ATLAS event data and simulated events indicates the model's readiness for application in a full analysis context. The method presented is the first machine learning tool to provide full reconstruction of the dileptonic decay channel available in the ATLAS Collaboration's official software repository. Application of these tools to current and future analyses is expected to substantially enhance the sensitivity and precision of searches and measurements.

Keywords: Top Quark, Dileptonic Decay, Event Reconstruction, Machine Learning, Transformer Networks

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W

1. Introduction

The top quark, being the heaviest known elementary particle, plays a crucial role in the Standard Model of particle physics. Due to its large mass, it stands out in two ways; for one, as having the highest Yukawa coupling to the Higgs boson and secondly, as being the shortest-lived elementary particle. While the Yukawa coupling makes it very interesting for Higgs physics, the short lifetime of the top quark presents it as the only window allowing us to study the bare coupling of quarks. In particular is its lifetime shorter than the timescale required to form hadrons or decorrelate the spin. The top quark decays almost exclusively into a W boson and a b-quark. Combined with the large cross-section, these properties make $t\bar{t}$-production one of the most studied processes at the Large Hadron Collider (LHC) at CERN.
Top quarks are predominantly produced in pairs ($t\bar{t}$) via the strong interaction in proton-proton collisions at the LHC. Each W boson coming from the top decay can decay either leptonically (into a charged lepton and neutrino) or hadronically (into a pair of quarks). The dileptonic decay channel, where both W bosons decay leptonically, results in a final state with two charged leptons, two b-jets, and two neutrinos. This channel, while having a lower branching ratio compared to other decay modes, offers a cleaner experimental signature due to the presence of leptons and reduced hadronic activity. Leptons, in particular, can be reconstructed with high efficiency in a hadron collider environment. However, the presence of two neutrinos in the final state poses significant challenges for event reconstruction, as they evade detection, resulting in missing transverse energy ($\not{E}_{\mathrm{T}}$).
Due to its clean signature and sensitivity to various top-quark properties, the dileptonic $t\bar{t}$ channel is widely used for precision measurements of top properties, particularly spin correlations. Two recent highlights in top-quark physics are the tests of spin correlations and quantum entanglement in pairs of top quarks [1], and the observation of a possible quasi-bound-state resonance in the $t\bar{t}$ invariant mass spectrum [2, 3]. Both effects are predominantly studied in the dilepton decay channel of the top quark in a mass range close to the on-shell production threshold of two top quarks $m(t\bar{t}) \gtrsim 2 \cdot m_t \approx 350$ GeV. Compared to hadronic decays, the leptons preserve a larger fraction of the top quark's spin information [4, 5].
Both analyses rely on measurements of angular distributions and correlations between the decay products of the top quarks. Probing this system requires a precise reconstruction of the top quarks from their decay products, which is complicated by the presence of the two neutrinos. While analytical neutrino regression strategies have been studied extensively, the problem of correctly assigning b-jets to their parent top quarks remains largely unstudied. In channels with hadronically decaying top quarks, various methods for jet assignment have been developed and employed successfully [6]. However, in the dileptonic channel, the jet assignment problem was often simplified or overlooked due to a focus on neutrino reconstruction.

However, many of the sensitive variables used to measure spin correlations depend critically on the correct assignment of the b-jets. Misassignments degrade the resolution of reconstructed observables, ultimately limiting measurement precision.
Furthermore, the jet assignment problem is not only relevant for the reconstruction of the top quarks but also for the neutrino regression itself. Many of the conventional neutrino regression methods rely on kinematic constraints that are applied to the decay products of each top quark. If the jets are misassigned, these constraints can lead to incorrect neutrino momentum estimates, further degrading reconstruction performance.
Consequently, this thesis seeks to address the jet assignment problem in close combination with the neutrino regression, with the goal of improving the overall reconstruction of dileptonic $t\bar{t}$ events. The properties of different model architectures are examined, and the impact of the multi-task learning is analysed. To account for the substantial degeneracy in evaluating the performance of the different methods, a detailed comparison across metrics and kinematic regimes is conducted. This analysis provides some additional insights into the kinematic structure of the events and allows a more nuanced description of each method's performance.
Lastly, how the performance of the different methods propagates to the sensitivity of a measurement is investigated using a simple binned likelihood fit. Whether the model behaves on event data as expected is assessed by comparing the distributions of some reconstructed variables between ATLAS data and simulated events.
Beginning with a review of the theoretical background in chapter 2 and the experimental setup in chapter 3, the thesis provides the necessary context for understanding the challenges of reconstructing dileptonic $t\bar{t}$ events. Following this, chapter 4 introduces the foundations of the machine learning techniques employed in this work. The general setup of $t\bar{t}$ analyses and the conventional reconstruction methods are outlined in chapter 5. Starting with chapter 6, the novel machine learning-based reconstruction methods are developed and evaluated. Following this, the performance of the proposed methods is assessed in chapter 7, and they are applied to a physics analysis in chapter 8 to demonstrate their in-situ performance and highlight their availability for future analyses.
Finally, chapter 9 summarises the findings of this thesis and discusses potential future directions for improving the reconstruction of dileptonic $t\bar{t}$ events.

2. Theoretical Background

2.1. The Standard Model of Particle Physics

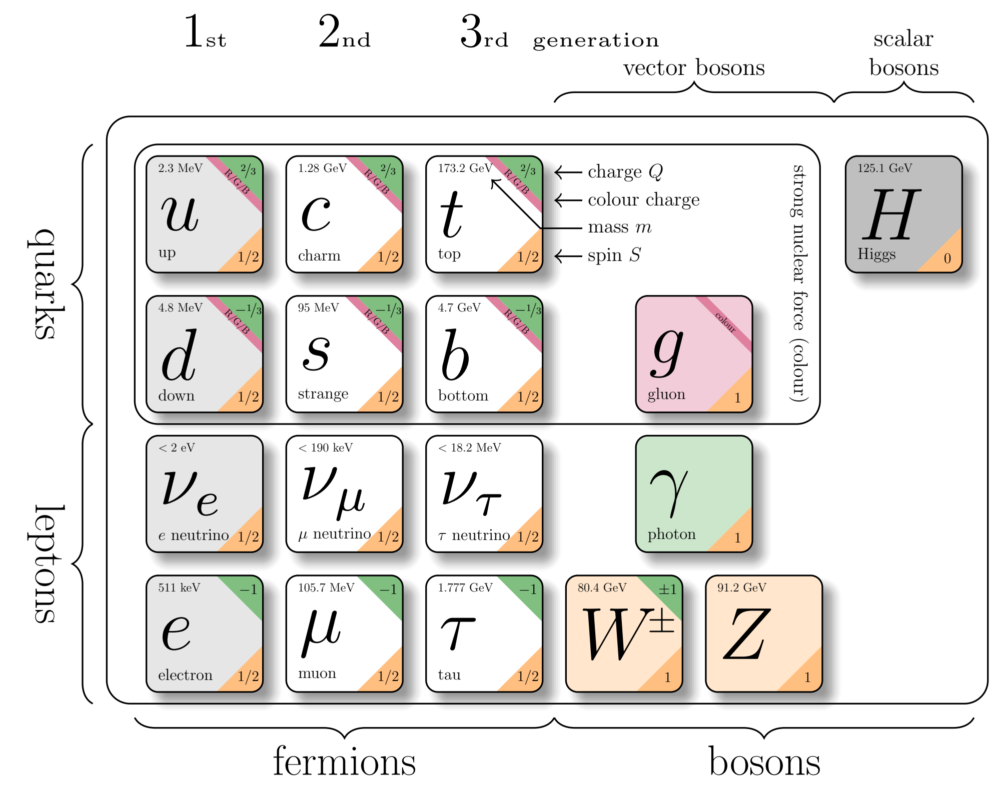


Figure 2.1.: The particles of the Standard Model. Particle properties taken from reference [7].

The Standard Model (SM) of particle physics is a quantum field theory describing three of the four fundamental interactions of nature—electromagnetic, weak and strong—and the elementary particles on which they act.
It is formulated as a renormalisable, Lorentz-invariant gauge theory with gauge group

$$SU(3)_C \times SU(2)_L \times U(1)_Y \tag{2.1}$$

and its structure has been established in a series of seminal works [8–13]. A summary of the particle content is shown in figure 2.1. The elementary particles fall into two categories, fermions and bosons. Fermions are spin-$\frac{1}{2}$ particles that constitute matter, organised into three generations of quarks and leptons. Quarks carry colour charge and participate in the strong interaction; leptons do not. Bosons are integer-spin gauge particles mediating the fundamental forces: the photon for electromagnetism, the $W^{\pm}$ and Z bosons for the weak interaction, and eight gluons for the strong interaction. The Higgs boson completes the SM and is responsible for electroweak symmetry breaking (EWSB), giving mass to the weak gauge bosons and, through Yukawa couplings, to the fermions. The SM Lagrangian can be written schematically as

$$\mathcal{L}_{\text{SM}} = \mathcal{L}_{\text{gauge}} + \mathcal{L}_{\text{fermion}} + \mathcal{L}_{\text{Yukawa}} + \mathcal{L}_{\text{Higgs}}, \tag{2.2}$$

where the first term contains the gauge kinetic terms and self-interactions, the second describes the fermions and gauge covariant derivatives, the third encodes the Yukawa couplings to the Higgs field, and the last term contains the Higgs potential and EWSB mechanism. Below, the three interaction sectors are briefly summarised, with an emphasis on aspects relevant to top-quark physics and LHC phenomenology.

2.1.1. Electromagnetic Interaction

The electromagnetic interaction is described by quantum electrodynamics (QED), the $U(1)_{\text{EM}}$ gauge theory that remains after EWSB. Its massless gauge boson, the photon, couples to electric charge and does not self-interact. Because QED is an Abelian theory, its coupling runs only mildly with energy and remains perturbative at all experimentally accessible scales.

2.1.2. Weak Interaction

The weak interaction, together with electromagnetism, forms the electroweak theory based on the gauge group $SU(2)_L \times U(1)_Y$. The vacuum expectation value of the Higgs field spontaneously breaks the symmetry to $U(1)_{\text{EM}}$, giving masses to the weak gauge bosons through the Higgs mechanism [14–16]. The resulting massive $W^{\pm}$ and Z bosons mediate a short-range force.

A defining property of the weak interaction is its *chiral* structure: charged-current interactions couple only to left-handed fermions and right-handed antifermions. This feature, predicted in [17] and confirmed in the Wu experiment [18], leads to maximal parity violation and characteristic angular distributions of weak decay products. The vertex factor of the charged-current interaction is given by

$$\frac{-ig}{2\sqrt{2}}\gamma^{\mu}(1-\gamma^5), \tag{2.3}$$

where g is the weak coupling constant and γ^{μ} are the Dirac matrices. The projection operator $(1-\gamma^5)$ leads to a coupling exclusively to the left-handed components. Based on this, the left-handed particles are arranged in $SU(2)_L$ doublets, while the right-handed particles are singlets.

For quarks, the weak eigenstates d', s', b' are not identical to their mass eigenstates d, s, b. Their mixing is described by the Cabibbo-Kobayashi-Maskawa (CKM) matrix [19, 20], which enables flavour-changing coupling in charged-current interactions.

2.1.3. Strong Interaction

The strong interaction is described by quantum chromodynamics (QCD), a non-Abelian gauge theory with gauge group $SU(3)_C$. Quarks carry colour charge, while gluons carry colour and anticolour index, giving rise to self-interactions through three- and four-gluon vertices. These features lead to two fundamental properties.

Asymptotic freedom The QCD coupling decreases at high momentum transfer [11, 21], allowing perturbative calculations of hard processes such as top-quark pair production. In the high-energy regime of the LHC, quarks and gluons effectively behave as quasi-free particles.

Confinement At low energies, the QCD coupling becomes large, preventing coloured states from existing in isolation. Instead, they form colour-neutral bound states–hadrons. The transition from short-distance partons to long-distance hadrons involves non-perturbative dynamics encapsulated through hadronisation models. These models cannot be obtained analytically and are a major source of uncertainty.
A crucial consequence for collider physics is that quarks and gluons produced in hard interactions undergo a cascade of QCD emissions before hadronising into jets. This process is typically described in terms of:

- **Parton distribution functions (PDFs)**, characterising the proton structure and entering via the QCD factorisation theorem [22].
- **Initial- and final-state radiation (ISR/FSR)**, arising from soft and collinear gluon emission.
- **Parton showers**, which resum logarithmically enhanced emissions and are implemented in Monte Carlo generators such as Pythia, first developed in [23, 24] and extended in modern formulations [25].
- **Hadronisation models**, such as the Lund string model [26, 27], which describe the confinement-driven formation of hadrons.

2.1.4. Monte Carlo Simulation at Colliders

Building on the QCD processes introduced in section 2.1.3, predictions at the Lhc are obtained almost exclusively via Monte Carlo (MC) simulations, in which event samples are generated by randomly sampling from the underlying probability distributions. The complex mathematical structure of the Standard Model, combined with the wide range of energy scales involved in collider processes, renders purely analytical predictions rarely feasible.

Hard scattering Events are simulated using a cascade of programs, each responsible for a distinct stage of the event. Matrix-element generators compute the hard-scattering amplitude for a given process and sample the kinematic phase space according to the

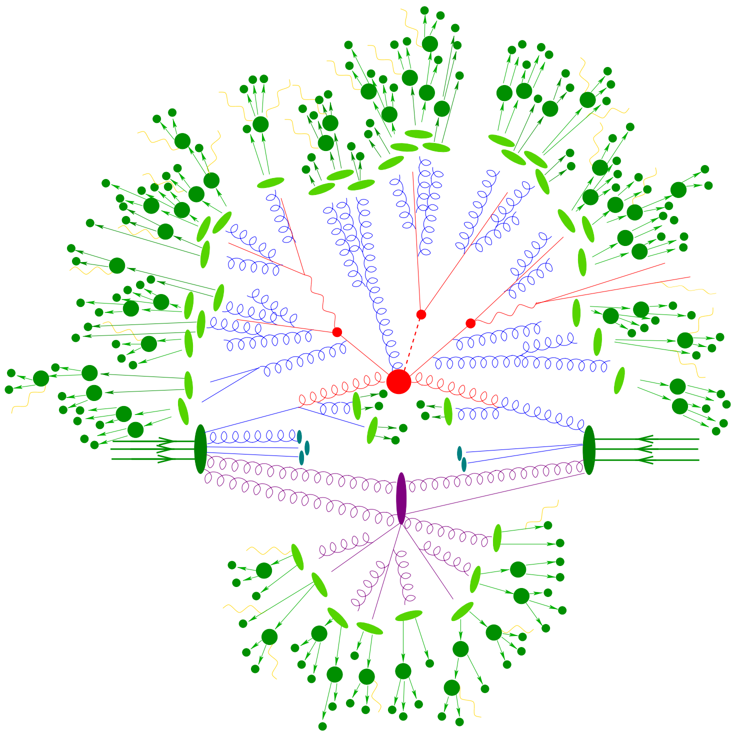


Figure 2.2.: Schematic representation of the Monte Carlo event generation process in particle physics. Colouring of vertices and particles indicates the event generation stage: hard scattering process (big red blob), QCD radiation (blue), hadronisation (light green), showering (dark green), and a secondary vertex (purple). Taken from reference [28].

differential cross-section, yielding the partonic final state of the hard interaction; events at this stage are referred to as *parton-level*.

Parton shower and hadronisation The parton-level events are subsequently passed through parton-shower and hadronisation algorithms, as outlined in section 2.1.3, which evolve the hard-scatter partons into colour-neutral hadronic final states. A schematic description of the full event generation chain is given in figure 2.2.

Detector simulation Finally, detector simulation programs model the interactions of stable particles with the detector material, producing realistic detector responses. This includes simulating energy deposits in calorimeters, hits in tracking detectors, and other relevant signals. For many applications, a simplified fast simulation is sufficient, involving smearing of particle momenta and energies to match the detector resolution, applying efficiency corrections, and accounting for pile-up—the contribution from additional inelastic pp interactions occurring within the same bunch crossing.

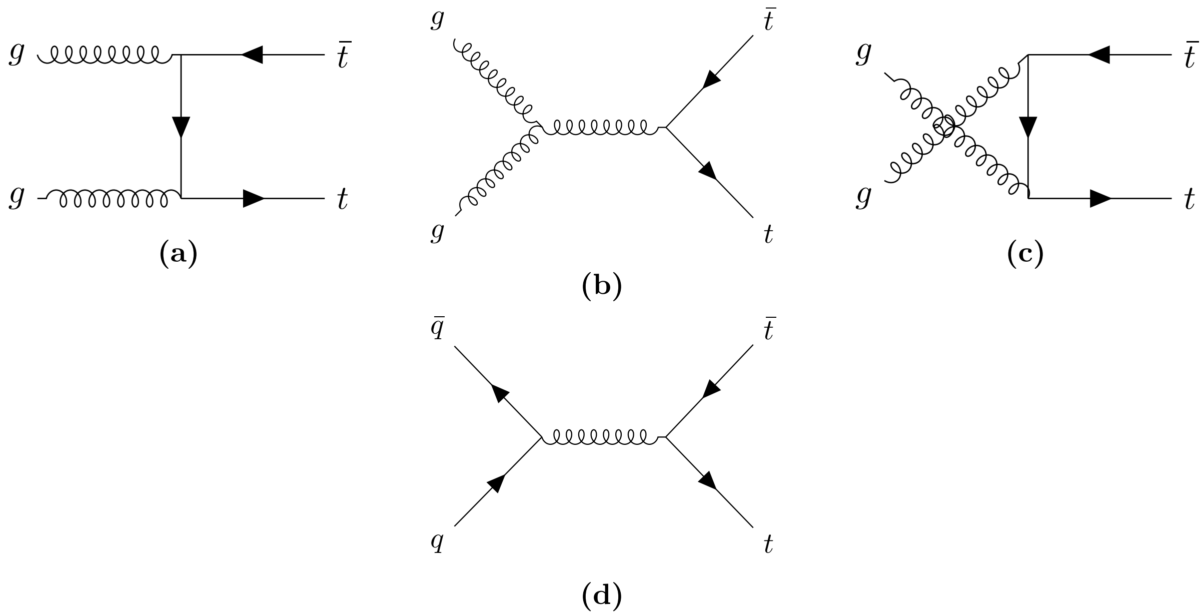


Figure 2.3.: Leading order Feynman diagrams for top-quark pair production in (d) (a), (b) & (c) gluon fusion (ggF) and (d) quark-antiquark annihilation.

2.2. Top-Quark Physics

The top quark is the heaviest known elementary particle. The prediction of its existence arose due to the already established CP-Violation in the kaon system, which required a third generation of quarks to be explained within the SM framework [20]. Furthermore, the discovery of the bottom quark in 1977 [29] implied the existence of its weak isospin partner, the top quark. The top quark was eventually discovered in 1995 by the CDF and DØ collaborations at the TEVATRON collider [30, 31]. Currently, the world average of the top-quark mass is measured to be [7]

$$m_t = 172.56 \pm 0.31 \text{ GeV}. \tag{2.4}$$

Along with the largest mass, the top quark also has the strongest Yukawa coupling to the Higgs boson, making it interesting for studies of the Higgs mechanism and electroweak symmetry breaking. Due to its large mass, the top quark also has the shortest lifetime of all quarks, about 5×10^{-25} s [7], which is shorter than the typical hadronisation timescale of 3×10^{-24} s [32]. Therefore, the top is the only quark that decays before it hadronises, allowing for direct studies of its properties through its decay products.
Notably, the top quark also decays before its spin can decorrelate, preserving spin information in the angular distributions of its decay products [33]. This unique feature enables precise measurements of spin correlations and polarisation of top quarks. At hadron colliders like the LHC, top quarks are predominantly produced in pairs via the strong interaction. The leading order (LO) Feynman diagrams for top-quark pair production are shown in figure 2.3. At the LHC, the dominant production mechanism is gluon fusion (ggF). This is due to the high gluon density in the proton at the relevant Bjorken-x

values for top-quark production and the low antiquark-contributions at a pp-collider. The CKM matrix element V_{tb} is close to unity [7], leading to a nearly exclusive decay of the top quark into a W boson and a b quark. The W can subsequently decay either leptonically into a charged lepton and a neutrino or hadronically into a pair of quarks, with branching ratios of about $\mathrm{BR}(W \to \ell, \nu) \approx 1/3$ and $\mathrm{BR}(W \to q\bar{q}) \approx 2/3$, respectively [7]. Based on these W decay modes, top-quark pair events are categorised into three channels. In the **dileptonic channel**, both W bosons decay leptonically ($W^+ \to \ell^+\nu_\ell$,$W^- \to \ell^-\bar{\nu}_\ell$), resulting in two charged leptons, two neutrinos and two b quarks in the final state. This channel has a branching ratio of approximately 1/9. The **semileptonic channel** occurs when one W boson decays leptonically and the other hadronically ($W \to q\bar{q}'$), yielding one charged lepton, one neutrino, and four quarks (including two b quarks) in the final state. Finally, in the **all-hadronic channel**, both W bosons decay hadronically, producing six quarks (including two b quarks) in the final state, which suffers from large multijet background contamination in the busy environment of hadron colliders like the LHC. Both hadronic decays have a similar branching ratio of 4/9.

2.2.1. Spin Correlations in Top-Quark Pairs

As elaborated upon before, the top quark decays before it can hadronise or its spin decorrelates. Therefore, the spin information of the top quark is directly transferred to its decay products. In top-quark pair production, the spins of the top and antitop quarks are correlated due to the production mechanism. Due to the chiral nature of the weak interaction, these spin correlations manifest in the angular distributions of the decay products, particularly the charged leptons in the dileptonic decay channel.

Recent studies at the LHC began to explore the potential effects of quantum entanglement in top-quark pairs [1]. Entanglement generally describes a quantum mechanical phenomenon where the quantum states of two or more particles become correlated in such a way that the state of one particle cannot be described independently of the state of the other(s), even when the particles are separated by large distances [34]. This property is expressed as the system having a non-separable density matrix. While a general description of this phenomenon is beyond the scope of this thesis, entanglement can be tested using specific criteria. One such criterion is the *Peres-Horodecki criterion* [35, 36], which states that if the partial transpose of the density matrix of a bipartite system has at least one negative eigenvalue, then the system is entangled. Considering the top-quark pair's spin as a bipartite system of two spin-$\frac{1}{2}$ particles (qubits), this criterion can be applied to test for entanglement in their spin states.

In reference [37], it was proposed to measure entanglement in top-quark pairs produced at the LHC by analysing the angular distributions of their decay products, specifically the charged leptons in the dileptonic decay channel. The study showed that by measuring the opening angle between the two leptons, one can construct an observable sensitive to the entanglement of the top quarks' spin system. Due to the kinematics of top-quark pair production, the entanglement is expected to be more pronounced in certain regions of phase space; one such region is at low invariant mass of the top-quark pair system.

One particular region of interest is the so-called *threshold region*, referring to the region in which the invariant of ht $t\bar{t}$-system is close to the production threshold of $m(t\bar{t}) \sim 350$ GeV, where the top-quarks are produced with low relative velocity. In this region, the top-quark pair system is produced as a spin singlet state in about $\sim$ 80% of the cases, leading to a strong entanglement signature [38]. The ATLAS and CMS collaboration recently reported the first experimental evidence of quantum correlations consistent with entangled spin quantum states in top-quark pairs [1, 39]. Measurements of this variable are particularly sensitive in the dilepton channel, because the lepton has the highest spin-analysing power [4, 5]. The spin-analysing power measures the extent to which the spin information of the parent is imprinted in the angular distributions of the decay products. Due to QCD effects, this is lower for light jets compared to leptons.

2.2.2. Top-Quark Pair Quasi-Bound-State Effects

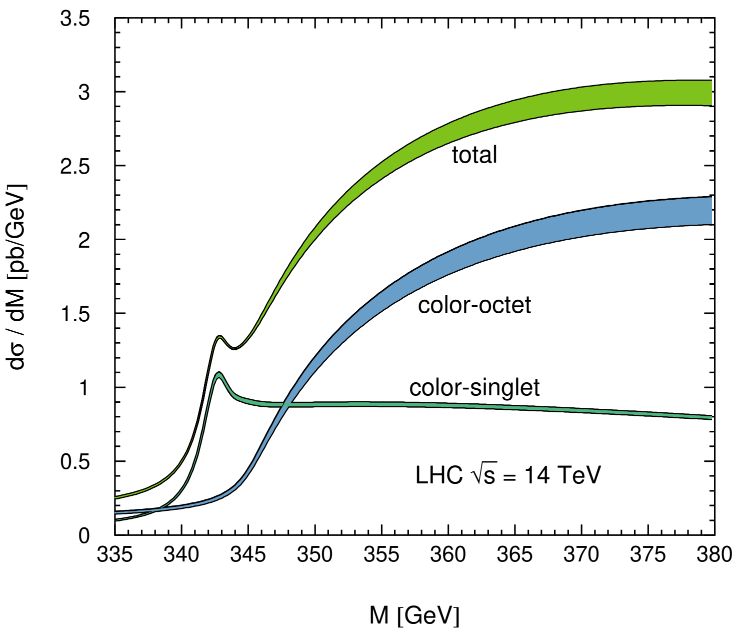


Figure 2.4.: Differential cross-section of top-quark pair production in a colour-singlet or -octet state. The Figure is taken from reference [38], which provides further details on the underlying modelling.

Even though the top quark decays before it can hadronise, near the production threshold of top-quark pairs, the strong interaction between the top and antitop quarks can lead to the formation of transient bound states, coined as *toponium* [38, 40, 41]. These bound state effects can influence the production cross-section and kinematic distributions of top-quark pairs near threshold. The CMS and ATLAS collaborations have recently observed an excess in the invariant mass distribution of top-quark pairs near threshold, consistent with the presence of toponium-like bound state effects [2, 3]. The modelling of these effects is based on non-relativistic QCD (NRQCD) calculations, which account for the strong interaction dynamics between the top and antitop quarks in the near-threshold region [42, 43]. The leading contributions arise from the formation of colour-singlet $^1S_0$, which enhances the cross-section due to an attractive QCD potential. The differential

cross-sections related to the different colour-states can be found in figure 2.4. A proper introduction to the physics behind this effect would exceed the scope of this thesis, which focuses on reconstruction techniques. Thus, for further reference, the reader is pointed to the aforementioned references [42, 43].
Understanding and accurately modelling these bound-state effects is crucial for precision measurements of top-quark properties and for searches of new physics in top-quark pair production. One notable example is the measurement of quantum effects, where the strength of the observed entanglement properties is influenced by bound-state effects and by their impact on the spin structure of the top-quark pair system. The measurement of these effects relies on reconstructing the top-quark system at the production threshold and uses angular correlations among the top-quark's decay products, similar to studies of quantum effects.

2.2.3. Dileptonic Decay Channel

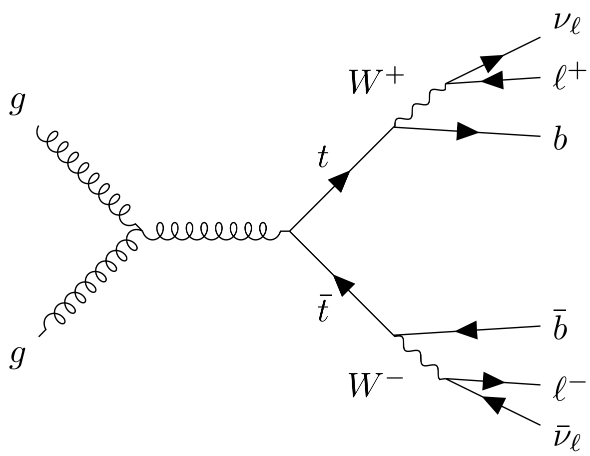


Figure 2.5.: Leading order Feynman diagram for top-quark pair production via gluon fusion with dileptonic decay.

While having the lowest branching ratio and thus the lowest event yields of all decay channels, the dilepton channel offers a particularly clean signature with reduced background contamination. Furthermore, leptons can be reconstructed and identified with a high efficiency in a hadron-collider environment and carry charge information, which enables assignment to top or antitop quark, respectively. This makes it very suited for precision measurements of top-quark properties. The presence of two neutrinos in the final state leads to missing transverse energy ($\not{E}_{\mathrm{T}}$) in the event, complicating the full reconstruction of the top quarks' kinematics. Advanced techniques, such as kinematic fitting and multivariate analyses, are often employed to address these challenges and extract maximum information from the dileptonic events.

Kinematic Constraints

Assuming both W bosons and top quarks are on-shell, the following kinematic constraints can be applied to the dileptonic decay channel,

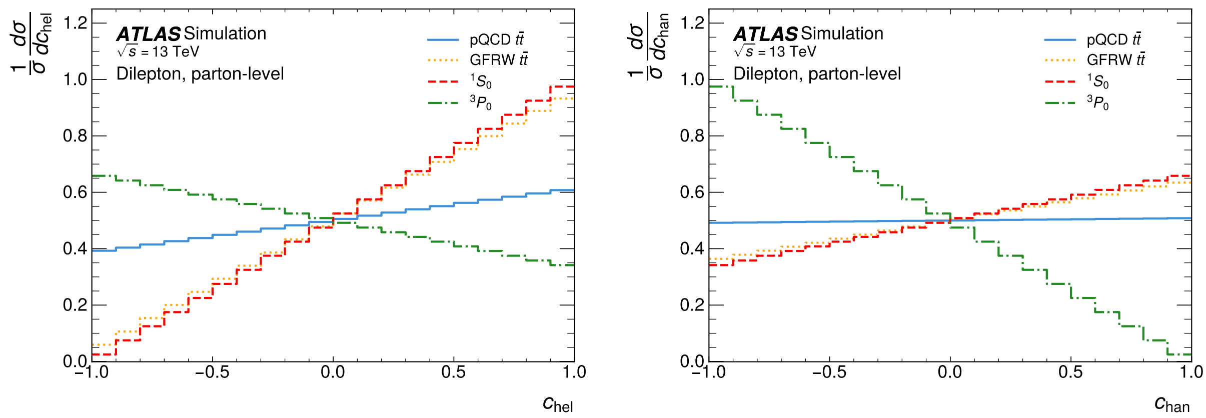
$$(\vec{p}_{\nu_1} + \vec{p}_{\nu_2})_x = \not{E}_{\mathrm{T}x}, \qquad (\vec{p}_{\nu_1} + \vec{p}_{\nu_2})_y = \not{E}_{\mathrm{T}y}, \tag{2.5}$$

$$(p_{\ell_1} + p_{\nu_1})^2 = m_W^2, \qquad (p_{\ell_2} + p_{\nu_2})^2 = m_W^2, \tag{2.6}$$

$$(p_{b_1} + p_{\ell_1} + p_{\nu_1})^2 = m_t^2, \quad (p_{b_2} + p_{\ell_2} + p_{\nu_2})^2 = m_t^2\,. \tag{2.7}$$

If one assumes the neutrinos to be massless, these six equations provide constraints on the six unknown components of the two neutrino momenta. Due to the algebraic nature of the constraints, multiple solutions can exist for a given event, leading to ambiguities in the reconstruction. Furthermore, the effects of detector resolution and off-shell particles can lead to the system of equations having no real-valued solution [44, 45]. This motivates the development of more sophisticated reconstruction methods.

2.2.4. Spin-Sensitive Angular Variables



(a) c_{hel} **(b)** c_{han}

Figure 2.6.: Simulation of the distributions of c_{hel} and c_{han} for dileptonic decaying top-quark pairs produced near threshold. Distributions are shown for different spin states, as well as calculations using perturbative QCD (pQCD) or a modelling of the quasi-bound-state effects using a Green's Function Reweighting (GFRW) [43]. The distributions are normalised to unity and computed at parton level. The Figures are taken from reference [2].

To study spin correlations and entanglement in top-quark pairs, a spin-sensitive angular variable is used. To measure quasi-bound-state effects, an additional angular variable is employed. Both variables are defined using the charged leptons from the dileptonic decay of the top-quark pair.
To compute the variables, top quarks and leptons are first boosted to the rest frame of the $t\bar{t}$ system.

Their momentum vectors are boosted once more into their respective parent top quarks' rest frame. The normalised vectors of these transformed momenta are labelled as $\hat{\ell}_t^+$ and $\hat{\ell}_t^-$. The inner product of these unit vectors then defines the variable

$$c_{\mathrm{hel}} = \hat{\ell}_t^+ \cdot \hat{\ell}_t^- \,. \tag{2.8}$$

Analogously, one can define the variable c_{han}, where either of the lepton momenta components parallel to the parent top-quark momentum in the $t\bar{t}$ rest frame is flipped before taking the inner product [46, 47]. This variable adds important context by enabling isolation of scalar states. The differential cross-section of the spin-states as a function of c_{hel} or c_{han} are obtained from simulation and found in figure 2.6. The distributions clearly show the distinctive shapes of the different spin states. Further, a stark difference between the slope of the perturbative QCD computation and the modelling of the quasi-bound state can be found, referred to as pQCD $t\bar{t}$ and GFRW $t\bar{t}$ in the plots, respectively. This highlights the variables' sensitivity to the spin structure of the $t\bar{t}$ system.

3. Experimental Setup

3.1. The LHC

The Large Hadron Collider (LHC) at CERN is a synchrotron designed to accelerate protons and lead nuclei. With a circumference of 27 km at a depth of 100 m, it passes through France and Switzerland. Its main operational goal was the discovery of the Higgs boson, which was achieved by the ATLAS and CMS Collaborations during Run 1 [48, 49] in 2012. Since then, the portfolio of measurements done at the LHC has grown significantly. This work focuses on the proton-proton collisions during Run 2. During Run 2 from 2015 to 2018, the LHC operated at a centre of mass energy of $\sqrt{s} = 13$ TeV with an integrated luminosity of $\mathcal{L}_{\text{int}} \approx 140\ \text{fb}^{-1}$ [50]. After completion of Run 3 in June 2026 and an operational pause, the LHC will operate with increased luminosity as the High-Luminosity Large Hadron Collider (HL-LHC). During this phase, it is planned to accumulate data with an integrated luminosity of around $\mathcal{L}_{\text{int}} = 3\ \text{ab}^{-1}$ [51].

3.2. The Detector

The ATLAS detector is a general-purpose particle detector located at the LHC at CERN. With a length of 46 m and a diameter of 25 m, it represents the largest detector constructed for a particle collider.

The cylindrical detector comprises three primary elements: the inner detector, the calorimeter, and the muon spectrometer [52]. These components are arranged concentrically around the beam pipe, each enclosing the one before it. The detector is further supported by a magnet system comprising two distinct magnets: a solenoid magnet surrounding the inner detector and a set of toroidal magnets that encompass the muon spectrometer and end-caps.

Each component of the detector fulfils a specific function. The inner detector enables tracking, identification, and vertex reconstruction of charged particles. The calorimeters determine the energy and position of charged particles, photons, and hadrons, thereby supporting their measurement. The muon spectrometer identifies muons and measures their momenta.

3.2.1. The Coordinate System

The interaction point marks the origin of the ATLAS coordinate system. The z-axis is orientated along the beamline, the x-axis points towards the centre of the LHC and the y-axis points upwards.

Using this convention, several kinematic variables are defined, such as the transverse momentum, defined as

$$p_T = \sqrt{p_x^2 + p_y^2} \tag{3.1}$$

Because of the ATLAS detector's cylindrical shape and the cylindrical symmetry of the interactions, the azimuthal angle is used as an additional variable. The polar angle relative to the z-axis can be used to define the pseudorapidity, which can also be expressed in terms of the momentum

$$\eta = -\ln\left(\tan\left(\frac{\theta}{2}\right)\right) = \ln\left(\frac{|\vec{p}| + p_z}{|\vec{p}| - p_z}\right), . \tag{3.2}$$

In the ultra-relativistic limit $m \ll |\vec{p}|$, the pseudorapidity is equal to the rapidity, defined as

$$y = \frac{1}{2}\ln\left(\frac{E + p_z}{E - p_z}\right), . \tag{3.3}$$

These variables are convenient for usage in the ATLAS experiment because p_T, ϕ, and differences of η are invariant under Lorentz-boosts along the z-axis.

3.2.2. Inner Detector

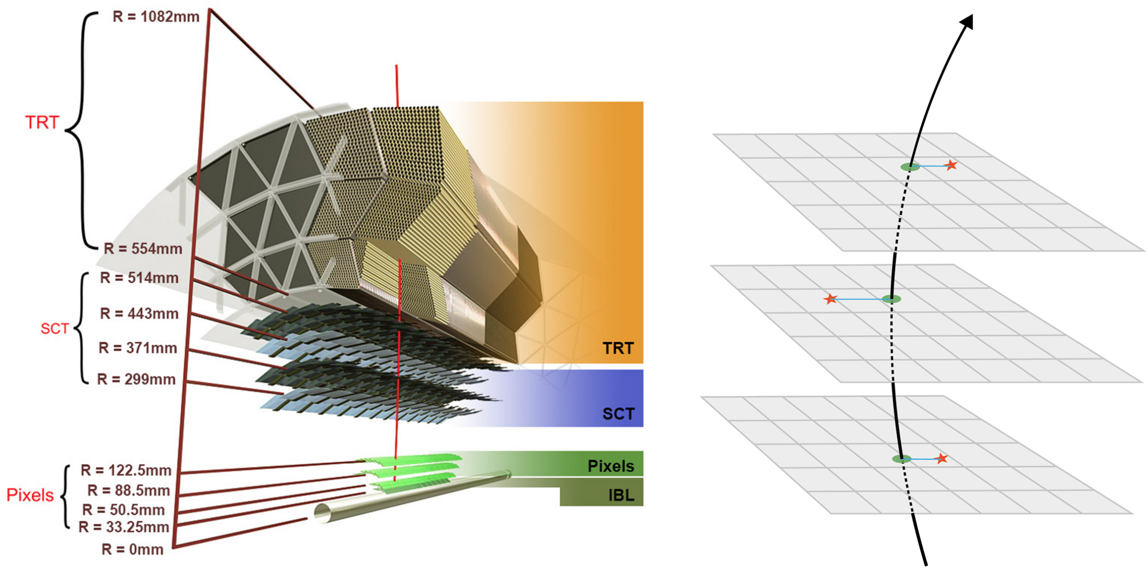


Figure 3.1.: Schematic cross-section of the inner detector of the ATLAS detector (© CERN).

The inner detector (ID) is the innermost part of the ATLAS detector. It is used for tracking charged particles, particle identification, and primary- and secondary-vertex determination [52]. The ID consists of three different sub-detectors, in addition to the Insertable

B-Layer (IBL), which was installed after Run 1 [53]. Their rough structure is depicted in figure 3.1. All the ID's components are encapsulated within a solenoid magnet, providing a 2 T axial magnetic field inside the ID. The magnetic field is crucial for measuring the momentum of a charged particle, as it enables the determination of the particle's transverse momentum via the curvature induced by the Lorentz force.
The detector part closest to the beam pipe is the pixel detector. It consists of 1744 modules arranged in three barrel layers. Each module hosts 47232 silicon pixels with a size of 50 µm × 400 µm. The pixels are semiconductor trackers used to detect the tracks of charged particles. An additional layer, the IBL, was installed after Run 1 closest to the beam pipe. It is made up of 14 starves, each comprising 32 modules of silicon pixel sensors with a pixel size of 50 µm × 250 µm. It mainly served to improve vertexing performance, thereby enhancing b-tagging performance.
The following part of the detector is the semiconductor tracker (SCT). It consists of 4088 modules arranged in four layers to ensure four-position measurements of charged particles. Each module consists of four silicon sensors. The sensors offer a 17 µm spatial resolution in-plane laterally and 580 µm in-plane longitudinally.
The outermost part of the inner detector is the transition radiation tracker (TRT). It consists of polyimide drift (straw) tubes with a 4 mm diameter that are arranged in a 528 mm thick cylindrical layer around the beam pipe. The straw tubes are interleaved with fibres for the readout. The transition radiation tracker utilises the transition radiation emitted by charged particles traversing the interface between two media with different refractive indices. The TRT provides velocity measurements of charged particles, thereby enabling electron identification.

3.2.3. Calorimeter System

The calorimeters are the detector layers following the inner detectors. Its structure is divided into the electromagnetic calorimeter (ECal), the hadronic calorimeter (HCal) and the forward calorimeter (FCAL).

Electromagnetic Calorimeter

The ECal is used for the energy and position measurement of electrically charged particles and photons. It utilises bremsstrahlung and pair production to create a cascade of charged particles that is measured. Apart from offering full azimuthal coverage, it is equipped with end caps along the beam pipe. The ATLAS ECal is a sampling calorimeter using lead as the passive medium and liquid argon as the active medium. The innermost part of the ECal is a presampler that detects whether the particle started showering before reaching the ECal.

Hadronic Calorimeter

The HCal measures the energy and position of baryons and mesons through strong interactions with the nuclei. In the range $|\eta| < 1.7$, it is a sampling calorimeter with steel as the passive medium and scintillators as the active medium. For the end cap, liquid argon is deployed as the active medium. The HCal operates on the same principle as the ECal

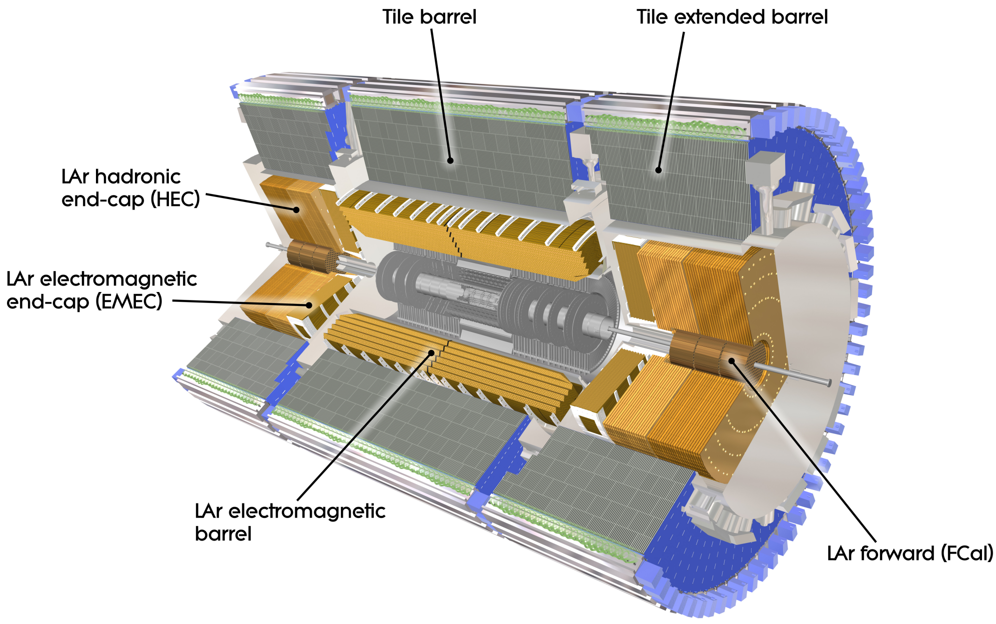


Figure 3.2.: Computer generated image of the ATLAS calorimeter (© CERN).

but provides less precise measurements. This is attributed to greater fluctuations in the shower shape, invisible energy losses, and coarser HCal sampling, which is necessary due to the larger length scale of hadronic showers.

3.2.4. Forward Calorimeter

In the range of $3.1 < |\eta| < 4.9$, ECal and HCal are substituted with the forward calorimeters (FCal), which are made up of three modules to fulfil the function of both calorimeters. In combination with the FCal, the calorimeters cover the full range $|\eta| < 4.9$.
A computer-generated image of the structures of the various calorimeters used in the ATLAS detector is shown in figure 3.2.

3.2.5. Muon Spectrometer

The muon spectrometer is the outermost part of the ATLAS detector. Its purpose is to detect charged particles exiting the calorimeters and measure their momentum in the range of $|\eta| < 2.7$. A detector dedicated to measuring muons is necessary because their mass makes them minimally-ionising particles in the energy range of LHC collisions. The energy loss due to bremsstrahlung is insufficient to produce the showers needed to measure their energy in the calorimeters. Like the inner detector, the muon spectrometer utilises a magnetic field to measure the particle's momentum. In the muon spectrometer, however, a toroidal magnet is used to allow for the measurement of the muon's momentum along a different direction. By combining measurements from the ID and the muon spectrometer, a full determination of the muon's momentum is possible, alongside high specificity in particle identification.

3.2.6. Trigger System

With a bunch spacing of 25 ns [50] collisions happen at a rate of 40 MHz. Each collision involves hundreds of particles, resulting in detector readout data rates that exceed the bandwidth of post-processing pipelines and data storage. Therefore, triggers are employed to target specific physics processes and events of particular interest.
Different layers of triggers operate either at the hardware or software level. The L1 hardware-level trigger serves as the first filter for events. It makes decisions in less than 2.5 µs, reducing the data rate to 100 kHz. The subsequent High-Level Trigger (HLT) is a software-level trigger that makes decisions in less than 200 µs. Combined, the trigger system reduces the event rate from 40 MHz to 1 kHz, with events stored at the CERN data centre for offline analysis.

4. Machine Learning in Particle Physics

Since this work uses supervised machine learning to improve the event reconstruction of dileptonic $t\bar{t}$ decays, this chapter provides an overview of the fundamental concepts and methodologies employed in machine learning.
From a computational perspective, machine learning can be viewed as the optimisation of a function, $f_{\boldsymbol{\theta}} : \mathbb{R}^n \to \mathbb{R}^m$, that maps input data points, $\mathbf{x} \in \mathbb{R}^n$, to output predictions $f_{\boldsymbol{\theta}}(x) \in \mathbb{R}^m$. The function $f_{\boldsymbol{\theta}}$ is parameterised by a set of learnable parameters $\boldsymbol{\theta}$, which are adjusted during the training process to minimise a predefined *loss function* $L(\mathbf{y}, f_{\boldsymbol{\theta}}(\mathbf{x}))$. The loss function quantifies the discrepancy between the outputs and the target values (labels), guiding the optimisation process. Its average over a dataset or batch will be referred to as batch loss and denoted as $\mathcal{L}$. Note, that bold letters and symbol, e.g. $\mathbf{y}$, $\boldsymbol{\theta}$, or $\mathbf{x}$ represent potentially multidimensional objects.

4.1. Supervised Learning

Machine learning can be broadly categorised into supervised and unsupervised learning. This thesis focuses on the former of the two. In supervised learning, the model is trained on a labelled dataset, where each input data point is associated with a corresponding label. The goal of the model is to learn a mapping from inputs to outputs, enabling it to make accurate predictions on unseen data.

4.1.1. Training with Monte Carlo Simulations

As mentioned before, supervised training requires labels associated with the input data. While this can be a major bottleneck for many machine learning applications, this limitation is largely absent in particle physics. In section 2.1.4, it was elaborated on how fundamental physics is simulated at colliders. Events are sampled from the underlying probability distributions of the produced particles and then further processed to match the observed detector-level structure. Thus, the necessary labels that are required for supervised training of the model are readily available for simulated events.
During the training phase, the model is presented with a set of input features derived from the reco-level event variables, along with their corresponding label values, which are typically derived from the parton-level event variables. The training process involves iteratively updating the model's parameters (each step is called an *epoch*) using an optimisation algorithm to minimise the loss function. Since the structure of the simulated events

closely resembles the data, the model is expected to smoothly generalise from simulation to data.

4.1.2. Validation and Testing

To evaluate the performance of the trained model, it is essential to validate and test it on independent datasets that were not used during training. This helps to assess the model's generalisation capabilities and ensures that it can make accurate predictions on unseen data. A common practice is to split the total available data into training, validation, and test sets. The validation set is used to tune hyperparameters and monitor the model's performance during training, while the test set is reserved for the final evaluation.
In particle physics simulated data is often used for training and background modelling. Therefore, one typically employs a k-fold strategy [54], in which the data for training and validation is divided into k subsets. The model is trained k times, each time using a different subset as the validation set and the remaining subsets for training. The test set, on the other hand, is disjunct from all trainings and validation sets.

4.2. Regression and Conditional Density Learning

While the mapping of inputs to outputs is frequently viewed as a one-to-one correspondence, the outputs and labels themselves are often random variables. That is, instead of a deterministic mapping, the outputs follow a conditional probability distribution $p(\mathbf{y}|\mathbf{x})$. This distinction is particularly relevant in reconstruction tasks, such as neutrino regression. Because only two of the six unknown neutrino components are constrained by the missing transverse energy, multiple distinct parton-level neutrino momenta can produce the exact same detector-level event. Consequently, any mapping from detector-level observables to parton-level kinematics is inherently ambiguous. In practice, because of high dimensionality and finite datasets, the probability of observing two identical detector-level events is virtually zero. However, this underlying phase-space ambiguity fundamentally alters the learning process and the behaviour of the loss functions. This section explores the underlying foundation of this.

4.2.1. Regression

For regression tasks the output $\mathbf{y}$ is a continuous variable, and the model is trained to predict a *point estimate*, that is, a specific value of $\hat{y}_{\text{pred.}} = f_{\boldsymbol{\theta}}(\mathbf{x})$ for each input $(\mathbf{x})$. In the framework of a stochastic output variable, the supervised learning of such a model can be formalised as a risk minimisation problem. The objective is to minimise the expected loss under the joint distribution of inputs and outputs. The expected risk can be expressed as:

$$R[f_{\boldsymbol{\theta}}] = \mathbb{E}_{\mathbf{x},\mathbf{y}}[L(\mathbf{y}, f_{\boldsymbol{\theta}}(\mathbf{x}))] \tag{4.1}$$

where L is the loss function–quantifying divergence of outputs and labels– and $\mathbb{E}_{\mathbf{x},\mathbf{y}}$ denotes the average over distributions of the random variables $\mathbf{x}$, $\mathbf{y}$ [1]. The goal of the training procedure is to find the parameters $\boldsymbol{\theta}$ that minimise this expected risk. The loss function commonly used for regression is the mean squared error (MSE), which measures the average squared difference between the predicted values and the true values:

$$\mathcal{L}_{\mathrm{MSE}}[f_{\boldsymbol{\theta}}] = \frac{1}{N}\sum_{i=1}^{N}(\mathbf{y}_i - f_{\boldsymbol{\theta}}(\mathbf{x}_i))^2\,, \tag{4.2}$$

where N is the number of events in the dataset. The model learns to minimise this batch loss. If the labels are, however, not deterministic, meaning that there is inherent uncertainty in the target values, one can describe the labels using a conditional probability distribution $p(\mathbf{y}|\mathbf{x})$. In this case, minimising the batch loss with respect to the MSE corresponds to learning the conditional mean of the distribution (see section B.1.1 for a proof):

$$f_{\boldsymbol{\theta}}(\mathbf{x}) = \mathbb{E}_{\mathbf{y}|\mathbf{x}}[\mathbf{y}] = \int \mathbf{y}p(\mathbf{y}|\mathbf{x})d\mathbf{y},\,. \tag{4.3}$$

Using different loss functions implies that the model will learn different statistics of the conditional distribution. For example, using the mean absolute error (MAE) loss function instead of the MSE loss function would cause the model to learn the conditional median rather than the conditional mean.

4.2.2. Conditional Density Learning

In contrast to regression, conditional density learning aims to learn the entire conditional probability distribution $p(\mathbf{y}|\mathbf{x})$ rather than just a point estimate. This is particularly useful when the target variable has inherent uncertainty or when multiple outcomes are possible for a given input. To quantify the difference between the true conditional distribution and the model's predicted distribution, the Kullback-Leibler (KL) divergence is often used as a loss function [55]

$$L_{\mathrm{KL}}(p, q_{\boldsymbol{\theta}}) = \int p(\mathbf{y}|\mathbf{x}) \log\left(\frac{p(\mathbf{y}|\mathbf{x})}{q_{\boldsymbol{\theta}}(\mathbf{y}|\mathbf{x})}\right) d\mathbf{y}\,, \tag{4.4}$$

where $p(\mathbf{y}|\mathbf{x})$ is the true conditional distribution and $q_{\boldsymbol{\theta}}(\mathbf{y}|\mathbf{x})$ is the model's predicted distribution. Minimising with respect to this loss function encourages the model to produce a predicted distribution that closely matches the true distribution, allowing for a more comprehensive understanding of the uncertainty and variability in the target variable. In practice, computing the KL divergence can be challenging because the true distribution may be unknown or difficult to estimate. In such cases, one relies on the data to provide an empirical estimate of the true distribution, and the loss function is approximated using events from the dataset. It can be shown that minimising the KL divergence is equivalent to maximising the logarithmic likelihood of the observed data under the model's

[1] In the case of a deterministic mapping between inputs and labels this joint average over both inputs and labels reduces to an average over the inputs only.

predicted distribution, which is a common approach in training probabilistic models; see section B.1.2 for details. One notable case is a discrete target variable, for which this corresponds to minimising the cross-entropy loss, a widely used measure in classification tasks [56].

4.3. Neural Networks

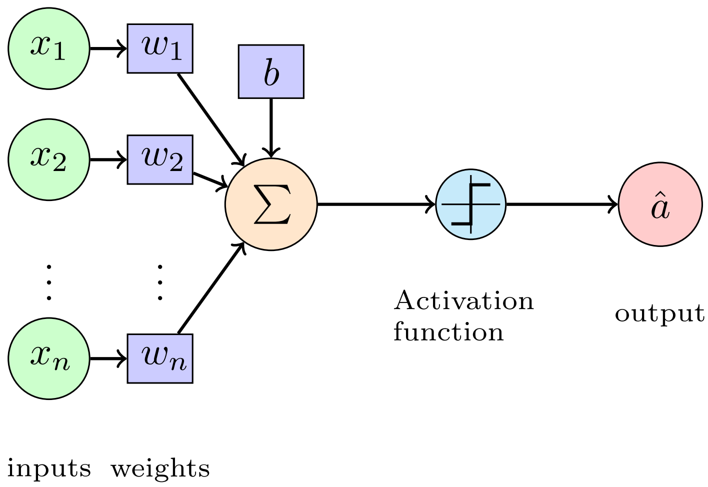


Figure 4.1.: Schematic of a single artificial neuron. The neuron receives multiple input signals, each associated with a weight, and computes a weighted sum of these inputs; a so-called bias is added. An activation function is then applied to this sum to produce the neuron's output.

As elaborated upon before, machine learning models can be viewed as parameterised functions that map input data to output predictions. Neural networks are a class of machine learning models that are inspired by the structure and function of biological neural networks in the human brain. They consist of interconnected artificial neurons that process and transform the input data to produce the desired output. The simplest unit of a neural network is the artificial neuron, also known as a perceptron [57]. An artificial neuron takes multiple input signals, each associated with a weight, computes a weighted sum of these inputs, and potentially adds a so-called bias. An activation function is then applied to this sum to produce the neuron's output. This structure is illustrated in figure 4.1. The general structure of an artificial neuron is inspired by biological neurons. The activation function introduces non-linearity into the model, allowing it to learn complex patterns in the data. Common activation functions include the sigmoid function, hyperbolic tangent (tanh), and rectified linear unit (ReLU) [58]. For this work, the ReLU activation function is used, which is defined as $f(x) = \max(0, x)$, meaning that it outputs the input directly if it is positive, and zero otherwise. This activation function has been shown to be effective in training deep neural networks by mitigating the vanishing gradient problem [58].

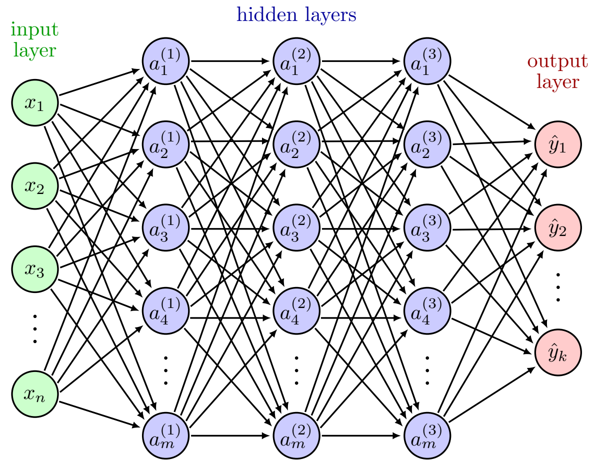


Figure 4.2.: Schematic of a dense neural network with multiple layers. Each layer consists of multiple neurons, and each neuron in one layer is connected to every neuron in the subsequent layer.

4.3.1. Dense Neural Network

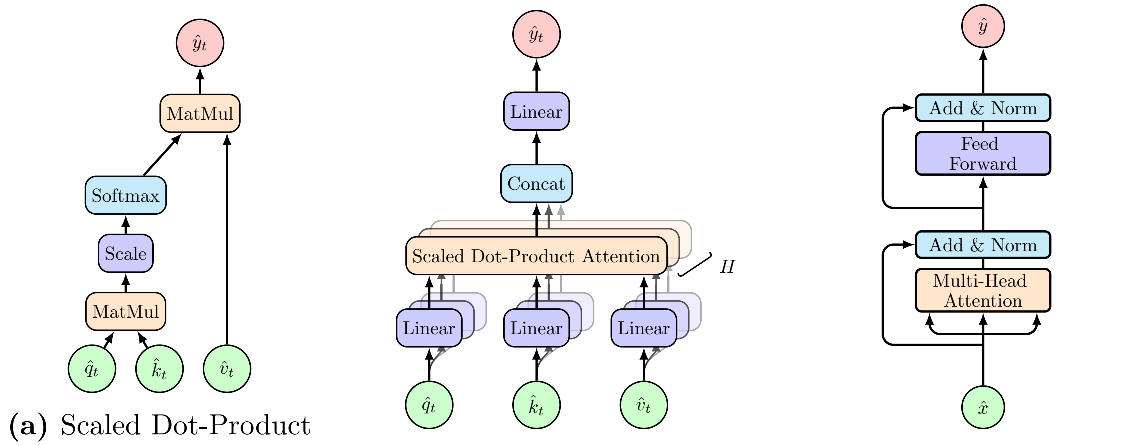
A dense neural network (DNN), also known as a fully connected neural network [59], is an artificial neural network in which each neuron in one layer is connected to every neuron in the subsequent layer. This architecture enables learning of complex relationships between input features and output targets. A schematic of a dense neural network is shown in figure 4.2.

A DNN consists of an input layer, one or more hidden layers, and an output layer. The input layer receives the input features, while the hidden layers perform a series of transformations on the data using the weights and biases associated with each neuron. The output layer produces the final predictions. Each layer applies an activation function to the weighted sum of inputs from the previous layer, enabling the network to learn non-linear mappings. The depth (number of layers) and width (number of neurons per layer) of the network can be adjusted and therefore optimised.

4.3.2. Transformer Networks

The transformer architecture deals with sequential data. The core component of the transformer architecture is the attention mechanism, which allows the model to focus on different parts of the input sequence when making predictions. This structure allows transformers to process information in parallel, making them efficient for training on large datasets. Additionally, transformers can effectively capture long-range dependencies in the data by relating all elements of the input sequence to one another through attention mechanisms.

The most commonly used attention mechanism is the *scaled dot-product attention* [60], illustrated in figure 4.3a. In this mechanism, three vectors are computed for each input element: the query (q), key (k), and value (v) vectors. The attention scores are calculated by taking the dot product of the query and key vectors, scaling the result by the square



(a) Scaled Dot-Product Attention **(b)** Multi-Head Attention **(c)** Transformer Layer

Figure 4.3.: Different components of the transformer architecture from smallest to largest unit: (a) Scaled Dot-Product Attention, (b) Multi-Head Attention, and (c) Transformer Layer. These structures were first outlined in reference [60].

root of the key vector dimension, and applying a softmax function to obtain attention weights. These weights are then used to compute a weighted sum of the value vectors, yielding the attention mechanism's output.

Multi-head attention, shown in figure 4.3b, extends the scaled dot-product attention by allowing the model to attend to different parts of the input sequence simultaneously. This is achieved by using multiple parallel attention heads, each with its own set of learned linear transformations for the query, key, and value vectors. The outputs of all attention heads are concatenated and linearly transformed to produce the final output.

Multi-head attention is embedded in a structure called the *transformer layer*, as depicted in figure 4.3c. This layer combines the attention mechanism with trainable normalisation layers and residual summation of the inputs. Lastly, the outputs pass through a feed-forward layer consisting of a dense layer with ReLU activation that projects into a space of twice the input dimension, followed by a linear projection back to the input dimension. These outputs are once more added to the inputs and finally scaled by a normalisation layer [60].

The key, query and value vectors are typically obtained by applying learned linear transformations for the input embeddings. Depending on the type of relational information one wants to capture, one can choose to derive these vectors from different input sources. For example, in the *self-attention mechanism*, all three vectors are derived from the same input sequence ($\hat{q}_t = \hat{k}_t = \hat{v}_t = \hat{x}_t$), allowing the model to relate different elements within the same sequence. In *cross-attention*, the query vectors are derived from one sequence, while the key and value vectors are derived from another sequence ($\hat{q}_t = \hat{x}_{1,t}$, $\hat{k}_t = \hat{v}_t = \hat{x}_{2,t}$), enabling the model to relate information between two different sequences.

Symmetries In many applications, including particle physics, the input data exhibits certain symmetries that can be exploited to improve model performance.

The attention mechanism has inherent permutation equivariance that can be exploited. Self-attention is permutation-equivariant to the input sequence, meaning that shuffling the input elements results in a corresponding permutation of the output elements. This property is particularly useful when dealing with sets of unordered data, where the order of the elements does not carry any inherent meaning. Cross-attention is also permutation-equivariant with respect to permutations of the query inputs and separately invariant to permutations of the key/value inputs. This allows the model to effectively relate information between two different sequences, regardless of their order.

4.3.3. Training Neural Networks

Training neural networks involves optimising the model's parameters (weights and biases) to minimise the batch loss. The number of parameters in a neural network can rise rapidly, especially in deep architectures, making the optimisation process computationally intensive. The most commonly used optimisation algorithm for training neural networks is stochastic gradient descent (SGD) [61], which iteratively updates the model's parameters based on the gradients of the batch loss with respect to the parameters. Modern variants of SGD, such as Adam [62], incorporate adaptive learning rates and momentum to improve convergence. The backpropagation algorithm [59] is used to efficiently compute the gradients of the batch loss with respect to the model's parameters. It involves propagating the error signal backwards through the network, allowing for the calculation of gradients at each layer. Based on the gradients, the optimisation algorithm updates the parameters towards the minimum of the batch loss. Regularisation techniques, such as dropout [63] and weight decay [64], are often employed during training to prevent overfitting and improve the model's generalisation capabilities. Dropout randomly deactivates a fraction of the neurons during training, forcing the network to learn more robust features. Weight decay adds a penalty term to the batch loss based on the magnitude of the weights, discouraging overly complex models and promoting generalisation.

5. Analysis and Reconstruction of Dileptonic Top-Quark Pair Events

5.1. Data and Simulation

This section provides a brief overview of the modelling and reconstruction of top-quark pair events. The simulated sample of $t\bar{t}$ events is generated using a Monte Carlo event generator as described in section 2.1.4.

While a complete analysis requires a detailed investigation of background processes contributing to the signal regions, the following chapters focus on developing and evaluating machine-learning methods applied to $t\bar{t}$ events. Therefore, only the process of $t\bar{t}$ production with dileptonic decays is described here. The background processes and their simulation are discussed in chapter 8, which investigates the tool's performance in a physics analysis.

All the samples used here are from the ATLAS MC simulation campaign. Thus, all plots and results are obtained with the courtesy of the ATLAS collaboration.

5.1.1. Simulation of Dileptonic Top-Quark Pair Events

The process of $t\bar{t}$ production with dileptonic decays is simulated using POWHEG [65] at next-to-leading order (NLO) in perturbative quantum chromodynamics (QCD). The POWHEG generator is interfaced with PYTHIA [25] for parton showering, hadronisation, and underlying event modelling. A more detailed description can be found in the reference [2].

To link the reco-level particles to the parton-level objects, a parton matching is performed. The parton-level objects are defined as top quarks and their decay products, after final-state radiation, before hadronisation. The matching is performed by iteratively associating reconstructed jets with partons based on angular-distance criteria. The decay products of the top quarks at parton-level are matched to the reco-level particles within a cone of $\Delta R \leq 0.3$ and $\Delta R \leq 0.1$ for jets and leptons (e,μ) respectively. If multiple reconstructed objects are found within this cone, the one with the smallest ΔR is chosen. In case no reconstructed object matches the criteria, the parton is considered unmatched.

5.1.2. Object Definition

The reconstruction of physics objects—electrons, muons, jets, and missing transverse energy ($\not{E}_{\mathrm{T}}$)—is performed using standard ATLAS reconstruction algorithms [52].

Electrons are required to have transverse momentum $p_{\mathrm{T}} > 10$ GeV and pseudorapidity $|\eta| < 2.47$, excluding the transition region between the barrel and endcap calorimeters ($1.37 < |\eta| < 1.52$). Electrons must satisfy the tight identification criteria and be isolated from other activity in the detector [66, 67].

Muons are reconstructed using information from both the inner detector and the muon spectrometer. They are required to have $p_{\mathrm{T}} > 10$ GeV and $|\eta| < 2.5$. Muons must satisfy the tight identification criteria and be isolated [68, 69].

Jets are reconstructed using the anti-k_t algorithm [70] with a radius parameter of $R = 0.4$. They are required to have $p_{\mathrm{T}} > 25$ GeV and $|\eta| < 2.5$. b-tagging is performed using the GN2 algorithm [71] to identify jets originating from b-quarks.

Missing Transverse Energy ($\not{E}_{\mathrm{T}}$) is calculated from the negative vector sum of the transverse momenta of all reconstructed objects in the event, including a soft term to account for energy not associated with reconstructed objects.

5.1.3. Event Selection

Events are selected to contain exactly two oppositely charged leptons (electrons or muons). One of the leptons must have $p_{\mathrm{T}} > 25$ GeV, while the other must have $p_{\mathrm{T}} > 5$ GeV. At least two jets are required, with at least one b-tagged at the 85% working point (WP).
For training and evaluation of the reconstruction methods, only events that meet the selection criteria and have successful parton matching are considered. Fully parton-matched events account for about 65% of the total sample. Jets or leptons from the $t\bar{t}$ decay that end up outside the detector acceptance are the main cause of an event being unmatched. Thus, training only on parton-matched events might introduce a bias towards certain event topologies if they are disfavoured by parton matching.

5.2. Reconstruction of Top-Quark Pairs

Reconstructing the kinematics of dileptonic $t\bar{t}$ events requires an assignment of reconstructed objects to the decay products of the $t\bar{t}$-system. This section describes the different reconstruction methods developed and evaluated in this thesis.
While the leptons are directly identified from the reconstructed objects based on their charge, the assignment of jets to the b-jets from the top quark decays is ambiguous due to the presence of multiple jets in the event. Additionally, the two neutrinos from W boson decays are not directly detected, resulting in missing information in the event reconstruction. Based on this, the reconstruction breaks down into two main tasks:

Jet Assignment Identifying the correct reconstructed jets as the jets originating from b-quark decays

Neutrino Regression Estimating the momenta of the two neutrinos using the measured $\not{E}_{\mathrm{T}}$ and other event kinematics.

5.2.1. B-Tagging

The identification of b-jets is crucial for the reconstruction of $t\bar{t}$ events. Due to the long lifetime of B-hadrons, they travel a measurable distance from the primary vertex before decaying, leading to displaced vertices and tracks that do not point back directly to the primary vertex. This displacement is quantified by the impact parameter, defined as the distance of closest approach between a track and the primary vertex (in the transverse plane, d_0, and along the beam axis, z_0); tracks originating from B-hadron decays are thus characterised by large impact parameters relative to those from the primary vertex. The GN2 algorithm [71] is used for b-tagging in this analysis, which combines various discriminating variables, including track impact parameters and reconstructed secondary vertices, using a deep (graph) neural network to achieve high efficiency and purity in identifying b-jets. The performance of the b-tagging algorithm is characterised by its efficiency (the probability of correctly identifying a b-jet) and its misidentification rate (the probability of incorrectly tagging a non-b-jet as a b-jet).

5.2.2. Jet Assignment Methods

This thesis evaluates several methods for jet assignment, ranging from traditional algorithms to machine learning approaches. While the leptons can be assigned to the W bosons based on their charge, the assignment of jets to the decaying parent top-quark is not straightforward due to the presence of multiple jets in the event. All the methods presented here utilise kinematic relationships to group jets and leptons stemming from the same top quark decay. From recent publications, e.g. [1, 2], two conventional methods have been implemented as baselines:

Jet-Selection Both methods start by selecting the two b-jet candidates in the event using b-tagging at the 77% WP. If more than two b-tagged jets are present, the two with the highest p_{T} are chosen. If only one b-tagged jet is found, the non-tagged jet with the highest b-tagging score is selected as the second candidate.

χ^2-Method

For both possible assignments of the two b-jets to the parent top quarks, the invariant masses of the reconstructed top quarks are computed,

$$\chi^2 = (m_{b_1,\nu,\ell^+} - m_t)^2 + (m_{b_2,\bar{\nu},\ell^-} - m_t)^2, \tag{5.1}$$

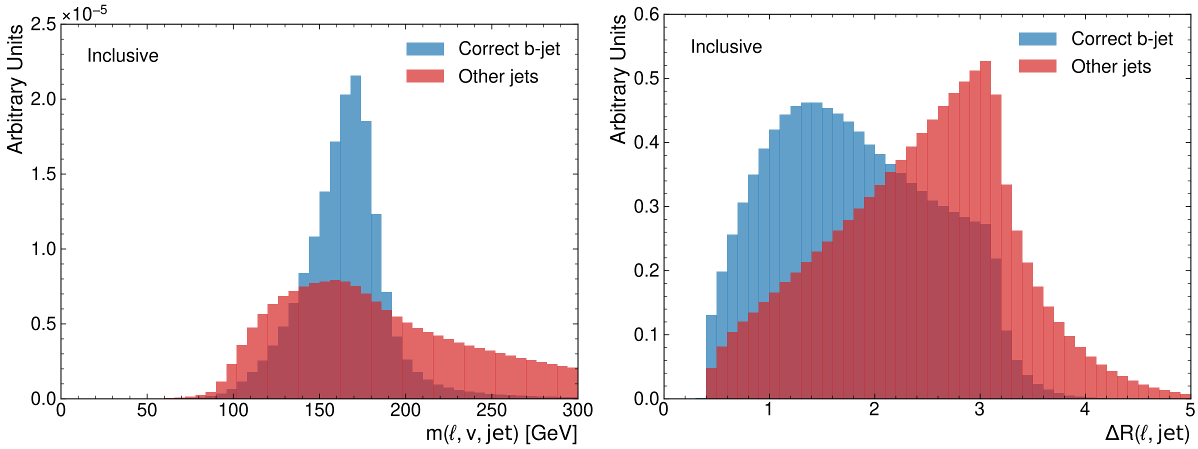
where $m_t = 172.5$ GeV is the top quark mass. The assignment with the smaller χ^2 value is chosen.

ΔR-Method

This method assigns the b-jet candidates based on their angular distance to the leptons.

$$\Delta R = \sqrt{(\Delta\phi)^2 + (\Delta\eta)^2} \tag{5.2}$$

The jet and the lepton that form the pair with the smallest dR are assigned to each other. The remaining jet is assigned to the other lepton.



(a) χ^2-Method **(b)** ΔR-Method

Figure 5.1.: Distributions for pairing up correct and incorrect jet-lepton pairs for the reconstructed invariant mass of the top quark (a) and the ΔR between the jet and lepton (b). The correct pairs are shown in blue, while the incorrect pairs are shown in red. The distributions are normalised to unit area. Note that the reconstructed invariant mass for the χ^2-Method includes the neutrino prediction from ν^2-Flows.

Note that the χ^2-Method relies on the reconstructed neutrino momenta to compute the invariant masses of the top quarks. Therefore, it is inherently linked to the neutrino regression method used. In contrast, the ΔR-Method is independent of the neutrino momenta.
Empiric evidence for the validity of both can be found in figure 5.1, which shows the distributions of the reconstructed invariant mass of the top quark and the ΔR between the jet and lepton for correct and incorrect jet-lepton pairs. The correct pairs show a clear peak around the top quark mass in the invariant mass distribution and a smaller ΔR compared to incorrect pairs, which supports the use of these variables for jet assignment. The ΔR-Method was used in previous ATLAS analyses [1, 2] for events that failed the kinematic reconstruction required for the χ^2-Method. However, in this thesis, both methods are evaluated on the full dataset for comparison.

5.2.3. Neutrino Regression Methods

Due to the limited availability of kinematic reconstruction methods in the ATLAS software framework, two methods for neutrino regression are implemented and evaluated in this thesis. In previous ATLAS analyses, the Ellipse method, which solves the kinematics constraints as introduced in section 2.2.3 was used. This method, however, is not evaluated here, due to the aforementioned software availability.

Neutrino-Weighter

The Neutrino-Weighter (ν-Weighter), first described in [72], is a method that assigns weights to different possible neutrino momentum configurations based on how well they satisfy the kinematic structure of the event using the measured $\not{E}_{\mathrm{T}}$ and the momenta of the leptons and jets to. These momentum configurations are obtained by sampling the neutrinos' pseudorapidities from a Gaussian distribution and computing the full 3-vector of the neutrinos by solving the system of equations derived from the constraints in section 2.2.3. The masses of the top quarks and W bosons are smeared around their nominal values, i.e. different values are considered and evaluated. For each configuration, a weight (called likelihood) is computed based on loose kinematic constraints associated with $t\bar{t}$ events – including on-shell requirements for the W bosons and top quarks. The configuration with the highest weight is then selected as the reconstructed neutrino momenta. This method is computationally intensive because it requires scanning through numerous possible neutrino momentum configurations. Also, the computation of the reconstructed masses for the top quarks and W-bosons requires identification of the jets with a lepton. In principle, this might be done by a sophisticated method such as the one to be developed in this thesis. However, the implementation used here selects the jets based on b-tagging at the 85% working point and the leading p_T if the number of b-tagged jets deviates from two. While this still leaves a twofold degeneracy in the assignment, both scenarios are evaluated, weighted, and considered to maximise the likelihood.

ν^2-Flows

The method ν^2-Flows [73] uses normalising flows to model the conditional probability distribution of the neutrino momenta given the observed event kinematics. The model is trained on simulated dileptonic $t\bar{t}$ events, in which the true neutrino momenta are known from parton-level information. The trained model can then be used to sample possible neutrino momenta for unseen events, enabling probabilistic reconstruction of the event kinematics. For each event, multiple samples of neutrino momenta are drawn from the model, and the one with the highest probability density is selected as the reconstructed neutrino momenta. A detailed outline of the ν^2-Flows method and its implementation can be found in the reference [73]. For this thesis, this method serves to establish a baseline for neutrino regression to evaluate the impact of improved jet assignment methods. Note, however, that this method has not yet been used in ATLAS analyses. Due to software compatibility issues, it was one of the few viable options to use as a baseline for this work. Since ν^2-Flows processes the full event information to predict the neutrino momenta, it does not require a jet-assignment. Therefore, it can be used in conjunction with both the χ^2-Method and the ΔR-Method for jet assignment
For this analysis, the regression from ν^2-Flows was found to deliver much better performance for the assignment. Thus, the χ^2-Method was ultimately used in conjunction with the prediction of ν^2-Flows, rather than the ν-Weighter, to provide a reasonable baseline.

6. ML-Based Reconstruction of Dileptonic Top-Quark Pair Events

This chapter presents the development of an ML-based reconstruction strategy for the dileptonic $t\bar{t}$ channel, aiming to improve upon the baseline reconstruction methods introduced in the previous chapter. In recent years, ML-based reconstruction techniques have become increasingly prominent, particularly in applications such as flavour tagging algorithms [74].

In the context of all-hadronic channels, where jet assignment involves high permutation multiplicity, SPANet [6] has demonstrated substantial improvements over traditional methods. These architectures employ attention mechanisms to enhance accuracy by leveraging complete event information and implementing permutation-equivariant designs to address the combinatorial complexity inherent in the problem. Building on these advancements, the present work investigates the adaptation of similar architectures for the dileptonic $t\bar{t}$ channel, which poses distinct challenges due to the presence of two neutrinos and a reduced number of jets. This area remains relatively unexplored in the literature, with most research concentrating on neutrino regression. Therefore, this study aims to address this gap by developing a machine learning model specifically designed for the jet assignment problem in the dileptonic channel.

Although jet assignment and neutrino regression are highly interrelated, they have predominantly been studied in isolation, and the potential synergies between these tasks remain largely unexplored. This thesis examines the integration of jet assignment and neutrino regression within a unified machine learning framework, facilitating a more comprehensive approach to event reconstruction in the dileptonic $t\bar{t}$ channel.

6.1. Jet-Assignment

Given the diversity of available neutrino regression methods, the jet assignment problem is initially examined independently, prior to exploring its integration with these methods within a unified framework. In general, the task the model is trained to solve is to associate jets and leptons originating from the same top-quark decay. To encode this, a matrix

$$P_{\text{pred}}(\ell^{\pm}, j_i) \in [0,1], \quad i \in 1, \ldots, N_{\text{jets}} \ , \tag{6.1}$$

is used as the network's output, with

$$\sum_i P_{\text{pred}}(\ell^{\pm}, j_i) = 1 \ , \tag{6.2}$$

was used. This ensures that each lepton is associated with exactly one jet. Since for the model training and evaluation, only fully truth-matched events are used, the labels fulfil

$$P_{true}(\ell^{\pm}, j_i) \in \{0, 1\}, \quad i \in 1, \ldots, N_{jets} \ , \tag{6.3}$$

with

$$\exists! j : P_{\text{true}}(\ell^{\pm}, j) = 1 \ . \tag{6.4}$$

Consequently, the task is formulated as a multi-class classification problem for each of the two leptons. The labels correspond to the one-hot encoded indices of the true matched jets for each lepton. For consistency, the labels are padded to a uniform length, N_{max}. Events in which the truth-matched jet index exceeds the selected padding length are excluded from training and evaluation, thereby ensuring that equation (6.4) is satisfied for all events.
In developing the model architecture, various neural network designs were evaluated. To accommodate the sequential nature of the jets, both recurrent and transformer networks were primarily considered. Jet-permutation-equivariant transformer networks consistently demonstrated superior performance.

6.1.1. Neural Network Architectures

In reference [6], it was investigated how permutation equivariance of transformer networks can greatly improve the performance and efficiency of the jet assignment in top-quark pair events. The SPANet architecture was devised to handle the large number of equivalent jet-assignment permutations, all of which yield the same reconstructed tops. While the number of equivalent permutations was at least 20 in the all-hadronic decay channel, it is only 2 for the dileptonic decay channel in the lowest jet-multiplicity. Even though it rises quadratically with the number of jets, this is reduced by using the available b-tagging information. Since this work investigates pairing jets with leptons, the only permutation equivariances are jet ordering and, potentially, lepton ordering. The latter arises from the fact that, to a good approximation, there is no difference between the decay and reconstruction properties of the top and anti-top, respectively.
To investigate this, three different transformer network architectures were considered and evaluated.

FeatureConcatTransformer (FCT)

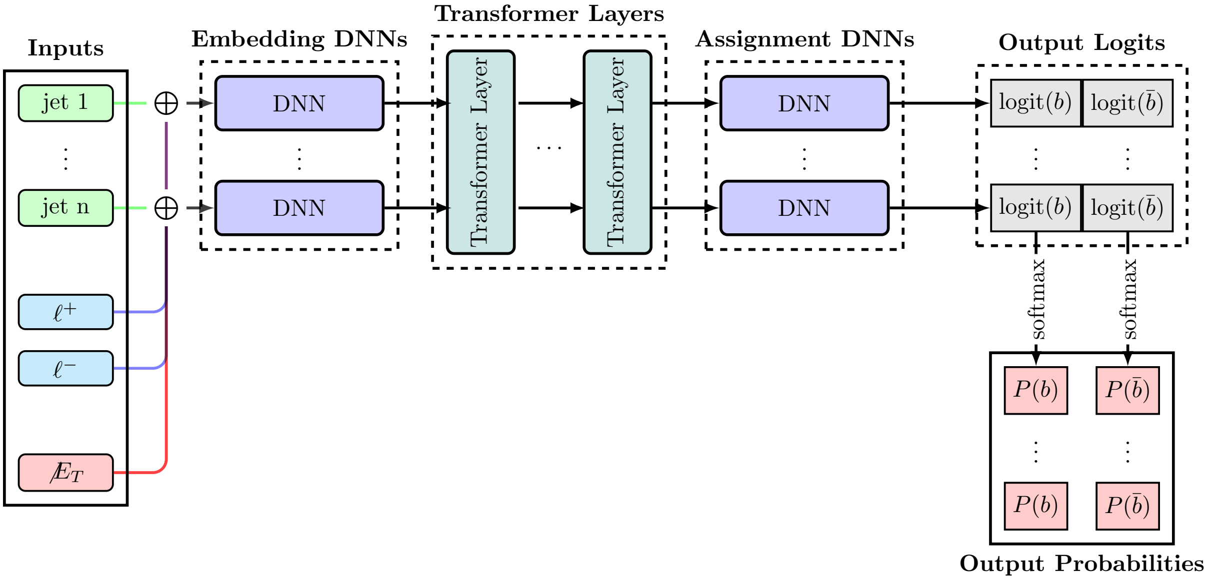


Figure 6.1.: This schematic visualises the general structure of the FeatureConcatTransformer architecture. A detailed description of the individual units making up this architecture is provided in this section. Note that $\oplus$ represents a concatenation of the corresponding features.

The FeatureConcatTransformer (FCT) architecture is outlined in figure 6.1. Its name-bearing component is the concatenation of the two leptons and the $\not{E}_{\mathrm{T}}$ features to each jet. This gives rise to a sequence of objects that resemble the jets observed in the event. By passing it through a shared DNN, this sequence is embedded into a fixed *Embedding Dimension* (ED), which is a model hyperparameter. The embedded sequence is passed through a number of transformer layers using self-attention [60]; the number of layers is another hyperparameter. The output elements of the central transformer are lastly processed by another shared DNN, which provides two assignment scores for each jet. These scores are normalised by applying a softmax function to each score across all jets. Due to the properties of the softmax function, this provides a score associated with the probability of each jet to be associated with the positive or negative lepton, respectively. The leptons are fed into the model with a fixed ordering based on their charge. The model inherits its permutation equivariance with respect to the jet-ordering from the self-attention mechanism. Regarding the lepton ordering, the model is not permutation equivariant. Since the underlying physics presumably is, the network can be expected to pick up this structure from the training data. In contrast to other architectures, the FCT also included the precomputed relational variables, $\Delta R(\ell, j_i)$ and $m(\ell, j_i)$ to provide additional context. While these were found to be speed up the training and add more transparency to feature evaluations, their impact on the performance was minimal[1].

[1]For models relying only on attention mechanism to relate jet and leptons, incorporating such variables is non-trivial. Therefore, these features were omitted in other architectures.

CrossAttentionTransformer (CAT)

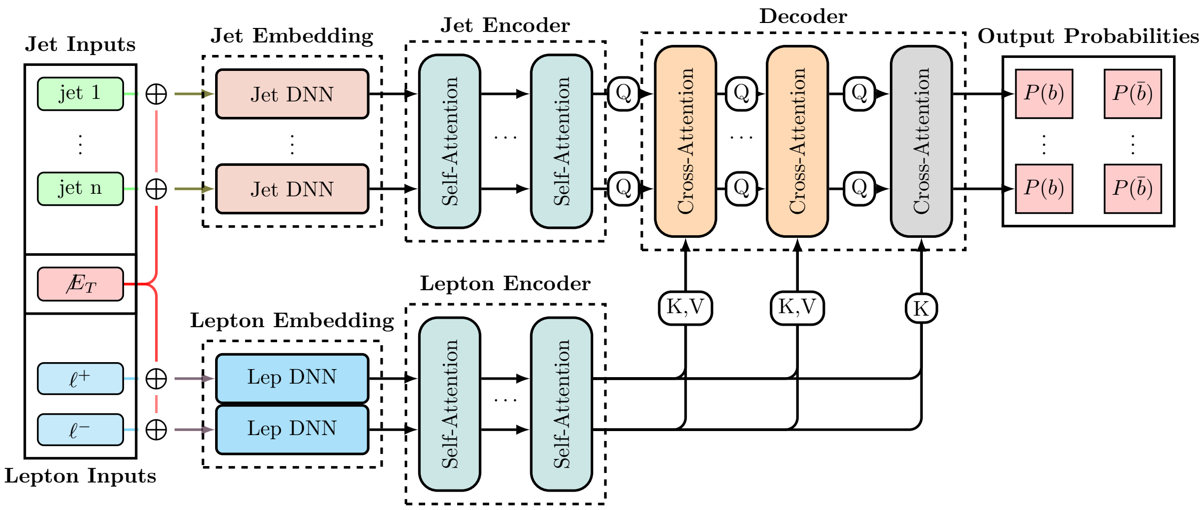


Figure 6.2.: This schematic visualises the general structure of the CrossAttentionTransformer architecture. A detailed description of the individual units making up this architecture is provided in this section. Note that $\oplus$ represents a concatenation of the corresponding features.

The CrossAttentionTransformer (CAT) architecture is outlined in figure 6.2. It is based on the idea of using cross-attention to directly model the interactions between the jets and leptons. Instead of concatenating their features, the leptons are treated as a sequence too. The $\not{E}_{\mathrm{T}}$ features are then concatenated to both the jets and the leptons. This architecture then encodes the jets and leptons separately, using shared DNNs to embed them into a fixed ED. The jets and leptons are then passed through separate transformer layers via self-attention, enabling the model to capture internal relationships within each set of objects. The outputs of the jet and lepton transformers are then combined in a decoder using cross-attention layers, enabling the model to learn interactions between jets and leptons. Finally, the last layer outputs the cross-attention scores for each jet-lepton pair using dot-product attention followed by a softmax function to provide the final probabilities for the jet-lepton assignments. For simplicity, the number of encoder and decoder layers is chosen to be the same. In addition to the FCT's permutation equivariance with respect to the jet-ordering, the CAT architecture is also permutation equivariant with respect to the lepton-ordering, due to the separate encoding and self-attention layers for the leptons. This allows the model to fully capture the underlying symmetries of the problem, potentially leading to improved performance.

CompactTransformer (CPT)

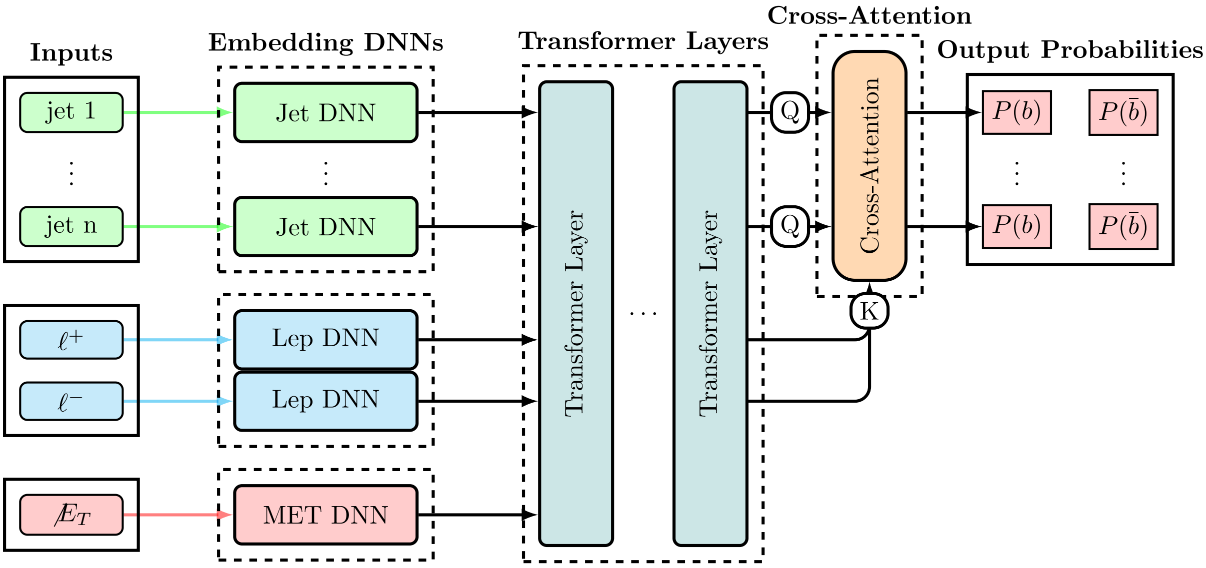


Figure 6.3.: This schematic visualises the general structure of the CompactTransformer architecture. A detailed description of the individual units making up this architecture is provided in this section.

The CompactTransformer (CPT) architecture is outlined in figure 6.3. It is designed to be a sleeker version of the CrossAttentionTransformer (CAT) architecture while maintaining permutation equivariance with respect to both jet and lepton orderings. The CPT embeds jets, leptons, and $\not\!\!E_{\mathrm{T}}$ features separately using shared DNNs, similar to the CAT architecture. However, rather than using separate transformer layers for jets and leptons, the CPT combines them into a single sequence and passes it through a shared self-attention transformer layer. This allows the model to capture both the internal relationships within each set of objects and the interactions between them in a more compact architecture. The final output layer provides the jet-lepton assignment scores using dot-product attention with a softmax function, similar to the CAT architecture. While the self-attention itself is permutation equivariant to the full sequence, the separate embeddings for the objects break this into permutation equivariance for the objects sharing the same embeddings—i.e. jets and leptons, respectively.

Common Properties

For all models, the input features for the jets and leptons include their kinematic properties such as transverse momentum (p_T), pseudorapidity (η), azimuthal angle (ϕ), and energy (E). Additionally, the b-tagging for the jets is included as the continuous quantile of the corresponding b-tagging working point, and the charge for the leptons is provided. The missing transverse energy (MET) features, including its magnitude and azimuthal angle, are also included as input to the model. Furthermore, global event features such as the number of jets and b-tagged jets, as well as the angular separation between the two leptons, are included to provide additional context for the model.

Each model includes a layer dedicated to normalising the input features, which helps to improve the training stability and convergence. These are set once on the training data and remain fixed during training. The normalisation is performed by computing the mean and standard deviation of each feature across the training dataset, and then using these statistics to standardise the features by subtracting the mean and dividing by the standard deviation. Further, quantities such as the p_T and energy of the jets and leptons are transformed using a logarithmic scale to better handle the wide range of values. The $E\!\!\!/_{\mathrm{T}}$ features are further complemented with additional global event variables, such as the number of (b-)jets and the angular-distance ΔR between the leptons.
Unless stated otherwise, all DNNs used here follow the same layout: it consists of a first layer that projects the input to twice the desired output dimension, applies a ReLU function, and once again applies a linear layer to the output dimension.
Using the model's forward pass, the jet-lepton assignment scores are computed for each jet and lepton pair. Next, each possible pairing of jets and leptons is scored by multiplying the probabilities for the positive and negative lepton. The pairing with the highest score is then selected as the model's prediction for the jet-lepton assignment. Solutions where the same jet is assigned to both leptons are discarded, as they are not physically meaningful. The maximum number of jets considered by the model is set to $N_{max} = 6$. This means that the model can handle events with up to 6 jets, which covers around 99.3% of dileptonic $t\bar{t}$ events. This cut-off was chosen to reduce the computational cost, which increases quadratically with the number of jets. Note, however, that for deploying a model in production, this can be easily adjusted and retrained with minimal impact on model performance. However, since this thesis employs investigations that require multiple retraining runs, this was found to be a reasonable trade-off.

6.1.2. Training Procedure

The training procedure for all three architectures is identical, as no major differences in training behaviour were observed. The models are trained using a cross-entropy loss function for each lepton, which measures the discrepancy between the predicted probabilities and the true labels. The total loss is computed as the sum of the losses for both leptons, and the loss for each event is scaled using sample weights. These weights are computed by combining the event weights from simulation with the inverse of the number of events per jet multiplicity, which helps mitigate potential biases in the training data and ensures the model learns effectively from all events, regardless of jet multiplicity. Model parameters are updated using the AdamW optimizer [64], a variant of the Adam optimiser that incorporates weight decay for regularisation. Furthermore, a dropout with a rate $r = 0.2$ is applied to all internal forward passes. The training is conducted over 100 epochs, with early stopping based on the validation loss to prevent overfitting. Additionally, learning rate scheduling adjusts the learning rate during training, which can improve convergence and overall performance. For the initial learning rate, 10^{-4} is chosen, and it is halved if the validation loss does not improve for 5 consecutive epochs. If the validation loss does not improve after 15 epochs, the training is terminated. The batch size is set to 4028, and the models are trained on a GPU to accelerate the training process.
During training, the model's performance is monitored on a separate validation set, com-

prising 20% of the initial training data, to ensure it does not overfit the training data. The best-performing model is selected based on the lowest validation loss, and its performance is further evaluated on a separate test set to assess its generalisation capabilities. In addition to the loss, other metrics, such as the accuracy of jet-lepton assignment, are reported during training to provide additional insights into the model's performance. If not reported otherwise, each model is trained on $5 \cdot 10^6$ fully parton-matched dileptonic top-quark pair events taken from the nominal $t\bar{t}$ sample described in section 5.1.1.

6.1.3. Hyperparameter Optimisation

To evaluate the performance of the different architectures, hyperparameter optimisation is performed for each architecture. The hyperparameters considered are the embedding dimension (ED) and the number of transformer layers. Other relevant hyperparameters, such as the learning rate, dropout rate, and batch size, were kept fixed across all models to ensure a fair comparison. The hyperparameter optimisation is conducted using a grid search, in which a predefined set of values for each hyperparameter is evaluated. For all architectures, the embedding dimension is varied in the range of $16, 32, 64, 128$, and the number of transformer layers is varied in the range of $4, 6, 8, 10$. The CAT architecture, uses the same number of layers for encoder and decoder layers is chosen to be equal. Each model-hyperparameter combination is trained and evaluated 5 times with different random seeds and k-fold training/validation splits to ensure the robustness of the results. The performance of the models is then compared using the average validation loss and the accuracy of jet-lepton assignment on a separate test set. From the variance of results across different random seeds and data splits, the statistical significance of performance differences between the architectures can be assessed. This study enables the identification of the most effective architecture and hyperparameter configuration, as well as insights into their scaling behaviour and robustness under different training conditions.
The individual results of the grid-search for each architecture are shown in figures A.10 to A.12 in the appendix. From these results, it can be observed that the performance of all architectures generally improves as the embedding dimension increases. Regarding the number of layers, performance appears more stable across different values, with a slight improvement observed for higher layer counts. However, in some instances, performance can degrade with very large numbers of layers, which may be due to overfitting or difficulties training deeper models. Overall, the results suggest that while both hyperparameters affect the model's performance, the embedding dimension appears to have a greater effect than the number of layers.
figure 6.4 compares the different architectures in terms of their (averaged) assignment accuracy as a function of the number of trainable parameters. Different aspects can be observed from these results. Regarding the individual architectures, the CPT architecture appears unstable at low embedding dimensions. While the other models consistently produce results across different random seeds and data splits, the CPT architecture exhibits large performance variance at low embedding dimensions, which decreases at higher embedding dimensions. This suggests that the CPT architecture may require a certain level of model complexity to perform well and may be more sensitive to hyperparameter choices than other architectures. This can be explained by the CPT architecture treating

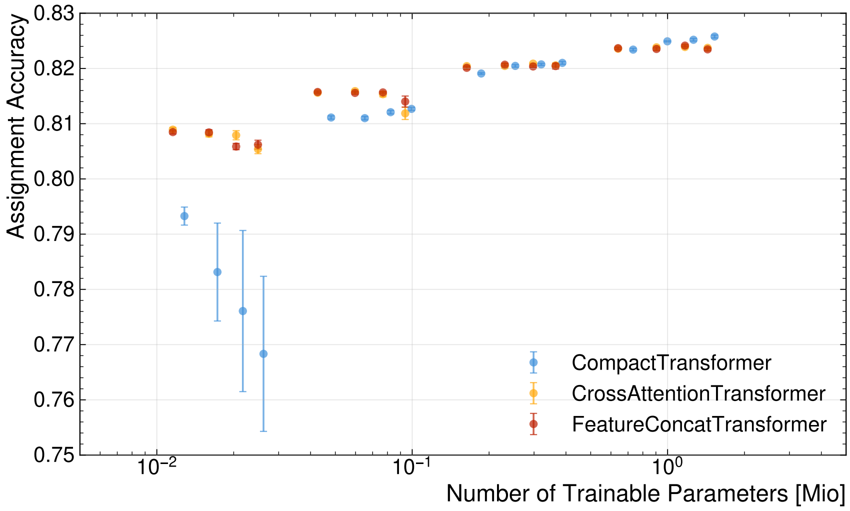


Figure 6.4.: Assignment Accuracy of all architectures as a function of the number of trainable parameters. Each point represents the average accuracy over 5 runs with different random seeds and k-folded training/validation data. The error bars indicate the standard deviation across the runs. Note that the scaling of the x-axis is logarithmic, which allows for a better visualisation of the differences in performance across different model sizes.

the jets, lepton, and $\not{E}_\mathrm{T}$ features as a single sequence, which may require a certain level of model complexity to effectively capture interactions among these feature types. For the FCT and CAT architectures, performance appears more stable across different hyperparameter configurations, with a clear trend toward improved performance as model parameter count increases and very little variation across training runs. This suggests that these architectures are more robust to hyperparameter choices and can achieve good performance even with smaller model sizes. Overall, it shows that the CPT architecture is more sensitive to hyperparameter choices, particularly the embedding dimension, than the FCT and CAT architectures. In contrast, the FCT and CAT architectures treat the jets and leptons separately, which may allow them to learn more effectively from the data even with smaller model sizes.

This plot also reveals why performance varies more across embedding dimensions than across layer counts: the embedding dimension has a much larger impact on the number of trainable parameters than the number of layers, which helps explain its more significant effect on the model's performance. Note that the number of parameters scales linearly with the number of layers and quadratically with the embedding dimension, which may explain the observed performance differences across hyperparameter configurations.

Apart from the instability of the CPT architecture for low embedding dimensions, the performance of the different architectures appears to be quite similar across different hyperparameter configurations. Generally, the performance seems to depend much more on the number of trainable parameters than on the specific architecture, with all architectures showing similar performance for a given number of parameters. This suggests that the choice of architecture may be less important than ensuring the model has sufficient capacity to learn the underlying patterns in the data, and that performance can be im-

proved by increasing model size rather than changing the architecture. Also, the layout of the different models (e.g. number of layers) appears to be less important than the overall model capacity.
figure 6.5 compares the different architectures in terms of their inference time as a function

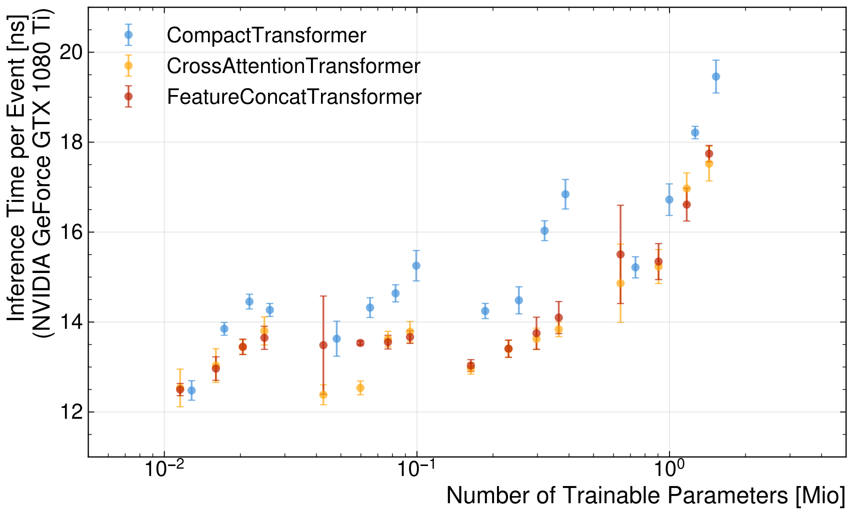


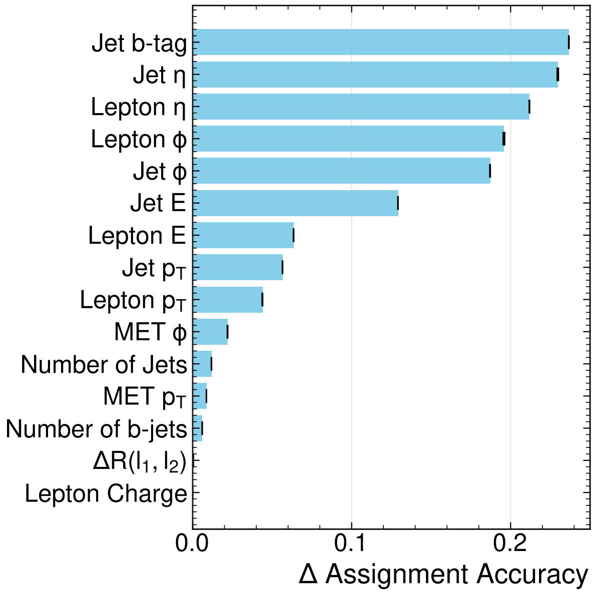
Figure 6.5.: Inference Time of all architectures as a function of the number of trainable parameters. Each point represents the average inference time over 5 runs with different random seeds and k-folded training/validation data. The error bars indicate the standard deviation across the runs. Note that the scaling of the x-axis is logarithmic, which allows for a better visualisation of the differences in performance across different model sizes.

of the number of trainable parameters. The inference time is measured as the average time to process an event during the evaluation phase and is an important metric for assessing the practical applicability of the models in real-time data processing scenarios. Note, that the events are batched, and the processing time for each batch is divided by the number of events per batch. From the results, it can be observed that inference time generally increases with the number of trainable parameters across all architectures, as expected, since larger models typically require more computational resources to make predictions. However, here, the number of layers appears to have a much greater impact on inference time than the embedding dimension, which can be explained by the fact that the number of layers directly affects the model's depth and thus the number of computations required per forward pass. In contrast, an increased embedding dimension has a less direct impact, as these computations can be parallelised more effectively, mitigating the increase in inference time. This is, however, only true for processing on highly parallelised hardware such as GPUs, and may not hold for CPUs, where increasing the embedding dimension can lead to a larger increase in inference time due to limited parallelisation. Thus, these observed differences in inference time are to be taken with a grain of salt, as they can be highly dependent on the specific hardware used for evaluation and may not generalise to other processing environments. Overall, the inference time increases most steeply for the CPT architecture. This can be explained by the CPT architecture, which combines jets,

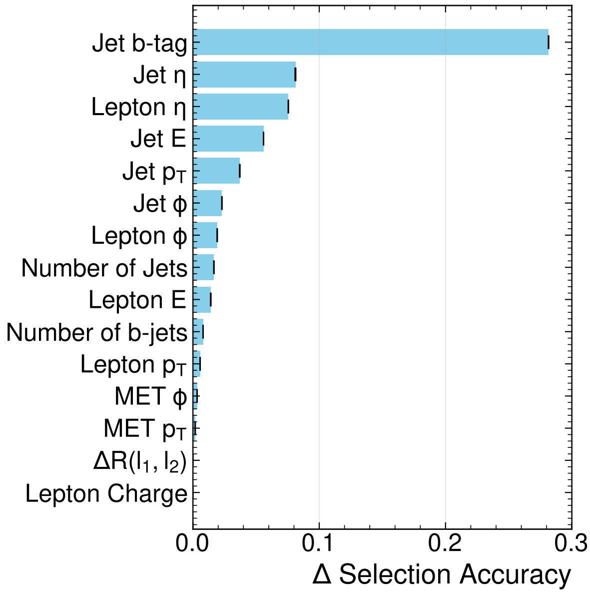
leptons, and $\not{E}_{\mathrm{T}}$ features into a single sequence and passes it through a shared transformer layer via self-attention. The self-attention mechanism has computational complexity that scales quadratically with sequence length, which can significantly increase inference time as the number of features and model size grow. Thus, the slightly longer sequence length of the CPT architecture compared to the FCT and CAT architectures contributes to its higher inference time, especially for larger model sizes. In contrast, the FCT and CAT architectures treat jets and leptons separately, which can allow them to process features more efficiently and reduce overall inference time, even as model size increases.
Combining the results from figure 6.4 and figure 6.5, it can be concluded that increasing the number of trainable parameters generally leads to improved performance in terms of assignment accuracy, but it also results in increased inference time. While the number of layers has a very limited effect on the assignment accuracy, it has a significant impact on the inference time. Thus, when selecting a model for practical applications, it is often beneficial to choose one with a higher embedding dimension and fewer layers, as this provides a good balance between performance and computational efficiency. Additionally, the choice of architecture may also play a role in determining the optimal configuration, as some architectures may be more efficient than others in terms of inference time for a given number of parameters. To strike this balance, the CPT architecture with an embedding dimension of 128 and 6 layers is chosen as the best-performing model, achieving high assignment accuracy while maintaining a reasonable inference time compared to other configurations. Note, that in the following chapters the CPT architecture will be used with a larger embedding dimension of 256. This was, however, not included in the hyperparameter-optimisation to reduce computational costs.

6.1.4. Feature Importance

Machine learning models are often regarded as black boxes, providing limited transparency into their internal computations and decision-making processes. This is mostly due to the abstract nature of the latent spaces. Nevertheless, several techniques have been developed to gain insights into the computational structure of these models. A prominent example is the measurement of *Feature Importance*, which quantifies the influence of each input feature on the model's computations and overall performance. The most robust framework uses a game-theoretic approach based on so-called SHAP values [75, 76]. Due to the computational complexity of this method, this thesis uses a simpler technique, called permutation importance [77]. This method quantifies a feature's impact on the model's performance by measuring the difference in performance when the feature is effectively eliminated from the inputs. The idea is to eliminate the feature by randomly shuffling its value among the events. Therefore, the feature's values become uncorrelated to the other features of the event, and the model cannot make use of this feature. This was found to provide a robust way of accessing the impact of a feature. As the transformer model processes features embedded as sequences, it was decided to permute the features across all events and sequence elements. For this analysis, each input feature was permuted, and the difference in the model's assignment and selection accuracy relative to the baseline performance was computed. This process is repeated with five permutations to reduce stochastic variation from individual permutations, and the differences are averaged across



(a) Assignment Accuracy



(b) Selection Accuracy

Figure 6.6.: The Plots (a) and (b) show the difference in assignment and selection accuracy, respectively, of the CPT-model when permuting the input features. For each feature, five different permutations are performed. The features are ranked in decreasing order of the observed reduction. The shown error bars represent the standard error of the performance difference over 5 permutations.

trials. The results are presented in figure 6.6, which shows the ranking and scores of the different features supplied to the model. Note that this analysis only evaluates the impact of these variables in these trained models. In principle, it would be possible for a model trained without this variable to attain the same performance by leveraging correlations of other variables. This is why the aforementioned SHAP values compare combinations of variables and left-out, and often analyse retraining a model without a certain feature. However, due to the high computational cost, such an investigation would exceed the scope of this thesis.

For both metrics, the b-tag of the jets is the most important feature, followed by the jets' and leptons' pseudorapidity η respectively. While the order of the three leading features is the same across the two metrics, their individual impact differs greatly. For the assignment accuracy, all three features impose a similar drop in performance upon elimination. For the selection accuracy, however, the performance difference caused by disabling the b-tag is more than three times as large as with the pseudorapidities. This highlights that selection accuracy is mainly carried by the b-tag, while for jet assignment, their kinematic relations to the leptons become more important. For the assignment accuracy, the four leading kinematic variables all relate to the angular features of jets and leptons and show quite some difference to the next important variable, which is the jets' energy. In terms of selection accuracy, energy and transverse momentum rank higher than the azimuthal angles. This can be explained by considering that the jets from the signal process tend to be more energetic. For both metrics, global event features, such as variables related to missing transverse energy, the number of jets, and the angular separation of the leptons, appear among the least impactful variables, with the azimuthal angle of the $\not{E}_{\mathrm{T}}$ leading for assignment accuracy and the number of jets for selection accuracy. Lastly, it can be observed that for both metrics, the charge of the leptons seems to be entirely irrelevant. This underlines the previous hypothesis that the assignment of jets to leptons is largely independent of the lepton charge. This makes permutation-invariant models such as the CompactTransformer favourable. Generally, the insights into the model's reliance on input features confirmed previous expectations. The model seems to utilise the variables in a way that is consistent with the baseline methods. Notably, the model exhibits only a weak dependence on the $\not{E}_{\mathrm{T}}$ variables, suggesting that information related to the neutrinos exerts minimal influence on its performance.

6.2. Combination with Neutrino Regression

To achieve a full reconstruction of the dileptonic $t\bar{t}$ events, the jet assignment model can be combined with a neutrino regression model to predict the momenta of the two neutrinos in the event. This combination enables a more comprehensive understanding of the event kinematics and improve overall reconstruction performance. This allows not only for the reconstruction of the full kinematics of the event but also for leveraging interdependencies between the jet assignment and neutrino regression tasks to improve the performance of both. For example, the jet assignment model can provide additional context for the neutrino regression model, helping it make more accurate predictions by considering relationships between jets and leptons in the event. Conversely, the neutrino regression model can provide additional information about event kinematics, helping the jet assignment model make more informed decisions about which jets are associated with which leptons. By combining these two models within a unified framework, it is possible to achieve a more holistic approach to event reconstruction in the dileptonic $t\bar{t}$ channel, potentially leading to improved performance and a better understanding of the underlying physics processes.

6.2.1. Unified Model Architecture

To combine the jet assignment and neutrino regression tasks into a unified model architecture, the CompactTransformer (CPT) can be extended by adding a neutrino-regression head. This new architecture, which we refer to as the MultiModalTransformer (MMT), retains the core structure of the CPT while incorporating additional components to handle the neutrino regression task. The structure for embedding, transformer layers, and assignment head remains the same as in the CPT architecture. This allows for a direct comparison of the performance of the jet assignment task before and after incorporating the neutrino regression, and ensures that any improvements in performance can be attributed to the addition of the neutrino regression head rather than to changes in the underlying architecture. For the regression head, the latent vectors of the $\not{E}_{\mathrm{T}}$ features after the shared transformer layers are concatenated with those of the leptons. Additionally,

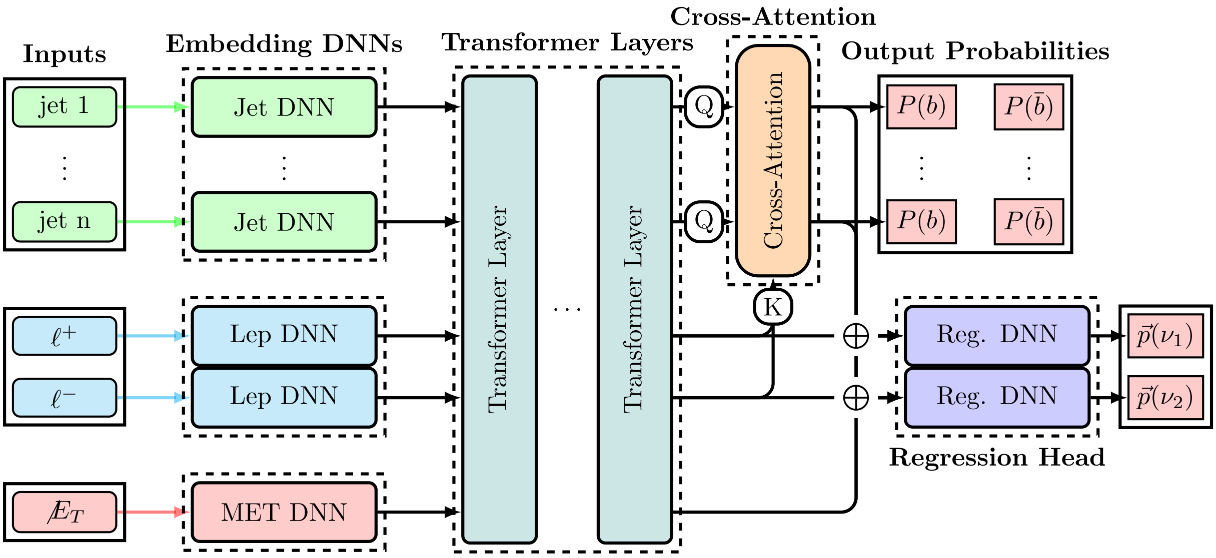


Figure 6.7.: This schematic visualises the general structure of the CompactTransformer architecture equipped with an additional head for neutrino regression. Note that $\oplus$ represents a concatenation of the corresponding features. This architecture is called MultiModalTransformer (MMT).

the jet features after the shared transformation are weighted by the computed assignment scores for each lepton-jet pairing; this projected jet-feature vector is then concatenated with the lepton's features. The regression head consists of a shared DNN that processes the concatenated features and outputs the predicted neutrino momenta in Cartesian coordinates for each neutrino. The DNN used for the regression here deviates from the layout used for the other: it features three linear layers with exponentially decreasing dimensions and ReLU activation to match the target output dimensions. While projecting the jet features was found to provide valuable context for the jets to the regression head, it introduced a side effect: propagating weight changes from the regression to the assignment. To disable this effect, one possible modification was to stop gradient propagation from the regression head to the assignment. The impact of this is investigated during the hyperparameter optimisation.

6.2.2. Training Procedure

The training setup follows closely the one outlined in section 6.1.2. The main difference is that the loss function now includes a term for the neutrino regression task – the total loss is computed as a weighted sum of both loss contributions. For the neutrino regression, different loss functions are investigated. All of them are based on some distance measure between the predicted and true neutrino momenta. An overview of the different loss functions can be found in table 6.1. The idea behind the different loss functions is to reflect that neutrino regression should be guided not only by deviations from the components but also in terms of the physical objects these components represent. Different loss functions can thus emphasise different aspects of the neutrino properties, which the transformer should aim to capture correctly. Note that for all loss functions, the output of the model remains the Cartesian coordinates of the two neutrinos. Only the method to compute the difference between truth and prediction is changed. Designing the most useful loss function may very well depend on the target of one's analysis. A thorough study of how to tune the difference measure is beyond the scope of this study. Thus, only three distinct instances are evaluated here to showcase their impact. The Cartesian-Loss computes just a regular MSE of the Cartesian coordinates of the neutrino momenta. In contrast, the PtEtaPhi-loss measures the difference between the true and predicted neutrino momenta in terms of their p_T, η, ϕ coordinates. Lastly, the MagDir-loss quantises the deviation from the truth of the vector's magnitude and cosine similarity. To measure the difference of p_T and $|\vec{p}|$, it was found to be beneficial to use the logarithmic difference with respect to the true value. This is because these variables span many orders of magnitude and would thus destabilise the training. Other options, such as a relative difference, were also investigated but were found to be less performant here.

Table 6.1.: Overview of different loss functions that were investigated.

Loss	Computation
Cartesian-Loss	$L = (\Delta p_x)^2 + (\Delta p_y)^2 + (\Delta p_z)^2$
PtEtaPhi-Loss	$L = \left(\log\left(\frac{p_{T,\text{true}}}{p_{T,\text{pred}}}\right)\right)^2 + (\Delta\eta)^2 + (\Delta\phi)^2$
MagDir-loss	$L = \left(\log\left(\frac{\lvert\vec{p}_\text{true}\rvert}{\lvert\vec{p}_\text{pred}\rvert}\right)\right)^2 - \frac{\vec{p}_{pred}\cdot\vec{p}_{true}}{\lvert\vec{p}_{pred}\rvert\lvert\vec{p}_{true}\rvert}$

Another related but conceptually different approach investigated is the so-called *Gaussian-Loss*. The rationale for using this loss is that the neutrino regression problem is inherently underconstrained. Therefore, the neutrino momenta can be more appropriately modelled using a conditional (posterior) probability distribution $P(\vec{p}(\nu), \vec{p}(\bar{\nu})|\text{Event})$. Instead of a single possible neutrino solution for each event, the neutrino momenta follow a conditional probability distribution, which more accurately represents the limited knowledge of the momenta. This provides a more coherent picture of the neutrinos' kinematic structure. In section 4.2, the methodological difference between the two approaches was elaborated. This approach was investigated for ν^2-Flows [73]. Since ν^2-Flows follow a very different pattern in their layout and training, it was decided to use a strongly simplified strategy here to model the posterior of the neutrino. For this, the neutrino momenta were modelled using a multivariate Gaussian distribution of uncorrelated variables. While this has obvious shortcomings and does not accurately capture multimodality in the distributions, it is computationally closely related to the structure of regression. The model architecture can be easily extended to provide $\mu_i(X), \sigma_i(X)$, which are the mean and standard deviation of each of the six neutrino momenta components, respectively. Note that X refers to the inputs of the model. This only requires raising the dimensionality by a factor of 2. The corresponding loss-function L_GL is then for an event with input variables X and parton-level momenta components $P_{i,\text{true}}$ given as

$$L_\text{GL} = \sum_{i=1}^{6}\left(\log(\sigma_i^2) + \frac{(\mu_i - P_{i,\text{true}})^2}{\sigma_i^2}\right), . \tag{6.5}$$

section B.1.1 provides a link between this loss function and the maximum logarithmic likelihood, which in turn is equivalent to minimising the Kullback-Leibler divergence. From the loss, it can be seen that this type of conditional density learning is closely related to, and computationally similar to, regression via MSE. Thus, this approach allows a very simple extension of the model to capture the inherent uncertainty in neutrino momenta. One final aspect to discuss is the behaviour of neutrino momenta during inference. Typically, only one value for the neutrino momentum is considered per event in the reconstruction and fitting. Methods like ν^2-Flows randomly sample 1000 solutions for the neutrino momenta and select the one with the highest probability. For the Gaussian-loss here, two modes are considered: either using the mean of the distribution, which is expected to behave similarly as the regression using MSE (see section B.1.1), or sampling one solution for the neutrinos' momenta from the predicted Gaussian distribution. It should be noted that other possible extensions to the regression structure were investigated but were not found to be beneficial. One of these approaches was to bin neutrino momenta and regress the probability of occupying a given bin using cross-entropy loss. A second idea was to use a tokenisation of the neutrinos' momenta. Both approaches were found to perform poorly and thus were discarded at an early stage. A major contributing factor is the high dimensionality of the output space, which quickly makes binning infeasible.

The later chapters further elaborate on how the optimal loss depends on the variables of interest of an analysis. So, depending on the future use cases of the tools presented here, it might be necessary to investigate and tailor this as needed.

For this chapter, however, the loss is taken as the mean-squared error (MSE) for each neutrino component. That is, because for investigating the architecture and its scaling, the explicit loss function was not found to have a significant impact. Apart from the loss function, the weighting of the two tasks and the scaling of the regression targets also have to be tuned. Here, it was found beneficial to scale the regression targets to units of 100 GeV. This helps to improve the training stability and convergence. The weighting of the two tasks was, after some experimentation, set to $w_{assignment} = 1$ and $w_{regression} = 0.2$, which provided a good balance and stable training behaviour. All other parameters of the training procedure, such as the optimiser, learning rate scheduling, and early stopping criteria, are kept the same as in the jet assignment training procedure to ensure consistency and allow for a fair comparison between the models. During training, performance on the validation set is monitored for both the jet assignment and neutrino regression tasks to ensure the model is learning effectively and not overfitting to the training data. To evaluate the performance of the combined model, separate metrics for each task are reported during training. For the jet assignment, assignment accuracy is monitored, whereas for neutrino regression, metrics such as the relative error in predicted neutrino momenta with respect to the true values are tracked. The relative error is computed as the L^2-Norm of the difference between the predicted and true neutrino momenta, divided by the L^2-Norm of the true neutrino momenta, which provides a measure of the accuracy of the predictions relative to the scale of the true values.

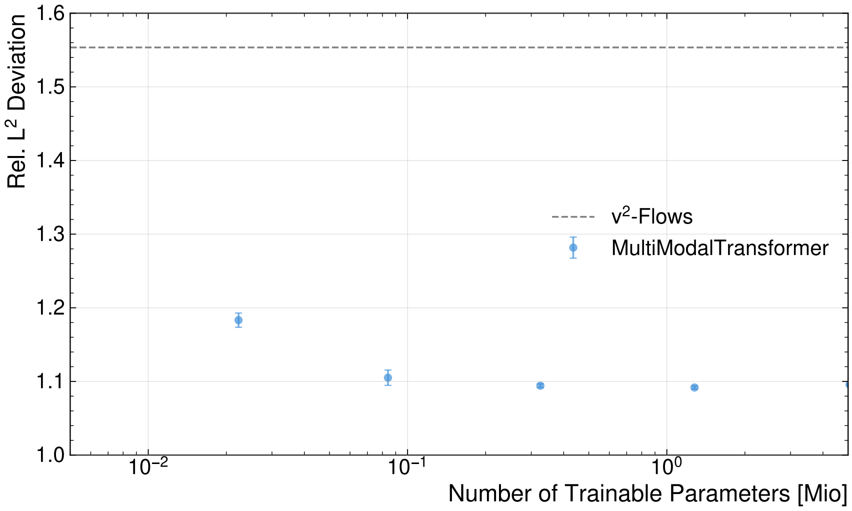
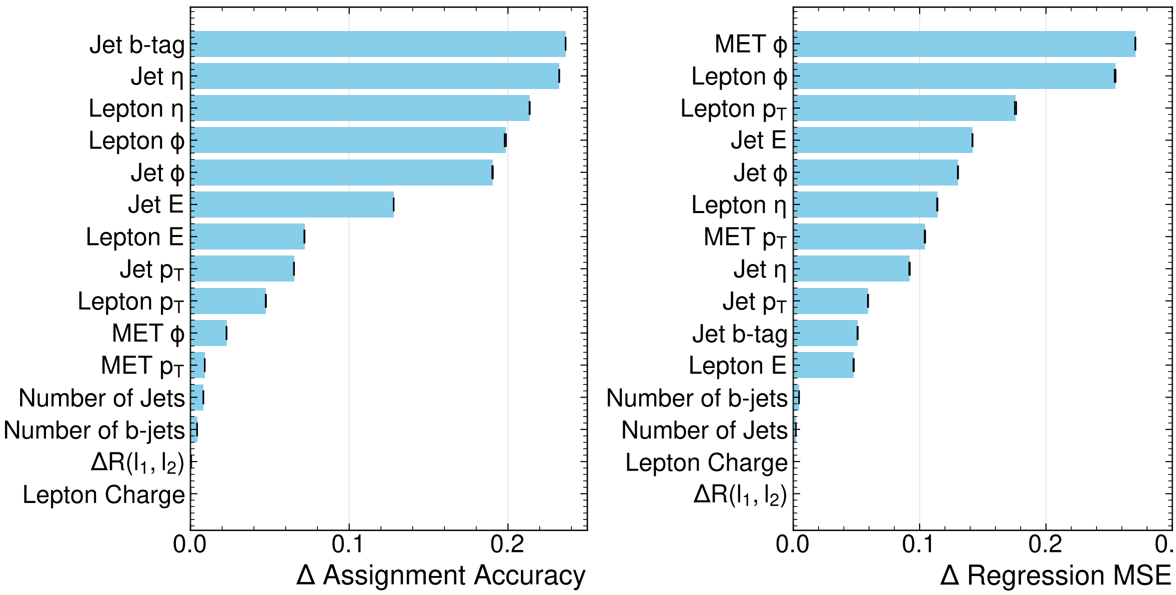


Figure 6.8.: Relative L^2 deviation of the predicted neutrino momenta compared to the true values as a function of the number of trainable parameters for the MultiModalTransformer (MMT) architecture.

6.2.3. Performance Evaluation

As the evaluation for the assignment models showed very minor differences in performance from different numbers of layers, for the combined model, the number of layers is kept fixed to 6, while the embedding dimension is varied in the range of $16, 32, 64, 128, 256$ to evaluate its impact on the performance of both tasks. Due to the increased complexity of the combined model, a higher embedding dimension may be required to achieve good performance on both tasks simultaneously. To keep the computational resources feasible, it was decided not to perform a full grid search for the combined model, but rather to focus on evaluating the impact of the embedding dimension while keeping the number of layers fixed. The figure 6.8 shows the performance of the MMT architecture in terms of the relative L^2-Norm deviation of the predicted neutrino momenta compared to the true values as a function of the number of trainable parameters. For comparison, the performance of the neutrino regression provided by ν^2-Flows is plotted as a horizontal line, serving as a baseline for evaluating the MMT architecture. From the results, two main observations can be made. First, the MMT architecture generally outperforms the ν^2-Flows baseline across different embedding dimensions. Second, there is a trend toward improved performance with increasing numbers of trainable parameters, indicating that the model benefits from sufficient capacity to learn the underlying patterns in the data. However, it is also observed that performance improvement tends to plateau at higher embedding dimensions, suggesting a point of diminishing returns beyond which further increases in model size yield no significant performance gains. Since no decrease in performance with higher parameter counts was observed, despite being trained on the number of events, the model does not appear to exhibit signs of overfitting at these parameter counts. This is most likely due to the aggressive dropout and regularisation applied during training.

6.2.4. Feature Importance

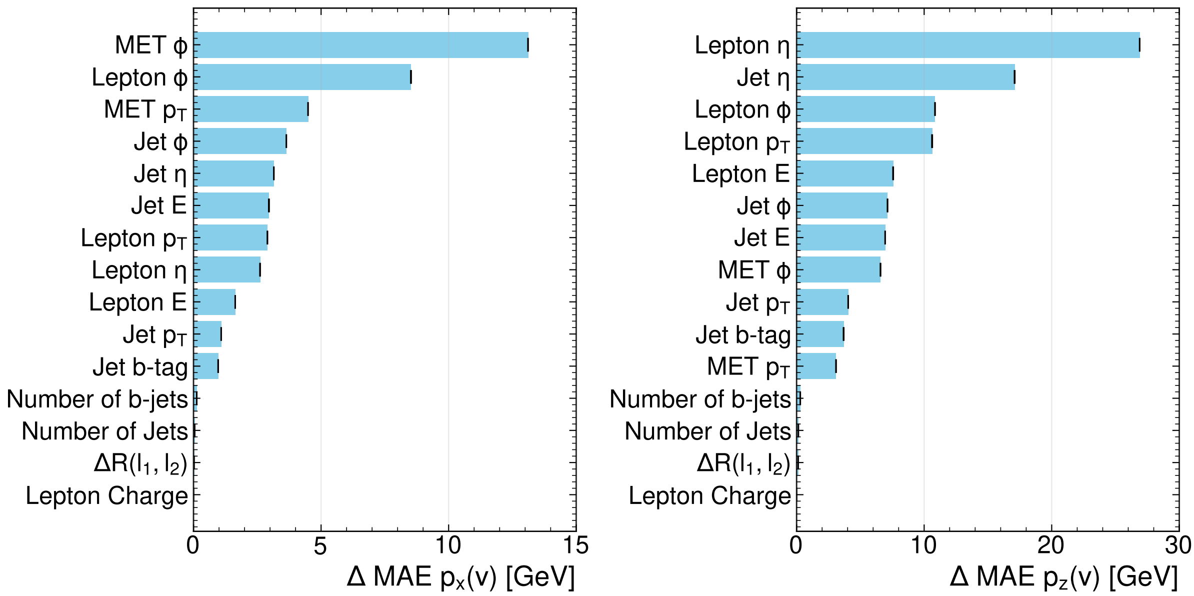


(a) Assignment Accuracy

(b) Regression MSE

Figure 6.9.: The Plots (a) and (b) show the difference in assignment accuracy and regression MSE, respectively, of the MMT-model when permuting the input features. For each feature, five different permutations are performed. The features are ranked in decreasing order of the observed reduction. Note that the mean-square error is again computed as the relative deviation of the L^2-norm as before.

Analogous to the models used for assignment, the model's reliance on the input features is investigated using permutation importance. The results are shown in figure 6.9. Despite the model being trained with the additional training target, the feature importance scores for assignment accuracy have not changed at all and closely match those in figure 6.6. The feature importance for the regression performance, however, shows a very different dependence on the input features. On the one hand, the jets' b-tag, which is the leading feature for the assignment, has very little impact and scores lower than all kinematic features except the energy of the leptons. Generally, it can be observed that while the pseudorapidity scores higher for the assignment than the azimuthal angles, this is completely reversed for the regression, where both the leptons' and jets' ϕ scores higher than their η. Also, features related to the leptons' energy, such as their transverse momenta, become much more important. In stark contrast to the assignment's feature importance, for the regression, the azimuthal angle ϕ of the $\not{E}_{\mathrm{T}}$ manifests as the most important variable, closely followed by the lepton's azimuthal angle. Given that the direction of the $\not{E}_{\mathrm{T}}$ is very strongly related to the neutrinos' momenta, this is unsurprising. Interestingly, the absolute value of the $\not{E}_{\mathrm{T}}$ scores rather low and seems to have much less impact. It should be noted at this point that the feature's score is expected to depend strongly on the metric used to measure the model's performance on the given task. To provide further insights, figure 6.10 shows the feature importance for the mean-absolute error of the neutrinos' p_x and p_z components. This allows us to disentangle the effect of the different kinematic variables on the components. Due to the rotational symmetry of both the detector and the physics around the beam-axis, the p_x represents the reconstruction within the transverse plane. One observation is that while the $\not{E}_{\mathrm{T}}$ features score very highly for reconstructing the transverse components, they score very poorly for the longitudinal component. This reflects that the $\not{E}_{\mathrm{T}}$ only constrains the neutrino momenta in the transverse plane. That this is accurately represented in the models provides a sanity check for the model's internal computation. Another difference between the feature importance for the two directions is that p_z seems to depend much more on the leptons than on the transverse components. First, all four leptons' kinematic variables are among the five leading variables. For the transverse components, however, only the azimuthal angle scores very high.



(a) MAE p_x **(b)** MAE p_z

Figure 6.10.: The plots (a) and (b) show the difference in MAE for the p_x and p_z components of the neutrinos, respectively, of the MMT-model when permuting the input features. For each feature, five different permutations are performed. The features are ranked in decreasing order of the observed reduction. Note that the mean-square error is again computed as the relative deviation of the L^2-norm as before.

Lastly, it should be noted that the b-tag is kinematically unrelated to the neutrinos. Thus, its non-zero score implies that the regression does depend on some internal representation of the jet-assignment or selection within the transformer's latent space.

6.2.5. Multitask Learning Benefits

One of the key prospects of combining the jet assignment and neutrino regression tasks in a unified model architecture is the potential for multitask learning benefits. The idea is that information exchange and additional context provided by the shared transformer layers can help both tasks learn more effectively from the data, thereby improving the model's overall performance. To investigate this, the same model architecture is trained separately for three different setups:

1. CompactTransformer: a MultiModalTransformer trained only for the jet assignment task, with a loss function that includes only the cross-entropy loss for the jet assignment and no contribution from the neutrino regression task ($w_{\text{assignment}} = 1$ and $w_{\text{regression}} = 0$).

2. MultiModalTransformer: a MultiModalTransformer trained for both the jet assignment and neutrino regression tasks, with a loss function that includes both the cross-entropy loss for the jet assignment and the mean-squared error loss for the neutrino regression, weighted by $w_{\text{assignment}} = 1$ and $w_{\text{regression}} = 0.2$. For this training setup, two variations were tested: including or excluding weight-updates propagating from the regression head to the cross-attention layers were investigated separately.

3. RegressionTransformer: a MultiModalTransformer trained only for the neutrino regression task, with a loss function that includes only the mean-squared error loss for the neutrino regression and no contribution from the jet assignment task ($w_{\text{assignment}} = 0$ and $w_{\text{regression}} = 1$).

The results of this comparison are shown in figures 6.11 and 6.12. They compare the regression accuracy and assignment between MultiModalTransformer and the CompactTransformer and RegressionTransformer, respectively, to evaluate the benefits of multitask learning in the unified model architecture. From this, it can be seen that the MultiModalTransformer shows no improvement in jet assignment accuracy compared to the CompactTransformer, suggesting that the inclusion of the neutrino regression task does not provide significant benefits for the jet assignment task. Further, a slight degradation in assignment accuracy is observed, which, although minuscule, exceeds the error bands. However, for the model with gradient stopping during weight updates from the regression head to the cross-attention layer, performance appears to be on par with that of the CompactTransformer. The effect becomes more pronounced at higher numbers of parameters. This indicates that updating the cross-attention layer based on which jet features are needed for regression is not favourable. On the other hand, it was found that disabling this gradient propagation did not reduce regression performance. Therefore, it was decided that the standard architecture, henceforth, includes stopping the gradients.
However, the MultiModalTransformer shows a minor improvement in neutrino regression performance over the RegressionTransformer, suggesting that the inclusion of the jet assignment task provides valuable context that helps the model make more accurate neutrino momentum predictions. Notably, this advantage is particularly pronounced for

higher embedding dimensions, indicating that the embedding dimension must be sufficiently large to capture the relevant information for both tasks. The impact of the included jet assignment on the regression performance is also indicated by the feature importance as discussed in the previous section 6.2.4, where it was pointed out that the non-vanishing impact of the b-tagging on the regression targets shows that the model does require an internal assignment of the jets to perform the regression. This can be further illustrated by considering the RegressionTransformer's assignment performance. Due to the weight updates from the jet-projection via the assignment scores, this model, despite being trained without any assignment labels (i.e. $w_{\text{assignment}} = 0$), was able to predict the assignment. A corresponding plot is shown in figure A.9 in the appendix. It compares the achieved accuracies for the RegressionTransformer and the MultiModalTransformer. While the MultiModalTransformer achieves a much higher performance, the RegressionTransformer can reach accuracies above 75%, which is not only far better than a random assignment, but also above the baseline methods (see section 7.1). This finding conclusively demonstrates that the information flow between the two tasks is inherently linked. It also indicates that a tool like ν^2-Flows, during feature extraction, already develops an understanding of the assignment. These scores could be extracted instead of relying on additional tools, such as the χ^2-Methods, for reconstruction. Not only would this result in superior accuracy in jet assignment, but it would also provide some transparency into the machine-learning method and its internal correlations across different objects.

Overall, these results highlight the potential advantages of multitask learning in this context and suggest that combining related tasks within a unified model architecture can improve performance on certain tasks by leveraging shared information and context across them. In particular, the jet assignment, which was often overlooked in previous works such as ν^2-Flows, can provide valuable context for the neutrino regression task, leading to improved performance when combined within a unified model architecture. Furthermore, the computational cost of training the MultiModalTransformer is not significantly higher than training the CompactTransformer or RegressionTransformer separately, because the shared transformer layers enable efficient learning of both tasks without substantially increasing model complexity. Similar arguments hold for inference time, which is also not significantly higher for the MultiModalTransformer than for the separate models, as the shared architecture enables efficient processing of both tasks without a substantial increase in computational requirements. As the computational cost for the ν^2-Flows method could not be evaluated, a direct comparison of the different methods in terms of computational efficiency is not possible. But in general, the benefits of multitask learning in terms of improved performance can be achieved without incurring high additional computational costs, making it an attractive approach for event reconstruction in the dileptonic $t\bar{t}$ channel. On the other hand, training and running separate models for both tasks is expected to approximately double the computational cost, as each model would require separate forward passes and training processes.

To conclude, while combining both tasks in a single model did not yield the expected improvements in assignment performance, regression performance increases beyond statistical uncertainty when the assignment is added as a training target. Furthermore, the model can perform both tasks without requiring a larger embedding dimension. That indicates that there is some information sharing and synergy in the embedding, and the model does not require additional dimensions to capture both. Thus, a model can be extended to perform the full reconstruction rather than a single task, while maintaining its size and runtime. This can significantly decrease computational costs for training and inference.

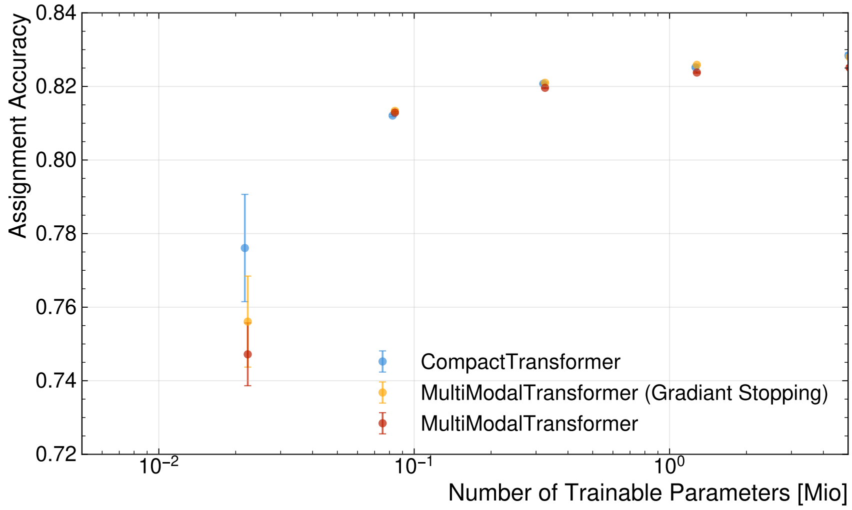


Figure 6.11.: Assignment Accuracy of the MultiModalTransformer–with and without Gradient Stopping from regression head to cross-attention– compared to CompactTransformer, which features the same architecture trained only for the jet assignment task, as a function of the number of trainable parameters.

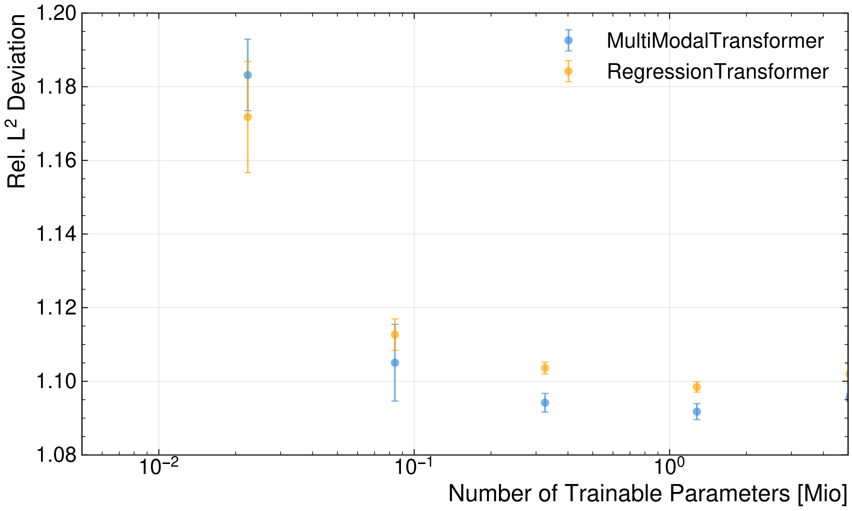


Figure 6.12.: Relative L^2 deviation of the predicted neutrino momenta compared to the true values for the MultiModalTransformer and the RegressionTransformer as a function of the number of trainable parameters.

7. Evaluation of the ML-Based Reconstruction

To evaluate the performance of the ML-based reconstruction method, several metrics and analyses are conducted. This chapter presents the results of these evaluations, focusing on the accuracy of particle assignments, the reconstruction of relevant physics variables, and performance across different kinematic regions. The ML-based methods are compared against the baseline reconstruction techniques, both overall and in specific regions of interest, such as near the $t\bar{t}$ production threshold.

7.1. Assignment Accuracy

Table 7.1.: Comparison of assignment and selection accuracies for different reconstruction methods. The uncertainties represent the statistical variations in the evaluation dataset.

Method	Assignment Accuracy	Selection Accuracy
$\Delta R(\ell, j)$-Method	$0.6460^{+0.0008}_{-0.0006}$	$0.9272^{+0.0003}_{-0.0003}$
χ^2-Method	$0.7501^{+0.0005}_{-0.0006}$	$0.9272^{+0.0003}_{-0.0003}$
CompactTransformer	$0.8325^{+0.0005}_{-0.0006}$	$0.9626^{+0.0003}_{-0.0003}$

Assignment accuracy is defined as the fraction of events in which the reconstructed particles are correctly assigned to their parent partons. The selection accuracy, on the other hand, measures the fraction of events where the correct assignment is among the selected candidates. Both metrics are crucial for evaluating the performance of reconstruction methods, as they directly impact the resolution of physics observables and the sensitivity of analyses. Furthermore, some variables, such as the invariant mass of the $t\bar{t}$ system, are sensitive only to the correct selection of candidates, making selection accuracy particularly important for certain analyses. The table 7.1 summarises the overall assignment performance of the reconstruction methods. The ML-based method, represented by the CompactTransformer, shows a significant improvement in the assignment accuracy compared to the $\Delta R(\ell, j)$-Method and the χ^2-Method. Concerning the baseline methods, the χ^2-Method performs roughly 16% more accurately than the $\Delta R(\ell, j)$-Method. For the selection accuracy, both baseline methods yield the same result within uncertainties, as they use the same jet selection criteria. The improvement in the CPT's selection accuracy is rather small, which can be attributed to b-tagging. Because baseline methods almost exclusively rely on b-tagging for the selection, and the GN2-tagger is very performant [71], the selection accuracy is already very high. Further, the b-tagging of the GN2-tagger already accounts for the full kinematic information of the event, making it difficult for the ML-based method to significantly improve selection accuracy. This is also supported by the model's reliance on the input features, as shown in section 6.1.4. The investigation revealed that the selection accuracy mainly depends on the b-tag. It does, however, utilise information from the kinematics of the jets and leptons, giving it an edge over baseline methods that rely only on b-tagging and jet p_T. Regarding assignment accuracy, the ML-based method shows a significant improvement, attributable to its ability to learn complex correlations among the kinematic variables of the reconstructed particles. These complex correlations can be much more efficient than the simple criteria used by baseline methods, allowing them to surpass the baseline methods' performance.

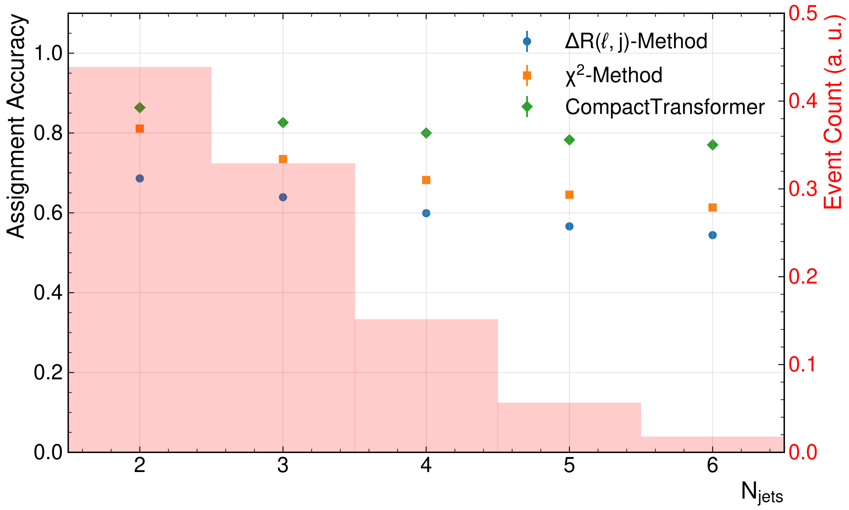


Figure 7.1.: Assignment accuracy as a function of the number of jets in the event for different reconstruction methods. The background histograms show the distribution of events as a function of the number of jets. The uncertainties represent the statistical variations in the evaluation dataset.

The figure 7.1 shows the assignment accuracy for events with different jet multiplicities for each method. The CompactTransformer outperforms the baseline methods across all bins. For all methods, performance drops as the number of jets increases. The performance drop, however, is relatively small compared to the number of permutations, which increases quadratically. This can be attributed to the b-tagging performance. This is in stark contrast to the methods for the all-hadronic channel, where performance drops sharply for events with many jets in the final state, as seen in reference [6]. This is, of course, a consequence of the much lower number of possible permutations for this decay channel. Even with additional jets, the chance of a jet being falsely b-tagged is quite low [71]. Thus, the performance of selecting the correct jets degrades much less than in the hadronic channels, where non-tagged jets must also be selected. Notably, the CompactTransformer shows the smallest performance drop, and the performance gap widens as jet numbers increase. This stems from improved jet selection accuracy, as shown previously. In the appendix, the figure A.14 shows the selection accuracy binned in the same way. It can be seen that the CompactTransformer does a much better job at selecting the correct jets for events with a higher jet multiplicity. It was found that this is the primary cause for differences in assignment accuracy with respect to the jet multiplicity. The efficiency of assigning correctly selected jets appears to be largely independent of jet multiplicity.

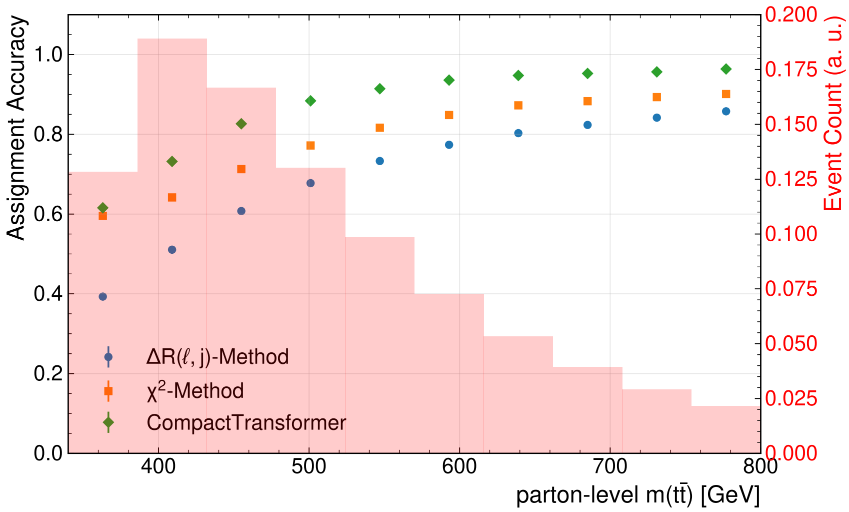


Figure 7.2.: This plot shows the assignment accuracy in bins of the true invariant mass of the top-quark pair.

figure 7.2 shows the assignment accuracy as a function of the parton-level $m(t\bar{t})$. All methods show similar behaviour: the accuracy drops at low invariant masses and appears to saturate at high invariant masses. For the CompactTransformer, the saturated performance in the high-invariant-mass regime approaches unity, indicating near-perfect reconstruction. At the other end of the spectrum, the performance drops to around 0.6 near the $t\bar{t}$ production threshold. For these low-mass events, the CompactTransformer and the χ^2-Method appear to be almost on-par. In general, this indicates that the χ^2-Method is the least sensitive to $m(t\bar{t})$, while the $\Delta R(\ell, j)$-Method seems to be most sensitive.
As was found earlier, the number of jets can affect accuracy. Interestingly, the double-binned distribution of events with respect to the invariant mass $m(t\bar{t})$ and number of jets in figure A.3 shows that events with lower energy have fewer jets on average. Naively, one would expect this to increase the assignment accuracy, which is the opposite of what is observed. To decouple the number of jets per event from the selection accuracy, one can consider the conditional probability of making a correct assignment, given that the correct jets were selected. For this, the assignment reduces to a binary problem in the dilepton case, as jets can only be assigned to bottom or antibottom quarks. Thus, the probability of correctly assigning at random is 50%. According to Bayes' theorem, the conditional probability for each method can be computed by dividing its assignment accuracy by its selection accuracy. The corresponding plot showing this probability for the different methods, binned in parton-level $m(t\bar{t})$, is displayed in figure 7.3. One thing to note is that the $\Delta R(\ell, j)$-Method has a performance below 0.5 for the lowest bin, meaning that it performs worse than a random assignment in this bin. This suggests that the particle topology in

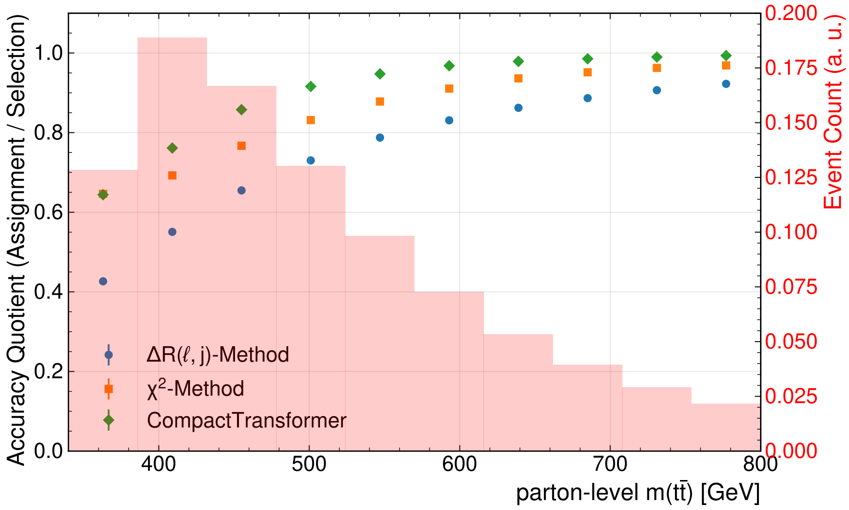


Figure 7.3.: Quotient of assignment and selection accuracy computed in bins of the parton-level $m(t\bar{t})$.

this region disfavours assigning jets based on proximity. figure A.7 in the appendix clearly shows the difference in angular distance distributions for correct and incorrect jet-lepton pairing between the inclusive sample and the near-threshold region. This explains the significant performance drop-off in this region. Going further, this behaviour can be directly linked to the underlying physics: At the production threshold, the top quarks carry very little energy. Thus, the decay products of the top quarks, i.e., the jets, the W-boson, and subsequently the lepton, are less collimated. This is because the bottom quark and W-boson are back-to-back in the top quark's rest frame and get collimated by the Lorentz boost of the top quark with respect to the lab frame. The strength of this boost is directly proportional to the top quark's energy, which itself is approximately proportional to the $m(t\bar{t})$. Note that this picture gets more complicated if one takes the chiral nature of the decays into account, which can affect the angular distributions. As the energy increases, decay objects become increasingly collimated. While this explains why the performance drops, it does not explain how performance can be worse than with random assignment of jets. To explain this effect more thoroughly, investigations into the angular structure of the events in this region would be necessary. Another observation is that in the lowest bin, the CompactTransformer and χ^2-Method are completely on par for this metric. Performance losses of the chi^2-Method can be explained by the figure A.8 in the appendix, which shows the different distributions of the reconstructed top-mass for the correct and incorrect assignment of the jets. For comparison, the distributions for the near-threshold and inclusive samples are shown. The correct pairing shows the typical Breit-Wigner mass peak, as expected, whereas the incorrect pairs show a washed-out distribution. For the low-energy events, the difference between the two distributions is smaller but still very pronounced. One reason is that the top quarks tend to be more off-shell, and faulty assignments do not change the kinematics as dramatically. But generally, the assignment based on the invariant mass seems relatively robust. For machine-learning methods, it is difficult to assess which specific features lead to performance degradation. The leading feature, i.e. the b-tag, is not found to have a varying performance in this range of invariant mass $m(t\bar{t})$– as seen in figure A.2. For the other features, assessing their effect across different regimes is non-trivial. One way is to examine whether certain features correlated with near-threshold events affect performance. A particular example for this is the angular separation $\Delta(\ell_1, \ell_2)$ of the leptons. The property of having less energy - and thus less boosted top-quarks, also significantly impacts the angular separation of the leptons themselves. While decays of high-energy top-quark pairs lead to back-to-back leptons, the opposite is true for low-energy top-quark pairs. The double-binned distribution of events in figure A.5 in the appendix shows that events with lower $m(t\bar{t})$ tend to have more collimated leptons, whereas they are more separated for more energetic events. Since the angular separation of the leptons is suspected to degrade event performance, its impact on the individual methods is investigated. As before, figure 7.4 shows the assignment

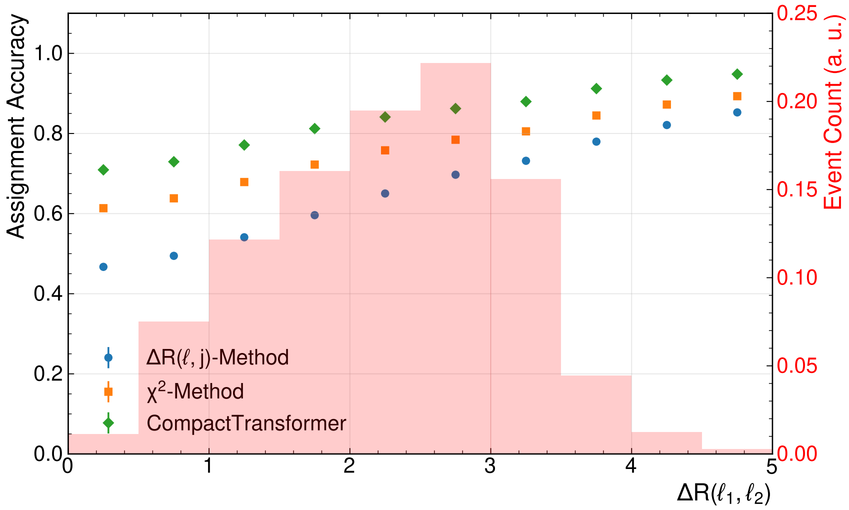


Figure 7.4.: Assignment accuracy binned in the angular distance between the two leptons of the events $\Delta R(\ell_1, \ell_2)$.

accuracy in bins of the angular distance $\Delta R(\ell_1, \ell_2)$ between the two leptons. All methods suffer performance losses for events with close leptons. Interestingly, however, for the smallest[1] angular separation, the performance is still above that of near-threshold events. This implies that the decreased performance due to the reduced angular separation in the near-threshold region is one contributing factor, but not the only one. A combined comparison of both variables' impact is shown in figure A.15 in the appendix. This distribution provides a more comprehensive picture by separating the effects of both variables. Also, it allows us to isolate the effects of each by varying one variable while holding the other constant. The double-binned accuracy reveals that the invariant mass $m(t\bar{t})$ has a much stronger impact on performance than the angular separation alone. Expectedly, the $\Delta(R, j)$ Method is the most sensitive to the angular separation. For the lowest bins in $\Delta R(\ell_1, \ell_2)$, the performance breaks down significantly and is nearly constant for varied $m(t\bar{t})$. The other methods' performances reduce much more in the rows of varied $m(t\bar{t})$,

[1]Due to the overlap-removal, which removes overlapping objects from the (simulated) events, there is a cut-off at $\Delta R < 0.3$. This technique accounts for the fact that in a detector, objects that coincide cannot be separated.

than they do in the columns of varied $\Delta R(\ell_1, \ell_2)$. This observation holds for all bins of energies and angular separations considered here. From this, it follows that the near-threshold region poses special challenges for jet assignment. For the traditional methods, explanations are quite clear, but even the transformer, which can leverage very complex correlations between variables, performs only slightly better than the baseline methods. While some impact factors could be identified, i.e. angular separation of the leptons and off-shell tops, the complete picture of why this region is particularly challenging remains elusive. Due to increased efforts to study quantum effects or quasi-bound-state effects at the production threshold, this remains a highly relevant question for future investigations. While it was found that other kinematic variables have regimes with quite different performance on assignment accuracy, none had an impact as severe as that of the invariant mass $m(t\bar{t})$. One to mention here is the transverse momenta of the top-quark pair system. This variable is highly sensitive to initial-state radiation. Thus, a high p_T is often accompanied by an increased number of jets, as seen in figure A.4. However, it was found that $p_T(t\bar{t})$ has no impact on the accuracy of the ML-based assignment, while it degrades both baseline methods. The corresponding binned accuracy evaluation is shown in figure A.13 in the appendix. This shows that the transformer seems to be particularly insensitive to initial-state radiation compared to the other methods. Another object of study was the aforementioned spin-sensitive variables. As these have been used in recent analyses, it is interesting to study whether particular regions are more or less sensitive to misassignments. To mimic the definition of signal regions used in [2], a double-binned plot, resembling the different signal regions in c_{hel} and c_{han}, shows the assignment accuracy in figure A.16 in the appendix.

7.2. Neutrino Regression

This section analyses and compares the neutrino regression of the MultiModalTransformer against existing baseline methods. Further, the impact of different loss functions for the MultiModalTransformer is investigated. For this study, only the regression models are considered. The model trained with the Gaussian-loss showed the same performance as the MMT with the Cartesian-loss when using the mean. However, when using random sampling, a consistent degradation in performance was observed, as expected, since it is equivalent to adding Gaussian noise to the reconstructed value. The appendix provides a summary of this in tables A.1 and A.2. Due to the ν-Weighter's non-unity efficiency of about $\epsilon = 94.0\%$, the evaluation of each method is done on the subset of events, where the ν-Weighter is available. Note that this can introduce biases in the evaluation. Various metrics are used to assess the method's quality. It should also be noted that evaluating the quality of the regression is much more nuanced compared to the assignment. As demonstrated below, ranking the performance of one method against the other is highly dependent on the metric one employs. This is because the reconstructed object is essentially six-dimensional, and its deviation must be projected in some way. But even quantifying a one-dimensional error comes with some degeneracy; different powers, for example, might put emphasis on larger or smaller deviations. Also, whether a relative or absolute deviation is considered can make a huge difference. To account for this, different metrics are used and compared.

Table 7.2.: Summary of the mean absolute error (MAE) of the different neutrino regression methods for the Cartesian coordinates of the neutrinos in the lab frame. For the MultiModalTransformer, the loss function used during training is shown in brackets. The uncertainties represent the statistical variation in the dataset.

Method	MAE (p_x) [GeV]	MAE (p_y) [GeV]	MAE (p_z) [GeV]
ν^2-Flows	$33.62^{+0.04}_{-0.04}$	$33.59^{+0.05}_{-0.02}$	$87.64^{+0.12}_{-0.05}$
ν-Weighter	$51.06^{+0.06}_{-0.05}$	$51.12^{+0.05}_{-0.06}$	$123.83^{+0.19}_{-0.10}$
MMT (Cartesian)	$24.144^{+0.018}_{-0.023}$	$24.119^{+0.016}_{-0.015}$	$63.48^{+0.09}_{-0.05}$
MMT (PtEtaPhi)	$24.81^{+0.03}_{-0.03}$	$24.769^{+0.015}_{-0.020}$	$66.03^{+0.08}_{-0.07}$
MMT (MagDir)	$26.420^{+0.019}_{-0.014}$	$26.495^{+0.025}_{-0.014}$	$60.94^{+0.10}_{-0.06}$

The table 7.2 shows the mean-absolute error of the neutrino momentum components for the different methods. Each MMT model, regardless of the loss function, achieves a smaller mean absolute error for each of the three components than the two baseline methods. Some impact of the loss functions used here can be identified. Notably, the Cartesian-Loss provides the lowest MAE for the transverse components. However, the performance is nearly on par with that of the model trained with the PtEtaPhi-loss. Interestingly, the model trained with the MagDir-loss achieves the smallest error for the longitudinal component. Even though this might seem counterintuitive, it can be attributed to the model with Cartesian loss being trained on squared error, which might disfavour larger errors more strongly than an absolute error would. This already highlights that very subtle changes in evaluation metrics might lead to different conclusions about the model's performance. Another noteworthy observation is that the difference between ν^2-Flows and ν-Weighter is much larger than the difference between the former and the MMT-architecture. As the only regression method that does not rely on machine learning, the ν-Weighter is found to perform significantly worse than all ML-based methods. The MultiModalTransformer is found to reduce the mean absolute error by a factor of 2, a significant improvement. This becomes even more impressive when one considers the lower computational cost of ML-based methods. Another trivial remark is that all methods show the same performance for the two transverse components x and y, as expected, since the detector and physics are rotation-invariant with respect to the beam direction.
To highlight the performance variance between different metrics, table 7.3 shows the mean absolute error for the p_T, η, ϕ, and E of the neutrinos. As for the neutrino momentum components, models based on the MMT, trained with different loss functions, outperform both baseline methods across all metrics. Further, for this set of variables, the ν-Weighter has the largest errors across all variables. However, for the pseudorapidity, it is on par with ν^2-Flows. For the differently trained models, variations in the performance can be found among all variables. The PtEtaPhi-loss provides the smallest error among p_T,η, and ϕ, which is, of course, consistent with those being its target for optimisation. Similarly, the model trained with the MagDir-loss has the smallest error in neutrino energy,

Table 7.3.: Summary of the mean absolute error of the neutrinos' p_T, η, ϕ, and E, evaluated for the different methods. Note that the computation of the error in ϕ was performed periodically.

Method	MAE (p_T) [GeV]	MAE (η)	MAE (ϕ) [GeV]	MAE (E) [GeV]
ν^2-Flows	$30.03^{+0.03}_{-0.02}$	$1.0027^{+0.0007}_{-0.0007}$	$0.9647^{+0.0008}_{-0.0004}$	$75.02^{+0.07}_{-0.07}$
ν-Weighter	$51.34^{+0.07}_{-0.06}$	$1.0677^{+0.0011}_{-0.0009}$	$1.0808^{+0.0005}_{-0.0007}$	$114.46^{+0.09}_{-0.11}$
MMT (Cartesian)	$24.497^{+0.016}_{-0.017}$	$0.9349^{+0.0003}_{-0.0004}$	$0.7897^{+0.0009}_{-0.0011}$	$55.024^{+0.028}_{-0.019}$
MMT (PtEtaPhi)	$19.965^{+0.015}_{-0.009}$	$0.7602^{+0.0006}_{-0.0007}$	$0.7584^{+0.0006}_{-0.0005}$	$57.71^{+0.06}_{-0.10}$
MMT (MagDir)	$23.125^{+0.012}_{-0.014}$	$0.7831^{+0.0011}_{-0.0008}$	$0.7515^{+0.0007}_{-0.0005}$	$49.45^{+0.03}_{-0.04}$

which is equivalent to their magnitude since they are massless. Both the PtEtaPhi- and MagDir-loss reduce the error in η about 20% with respect to the Cartesian-loss. To sum up, this reveals a strong dependence on the choice of loss function, despite using the same architecture. As different loss functions can reduce errors for some variables while increasing them for others, choosing the most appropriate loss function depends on the variable of interest in the analysis.

To avoid showing each loss function for every one of the following analyses, the main loss function considered here is the Cartesian loss. Unless stated otherwise, the MMT is always trained with the Cartesian loss. The figure 7.5 shows the angle between the true and

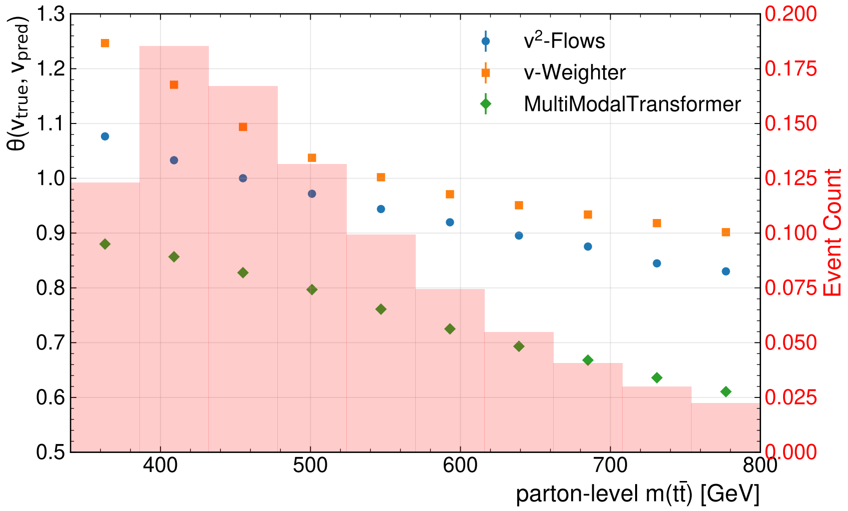


Figure 7.5.: Mean angle between the true and predicted neutrino vectors binned with respect to the parton-level invariant mass of the top-quark pair system.

predicted vectors of the neutrinos. This metric was chosen for comparisons across different energy regimes because it is scale-free and independent of the neutrino masses involved. As the energy of the neutrinos involved is strongly correlated with the invariant mass of the $t\bar{t}$-system, choosing a solid metric for this comparison was particularly difficult. The first observation is that the MMT consistently outperforms the other methods in providing, on average, a smaller directional deviation of the predicted neutrinos. Notably, this difference is substantial across the entire spectrum and much larger than the difference between the two baseline methods. Another notable feature is that all methods show a nearly linear dependence on the invariant mass $m(t\bar{t})$ as the angular distance increases at lower masses. While this trend appears to be very similar across all methods, the slope of the ν-Weighter seems to increase further in the lowest-mass bins, leading to a growing difference between the two baseline methods. One major impact factor for this behaviour is the decay angle $\theta(\ell, \nu)$ between lepton and neutrino. figure A.6 in the appendix shows the double-binned distribution of neutrinos with respect to the event's invariant mass $m(t\bar{t})$ and the angle between lepton and neutrino at parton level. It is clear that the distribution of angles between the lepton and neutrino is broader and peaks at a higher value in the low invariant-mass regime. This can, as before, be attributed to the lower energy of top quarks and W bosons in this region, which leads to decreased collimation of W-boson decay products. To investigate the methods' dependences on the angle between

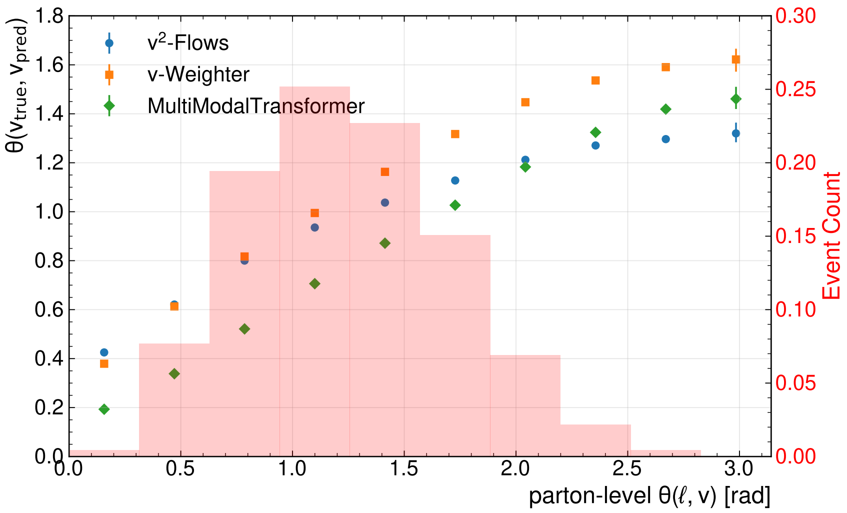


Figure 7.6.: Mean angle between the true and predicted neutrino vectors binned with respect to the parton-level angle $\theta(\ell, \nu)$ between lepton and neutrino.

lepton and neutrino, figure 7.6 shows the binned evaluation of the angular discrepancy. All methods show a significant increase in the angle between truth and reconstruction as the angle between the lepton and neutrino increases. The range covered by this variation is far larger than the variation seen for different invariant masses $m(t\bar{t})$. It can further be noted that the MultiModalTransformer appears to be most sensitive to the angle $\theta(\ell, \nu)$, since it exhibits the largest difference in performance between the largest and smallest bin of the variable. While for the bins corresponding to collinear leptons and neutrinos, the MMT can provide the best performance in terms of angular deviation between truth and prediction, ν^2-Flows provides the best performance for events for large angle $\theta(\ell, \nu)$. figure A.19 in the appendix provides a combined investigation of the impact of both variables. The double-binned evaluation of the angle between true and prediction shows that the impact of the angle between lepton and neutrino dominates significantly compared to the invariant mass of the top-quark pair system. All methods, but the ν-Weighter in particular, seem to be largely independent of the $m(t\bar{t})$ for fixed bins of $\theta(\ell, \nu)$. Thus, the difference observed in figure 7.5 seems to be largely explained by the correlated variable $\theta(\ell, \nu)$.

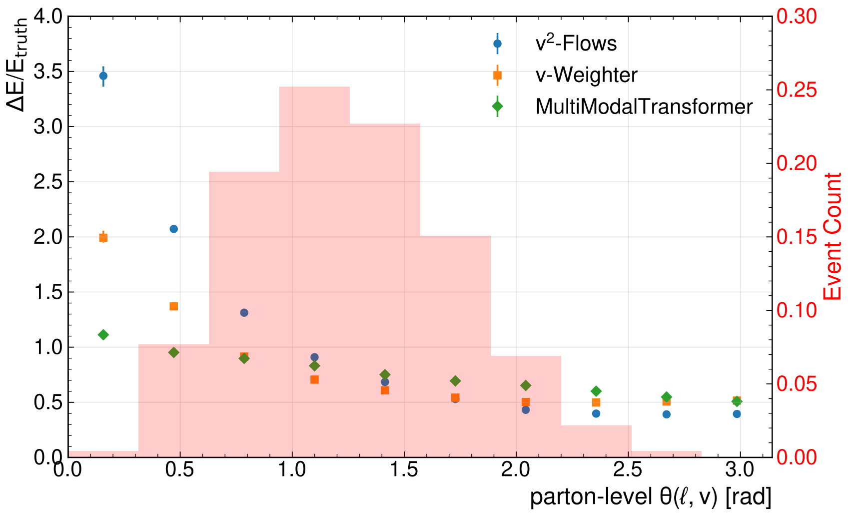


Figure 7.7.: Relative deviation of the neutrinos' energies (magnitudes) binned with respect to the decay angle between lepton and neutrino $\theta(\ell, \nu)$ at parton level.

The impact of $\theta(\ell, \nu)$ on the magnitude and scaling of neutrinos was examined by analysing the relative deviation of neutrino magnitudes, as shown in figure 7.7, binned according to the decay angle $\theta(\ell, \nu)$. Although the performance regarding the angle between truth and prediction improved for all methods as leptons and neutrinos became more collinear, the relative deviation in reconstructed neutrino magnitudes exhibited the opposite trend. For all methods, deviation increased as leptons and neutrinos became more collinear. However, significant differences in sensitivity were observed among the methods. For ν^2-Flows, the increase in relative deviation between the lowest and highest bins exceeded a factor of 7, whereas for the MultiModalTransformer, it was approximately 2. Both baseline methods demonstrated a much stronger decline in performance compared to the MMT model. While a detailed explanation for this behaviour cannot be provided at present, it is suspected to result from differences in the computational structures underlying the methods. Further investigation is required to substantiate this hypothesis. Notably, the MMT model trained with the MagDir loss exhibited the same dependence on the decay angle of lepton and neutrino as the ν^2-Flows method, underscoring the substantial influence of the chosen loss function on model behaviour.
No significant sensitivities were identified for other event variables examined in this thesis.

7.3. Variable Reconstruction

The primary interest in providing jet-assignment and neutrino regression lies in the subsequent reconstruction of event variables. Therefore, studying the impact of the proposed methods on the quality of event reconstruction is a critical assessment. To evaluate this, various variables of interest commonly used in analysis settings are examined. The variables investigated are primarily the invariant mass $m(t\bar{t})$ and the spin-sensitive markers

c_{hel} and c_{han}. As these were the main variables used in recent ATLAS analyses. Since the neutrino reconstruction of the ν-Weighter was found to offer the poorest resolution here, it was dropped to evaluate the variable reconstruction. Further, it is the only method that yields non-unity efficiencies; thus, the other variables can be evaluated without any subsampling, allowing a less biased evaluation.
The reconstruction performance is mostly evaluated in terms of the

$$\text{Mean Deviation} = \mu\left(x_{\mathrm{reco}} - x_{\mathrm{true}}\right), \tag{7.1}$$

$$\text{Relative Deviation} = \mu\left(\frac{x_{\mathrm{reco}} - x_{\mathrm{true}}}{x_{\mathrm{true}}}\right) \tag{7.2}$$

and

$$\text{Resolution} = \sigma\left(x_{\mathrm{reco}} - x_{\mathrm{true}}\right) \tag{7.3}$$

$$\text{Relative Resolution} = \sigma\left(\frac{x_{\mathrm{reco}} - x_{\mathrm{true}}}{x_{\mathrm{true}}}\right) \tag{7.4}$$

where x_{reco} and x_{true} denote the reconstructed and true value of the variable; the operators μ and σ are the mean and standard deviation computed over the sample of events, respectively. Unless stated otherwise, the true value of a variable is always computed at the parton level and after final state radiation. While the mean (relative) deviation is a measure of any biases and skew in the reconstructed distributions, the (relative) resolution is a measure of the spread of the reconstructed variable around the true value. Both of these are crucial to assess the quality of the reconstruction. For fits, however, resolution tends to be more important, as non-zero deviations and skew are corrected by the fit, provided they are consistent in both data and simulation. A distinction is made between the difference attributed to the jet-assignment and the neutrino regression. Therefore, improvements due to increased assignment performance are studied first.

7.3.1. Impact of Novel Assignment Methods

First, the effect of he improved jet-assignment is studied alone. To isolate the impact the assignment has, all methods are evaluated in conjunction with ν^2-Flows. The first variable to be investigated is the invariant mass $m(t\bar{t})$. The figure 7.8 summarises relative deviation and relative resolution of the invariant mass $m(t\bar{t})$. Note that the invariant mass is only sensitive to the selection of the jets, and not their assignment, as the computation is permutation invariant to the jet-ordering. As expected, both baseline methods achieve the same performance within the statistical uncertainties. While the mean relative deviation is positive and close to zero for the baseline methods, the CompactTransformer yields a much larger negative mean relative deviation. This means that the reconstruction with the CompactTransformer is biased towards smaller invariant masses. For the resolution, on the other hand, one can see that the CPT achieves a resolution, smaller by around 17%. The table A.3 in the appendix summarises the mean deviation and resolution achieved for this variable and compares them to the values for a perfect assignment of the jets in conjunction with ν^2-Flows. It clearly shows that the bias in the reconstructed $m(t\bar{t})$ induced by the CPT's assignment is very close to the one with the true assignment of

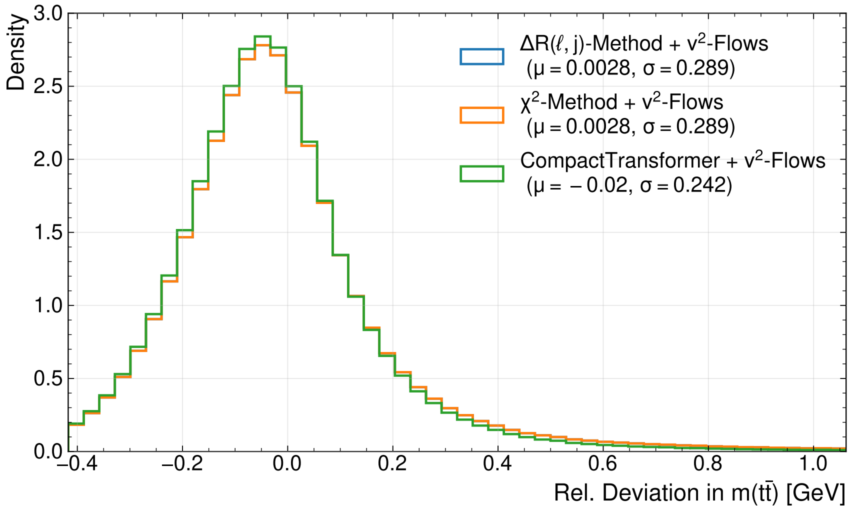


Figure 7.8.: Plot showing the relative deviation of the invariant mass reconstruction. The legend includes the obtained mean and standard deviation, which correspond to the resolution, for each method.

the jets. Thus, it can be assumed this is an effect from other parts of the reconstruction, which is cancelled out by the other methods, rather than a shortcoming of the Transformer. Furthermore, the true assignment does not show an improvement exceeding the statistical uncertainties. While this is hardly surprising, given the transformer's selection accuracy of about 96.6%, it implies that no measurable improvement in the resolution of the $m(t\bar{t})$ is expected from further optimisation of the jet selection. The mean deviation and resolution

Table 7.4.: Summary of the reconstruction metrics for c_{hel} for the different jet-assignments methods in conjunction with ν^2-Flows.

Method	Mean Deviation c_{hel}	Resolution c_{hel}
$\Delta R(\ell, j)$-Method + ν^2-Flows	$0.0295^{+0.0005}_{-0.0005}$	$0.4496^{+0.0005}_{-0.0005}$
χ^2-Method + ν^2-Flows	$-0.0024^{+0.0005}_{-0.0004}$	$0.4300^{+0.0003}_{-0.0002}$
CompactTransformer + ν^2-Flows	$0.0254^{+0.0006}_{-0.0004}$	$0.4117^{+0.0007}_{-0.0004}$
True Assignment + ν^2-Flows	$0.0166^{+0.0003}_{-0.0004}$	$0.38595^{+0.00017}_{-0.00033}$

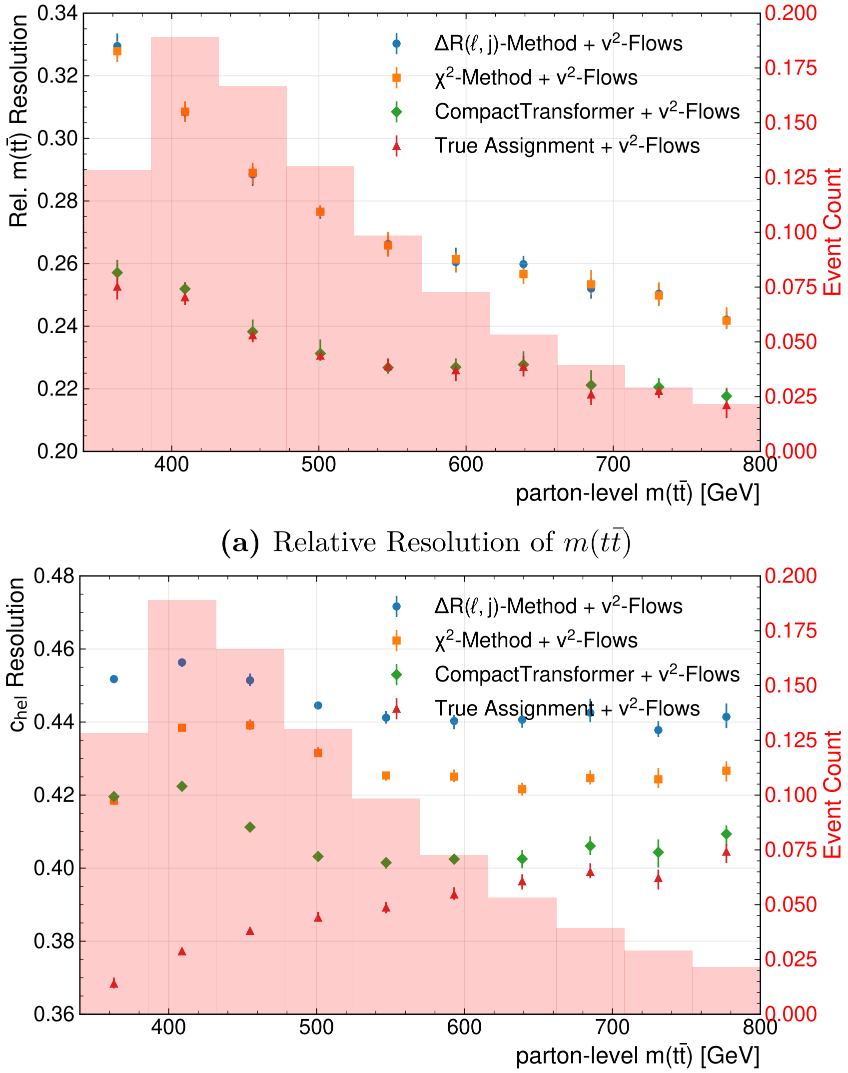
Table 7.5.: Summary of the reconstruction metrics for c_{han} for the different jet-assignments methods in conjunction with ν^2-Flows.

Method	Mean Deviation c_{han}	Resolution c_{han}
$\Delta R(\ell, j)$-Method + ν^2-Flows	$0.0476^{+0.0009}_{-0.0006}$	$0.5662^{+0.0006}_{-0.0004}$
χ^2-Method + ν^2-Flows	$0.0040^{+0.0006}_{-0.0007}$	$0.5342^{+0.0005}_{-0.0006}$
CompactTransformer + ν^2-Flows	$0.0118^{+0.0007}_{-0.0007}$	$0.5174^{+0.0005}_{-0.0005}$
True Assignment + ν^2-Flows	$0.0121^{+0.0008}_{-0.0007}$	$0.4874^{+0.0007}_{-0.0007}$

for the two spin-sensitive marker c_{hel} and c_{han} are summarised in the table 7.4, respectively. As before, the χ^2-Method attains the lowest mean deviation for both variables, indicating

that it again provides the least biased reconstruction. Previously, it provided the same unbiased reconstruction; however, for this variable, the ΔR-Method exhibits the highest bias, with the mean deviation for the c_{han} being roughly four times that of the next-largest method, which notably is the combination of true assignment with ν^2-Flows. As this variable largely depends on the angular relation between the leptons and the jets, it is expected to be substantially more biased for a Method that enforces the proximity of jets and leptons. The CPTs show significantly larger biases than the χ^2-Method for both variables. However, the same is true for the reconstruction with the true assignment interfaced with ν^2-Flows. As before, this implies that inherent biases in other aspects of reconstruction (such as ν^2-Flows itself) give rise to deviations at the parton level that are cancelled by the χ^2-Method.
Regarding the resolution, it is worth noting that it is generally much poorer than that of the invariant mass. Further, both variables yield the same results across methods, with the CompactTransformer achieving the highest resolution. The χ^2-Method comes in second, and its difference in resolution is about the same towards both methods for the two variables. While no significant improvement in resolution was observed for the reconstruction of the $m(t\bar{t})$ with the true assignment, this is substantially different for these variables. This highlights their strong dependence on the assignment in contrast to the invariant mass. For both variables, the difference between the perfect assignment and CompactTransformer is larger than its advantage over the χ^2-Method. Thus, for these variables, significant increases are expected as jet assignment improves, motivating future research in this area.
As before, the resolution of the reconstructed variables was examined across different event variables. For most of the considered event features here, the behaviour of the resolution followed closely that of the (selection) accuracy discussed in section 7.1. One notable exception is the invariant mass of the $t\bar{t}$-system, for which the binned resolution plots for the variable itself and c_{hel} are shown figure 7.9. For the resolution of the reconstructed $m(t\bar{t})$, the plot clearly shows that, across all methods, resolution degrades at low invariant mass. Notably, this increase is much more pronounced for the baseline methods, despite the selection accuracy being constant across the invariant mass spectrum for all methods. So this difference cannot be explained by a simple dependence on the jet selection, but reflects systematic effects in the misassignments across the different methods. As before, the resolution achieved by the CompactTransformer is on par with the perfect assignment across the full spectrum of invariant masses considered here. Thus, the drop-off in performance for the CompactTransformer can be attributed to other aspects of the reconstruction. Most notable is the error of the neutrino-regression as provided by ν^2-Flows, which was shown in section 7.2 to be larger for lower invariant masses. For the resolution of the second variable, c_{hel}, a different picture emerges from figure 7.9. The resolution achieved by the perfect assignment shows a steady improvement towards lower masses, indicating that the error of the neutrino regression is less important in this region. This finding already contrasts with the observation for the invariant mass $m(t\bar{t})$. For the assignment methods investigated here, the resolution is most stable at higher invariant masses; it shows the worst performance across all methods at the bulk of the distribution around $m(t\bar{t}) \sim 450$ GeV. Towards the lowest mass bin, all methods show again an increase in performance, which is most pronounced for the χ^2-Method. In this single bin,

it outperforms the resolution achieved by the CompactTransformer. This improvement in resolution is inconsistent with the steadily decreasing assignment accuracy shown in figure 7.2. Therefore, the effect seems to stem from a combination of improved performance in neutrino regression and reduced assignment accuracy. However, the significantly larger improvement of the χ^2-Method remains elusive. This underscores that not only the quantity of missassignments of the different methods but also their structure is decisive for the resolution of a variable.



(a) Relative Resolution of $m(t\bar{t})$

(b) Resolution of c_{hel}

Figure 7.9.: (Relative) Resolution of $m(t\bar{t})$ (a) and c_{hel} (b) computed in bins of the parton-level invariant mass of the $t\bar{t}$-system. The different methods, as well as the expected resolution with perfect efficiency for the assignment, are compared.The displayed error bars represent the statistical uncertainties of the dataset obtained via bootstrapping.

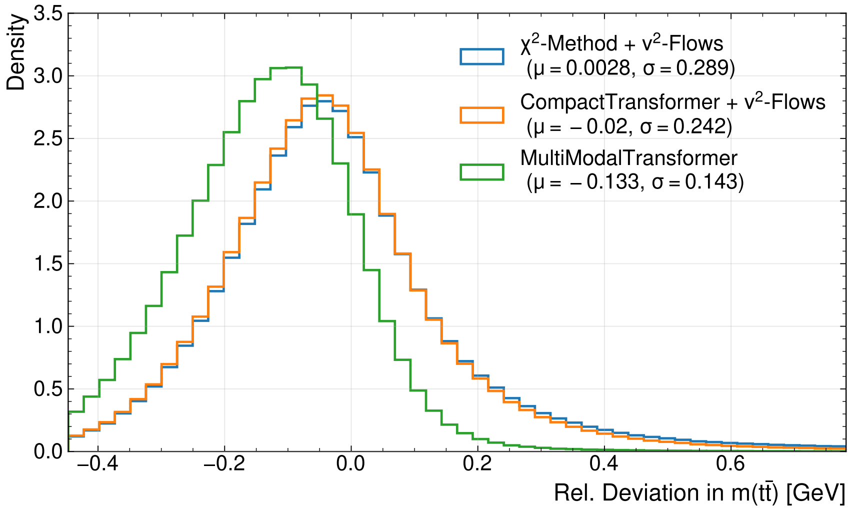


Figure 7.10.: Relative deviation of the invariant mass of the $t\bar{t}$-system, $m(t\bar{t})$, compared to the parton-level. The legend shows the mean relative deviation and the relative resolution for all methods.

7.3.2. Impact of Novel Neutrino Regression

To keep the investigated combinations of analysis setups feasible, the ΔR-Method is omitted from further investigations. The figure 7.10 shows the relative deviation of the invariant mass as before, including the reconstruction using the MultiModalTransformer. The first observation is that the MMT deviations show a very large negative shift-—meaning the reconstructed mass is smaller than the true mass. This can also be seen in the mean relative deviation for the MMT, which is substantially more negative at about -0.133 compared to the CPT with ν^2-Flows at -0.02. For the relative resolution, on the other hand, a clear improvement over the reconstruction with ν^2-Flows is evident. With a reduction of about 40% compared to the CPT with ν^2-Flows, this yields an improvement by a factor of 2 over the baseline χ^2-Method.

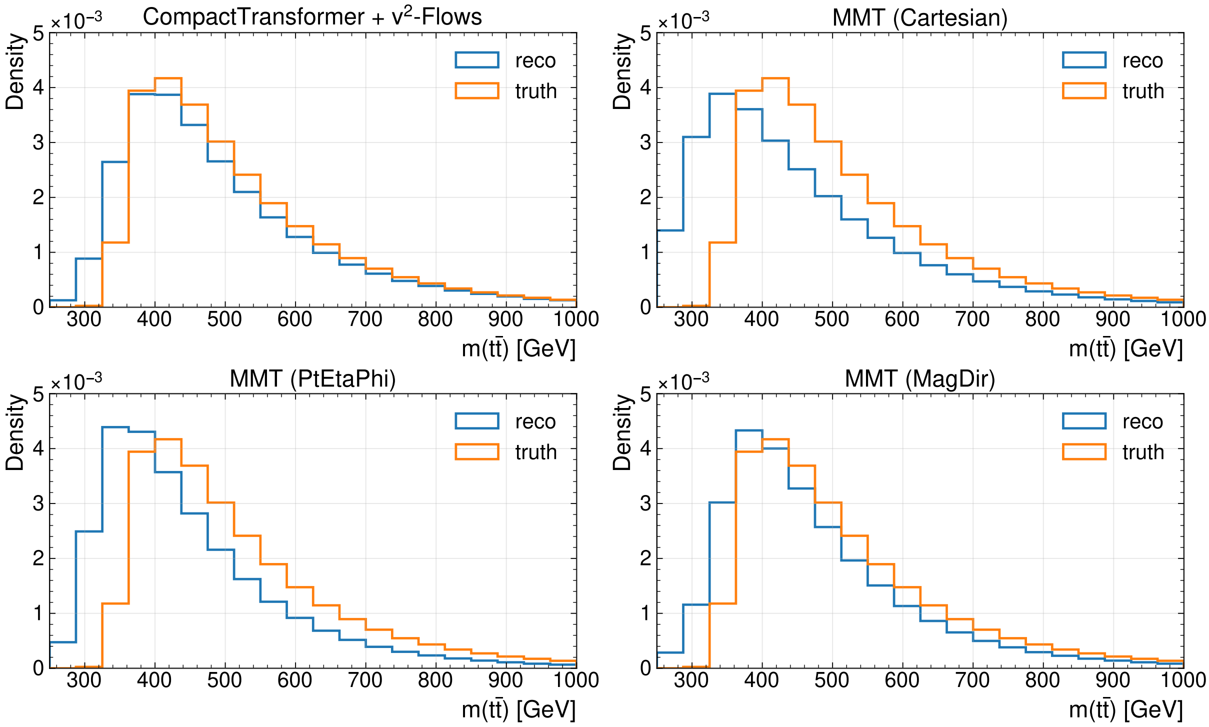


Figure 7.11.: Comparison of distributions of reconstructed and true $m(t\bar{t})$ for different reconstruction methods. Note that true refers to the parton level.

To examine the biases' impact on the shape of the distributions of the invariant mass, figure 7.11 compares the distributions at parton- and reconstruction-level for the different methods. The Figure compares the different losses used with the CPT interfaced with ν^2-Flows. The last reconstruction setup produces a reasonable distribution at the reco-level that matches the parton-level features, with the most notable discrepancy in the low-mass region. Two of the losses investigated, Cartesian- and PtEtaPhi-loss, show strong deviations in the shape of the distribution, which is also in good agreement with the observation of the bias for the relative deviation. Both reconstructions show a significant shift in the distribution's peak. Further, the bins below the production threshold of $m(t\bar{t}) \sim 2 \cdot m_t$ are strongly populated. The distribution of the model trained with the MagDir-loss shows a closer resemblance to the distribution, but displays stronger deviations in the shape compared to the reconstruction based on ν^2-Flows. table A.4 summarises the overall performance of the different tools for the invariant mass $m(t\bar{t})$. It shows that the MagDir-loss achieves the best resolution among all methods and has the lowest bias among all MMT-models, which is about a factor of 2 smaller than that of the other models. Thus, it is consistent with the distributions in figure 7.11.
The tables 7.6 and 7.7 summarise the comparison of the different MMT models with ν^2-Flows for c_{hel} and c_{han}, respectively, by showing mean deviation and resolution. Despite small differences in the mean deviation and resolution of the variables, the overall pattern remains the same: the MMT-based reconstruction provides the lowest resolution but incurs larger biases. This observation is valid across all losses. While the MagDir-loss could unambiguously provide the best performance for the invariant mass of the $t\bar{t}$-system, this

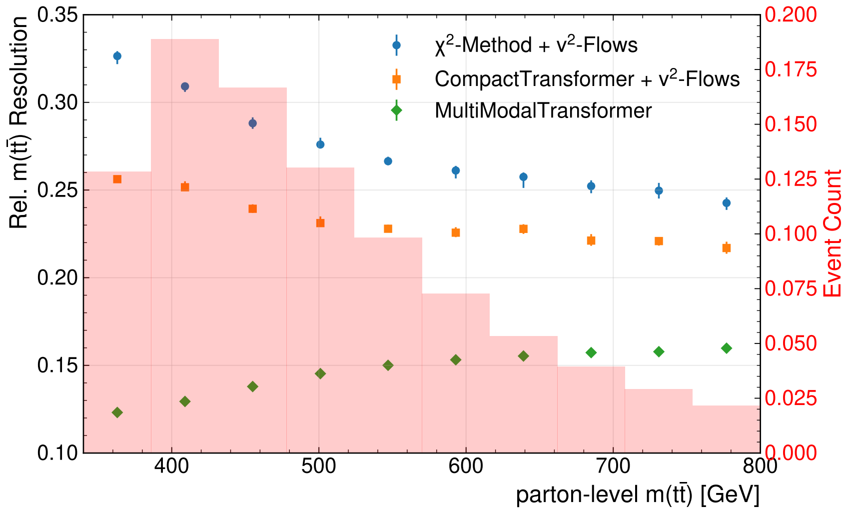
Table 7.6.: Summary of the reconstruction metrics for c_{hel} for MMT models with different models, compared to a reconstruction based on ν^2-Flows

Method	Mean Deviation c_{hel}	Resolution c_{hel}
CompactTransformer + ν^2-Flows	$0.0252^{+0.0007}_{-0.0004}$	$0.4115^{+0.0003}_{-0.0004}$
MMT (Cartesian)	$0.0697^{+0.0004}_{-0.0005}$	$0.3538^{+0.0006}_{-0.0005}$
MMT (PtEtaPhi)	$0.05674^{+0.00020}_{-0.00018}$	$0.3495^{+0.0007}_{-0.0005}$
MMT (MagDir)	$0.0719^{+0.0004}_{-0.0005}$	$0.3499^{+0.0005}_{-0.0004}$

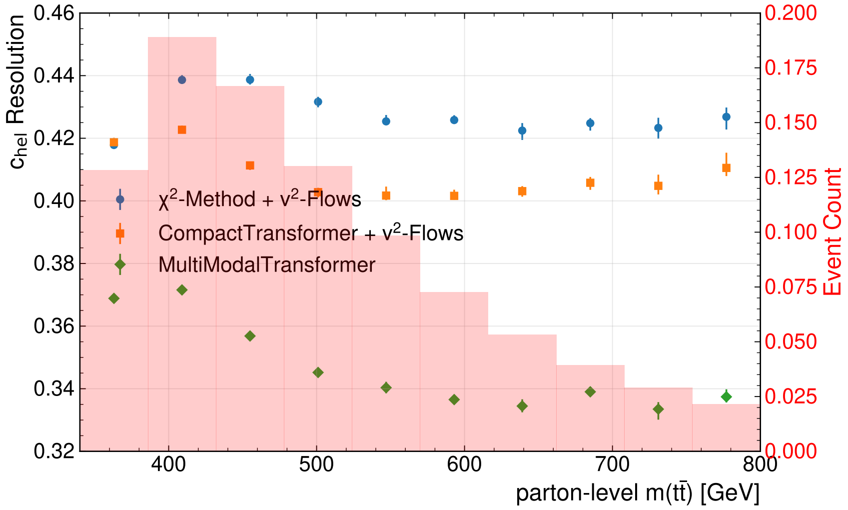
Table 7.7.: Summary of the reconstruction metrics for c_{han} for MMT models with different models, compared to a reconstruction based on ν^2-Flows

Method	Mean Deviation c_{han}	Resolution c_{han}
CompactTransformer + ν^2-Flows	$0.0120^{+0.0008}_{-0.0006}$	$0.5174^{+0.0006}_{-0.0006}$
MMT (Cartesian)	$0.0336^{+0.0004}_{-0.0004}$	$0.4585^{+0.0007}_{-0.0005}$
MMT (PtEtaPhi)	$0.0235^{+0.0006}_{-0.0006}$	$0.4858^{+0.0005}_{-0.0006}$
MMT (MagDir)	$0.0231^{+0.0005}_{-0.0004}$	$0.4791^{+0.0005}_{-0.0005}$

is not the case for the spin-sensitive variables. For these, no clear pattern emerges, making one loss preferable; for c_{hel} all loses show a very similar resolution and small differences in the mean deviation, and for c_{han} the Cartesian-loss has the best performance for the resolution, but also comes with the largest mean deviation. Generally, the Transformer's regression structure employed here yields results that are competitive with the baseline methods. In particular, the resolution of the variables investigated here could be reduced significantly. However, it was found that the regression, which does not provide any probabilistic modelling, unlike ν^2-Flows, incurs much larger biases that also lead to skewness in the distributions.



(a) Relative Resolution of $m(t\bar{t})$



(b) Resolution of c_{hel}

Figure 7.12.: (Relative) Resolution of $m(t\bar{t})$ (a) and c_{hel} (b) computed in bins of the parton-level invariant mass of the $t\bar{t}$-system. Different setups for assignment and regression are compared.

As before, some kinematic regions are of particular interest and have been shown to be challenging in previous analyses; the effects of certain event variables are to be investigated. The plots in figure 7.12 show the resolution for the reconstruction using the MultiModalTransformer across different $t\bar{t}$-mass bins. For the resolution of the $m(t\bar{t})$ itself, an interesting pattern emerges; whereas before the resolution decreased towards lower invariant masses, the opposite is true with the MMT-model. Its performance is mostly constant, with a slight improvement towards low masses. Therefore, the gain in resolution in the lowest-mass bin, which is of interest in many analyses, is approximately a factor of 3 over the baseline method. This is expected to significantly improve the sensitivity of analyses in this region. For the spin-sensitive variables, c_{hel} in particular, such a strong dependence on the mass of the $t\bar{t}$-system can not be observed. Here, the improvement in resolution offered by the MMT architecture appears to be very consistent. All methods seem to have the lowest performance around 400 GeV, which marks also the peak of the $t\bar{t}$-distribution.

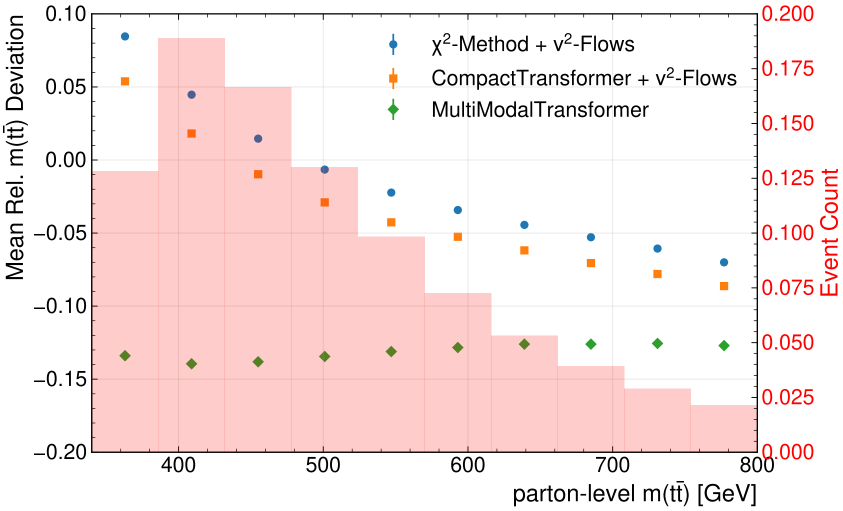


Figure 7.13.: Mean Relative Deviation of $m(t\bar{t})$

Since the mean deviation is also crucial for recovering the shape of a distribution, its behaviour for the invariant mass $m(t\bar{t})$ binned with respect to $m(t\bar{t})$ can be found in figure 7.13. While it was discussed earlier that the MMT has a strong bias towards lower invariant masses, the plot reveals that the bias's behaviour across the mass range differs markedly between the methods. The reconstruction based on ν^2-Flows yields a mean relative deviation ranging from -0.1 to $+0.1$, depending on the mass regime. While its modulus is smaller in all bins, this shows that the difference in the bias is much smaller than observed before, when averaging over the whole spectrum. Thus, it can be said that the methods using ν^2-Flows are also strongly biased in certain regimes, but these biases cancels out when considering the full invariant mass spectrum. For the MultiModalTransformer, on the other hand, the mean relative deviation remains very consistently at around -0.13. Thus, it might be possible to correct for the bias in the distribution at the reco-level for the MMT.
The mean deviations of c_{hel} and c_{han}, as depicted in figure A.20, show a different picture. It peaks at low masses for all methods. But the different methods have a very consistent difference in their bias across the different $m(t\bar{t})$-regimes.

7.3.3. Impact of Gaussian sampling

Lastly, the effect of the Gaussian-loss on the variable reconstruction, and their distributions in particular, has to be discussed. As pointed out before, no significant difference between using the mean of the Gaussian-loss-trained MMT and the nominal MMT was found. Therefore, this section compares the Gaussian-loss for reconstructing neutrinos using the mean and random sampling. For the deviation of the neutrinos' momenta, it was found that the Gaussian sampling consistently leads to larger errors. A similar observation can be made for its impact on the resolution of reconstructed variables.

Table 7.8.: Summary of mean relative deviation and relative resolution for the reconstructed mass of the $t\bar{t}$-system for the MMT trained with the Gaussian-loss with Mean and Gaussian sampling.

Method	Mean Rel. Deviation $m(t\bar{t})$	Rel. Resolution $m(t\bar{t})$
CompactTransformer + ν^2-Flows	$-0.02027^{+0.00019}_{-0.00020}$	$0.2419^{+0.0010}_{-0.0013}$
MMT (Mean)	$-0.12839^{+0.00013}_{-0.00008}$	$0.1408^{+0.0004}_{-0.0003}$
MMT (Gaussian Sampling)	$(0.3^{+0.1}_{-0.2}) \times 10^{-5}$	$0.2109^{+0.0005}_{-0.0004}$

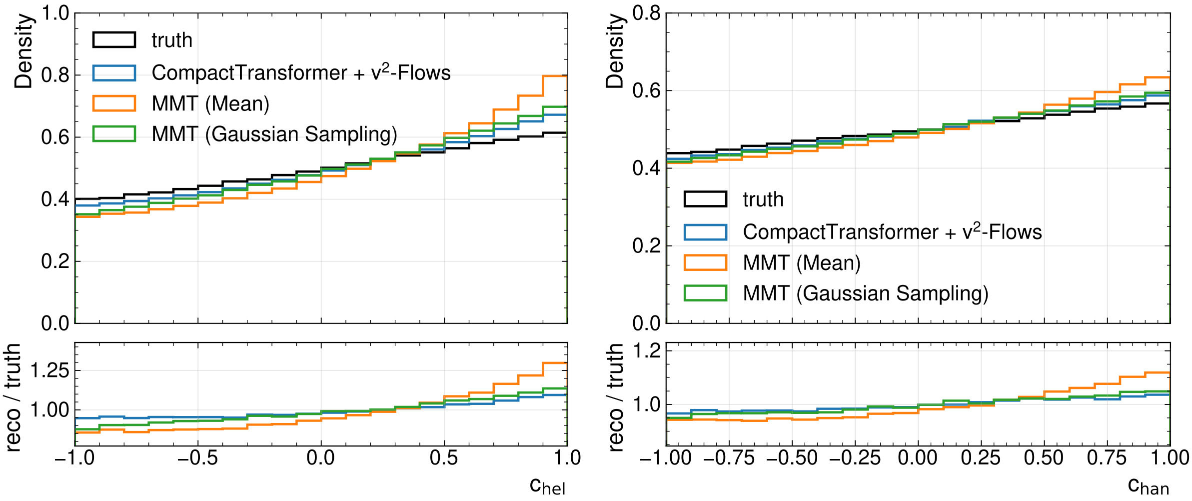
Table 7.9.: Summary of c_{hel}-reconstruction performance

Method	Mean Deviation c_{hel}	Resolution c_{hel}
CompactTransformer + ν^2-Flows	$0.0251^{+0.0004}_{-0.0003}$	$0.4114^{+0.0003}_{-0.0006}$
MMT (Mean)	$0.0705^{+0.0005}_{-0.0004}$	$0.3504^{+0.0004}_{-0.0003}$
MMT (Gaussian Sampling)	$0.0414^{+0.0006}_{-0.0005}$	$0.4339^{+0.0004}_{-0.0005}$

Table 7.10.: Summary of c_{han}-reconstruction performance

Method	Mean Deviation c_{han}	Resolution c_{han}
CompactTransformer + ν^2-Flows	$0.0118^{+0.0005}_{-0.0005}$	$0.5172^{+0.0005}_{-0.0005}$
MMT (Mean)	$0.0315^{+0.0004}_{-0.0004}$	$0.4559^{+0.0006}_{-0.0003}$
MMT (Gaussian Sampling)	$0.0166^{+0.0005}_{-0.0008}$	$0.5407^{+0.0008}_{-0.0003}$

The tables 7.8 to 7.10 provide a summary of the three reconstruction variables considered here. An overall pattern across all the variables can be identified: introducing the Gaussian sampling leads to a decrease in the mean (relative) deviation for all variables, i.e. reduces biases. On the other hand, the (relative) resolution decreases compared with mean-based regression. This loss of resolution varies across variables: while for $m(t\bar{t})$ it still exceeds that of the reconstruction with ν^2-Flows, the resolution of the spin-sensitive variables degrades significantly, falling below the performance of ν^2-Flows. The reduction of the bias for the $m(t\bar{t})$ is quite substantial with a factor of about 10^3, making the bias smaller than that of ν^2-Flows. For the other variables, such a strong reduction could not be observed, with the reduction being around a factor 2, and the bias still larger than for reconstruction with ν^2-Flows. While these metrics summarise the behaviour of the Gaussian-loss on the reconstruction, its impact on their distributions can also be studied. Considering the skew and distortions in the shape discussed earlier, this is one of the main points the Gaussian-loss seeks to address. For the mass of the $t\bar{t}$-system, plots comparing the distributions at the reco- and truth-levels are shown in figure A.21 in the appendix. The change in the distributions reflects a less biased reconstruction, which eliminates the previously observed shifted peak in MMT. It should be noted that while the peak position seems to align better with the truth, its height with respect to the truth level is substantially reduced. Comparisons of the distributions of the spin-sensitive variables are shown in figure 7.14, which presents the different reconstruction setups, the truth-level, and the ratios of the corresponding histograms. For both variables, the same behaviour of



(a) c_{hel} **(b)** c_{han}

Figure 7.14.: Distributions and density ratio between reco-level and parton-level for the MultiModalTransformer with Gaussian-loss for c_{hel} and c_{han} in (a) and (b), respectively.

the reconstruction methods is observed: while the reconstruction with ν^2-Flows provides the most faithful distribution, the MMT provides strong deviations in the tails of the distributions at ± 1. These effects are partly mitigated by the Gaussian sampling, which provides a much shallower ratio and a match to the truth-level that is comparable with what is obtained by ν^2-Flows.

While distributions of reconstructed masses of the intermediate particles, i.e. W boson and top quark, are less interesting for the aforementioned analyses, they can be considered a measure for the physical plausibility of the reconstructed neutrino momenta. For these two variables, two things are observed: the reconstructed masses of the MMT are much spread out and are significantly displaced in their peak with respect to the nominal value and true distributions. The corresponding distributions can be found in figures A.22 and A.23 in the appendix. Notably, the random sampling of the Gaussian-Loss seems to, at least partly, correct the shift in the distributions' peaks. This comes with the limitation of the peaks being even further spread out. Even though the mass constraints are not explicitly enforced in ν^2-Flows, the model seems to capture the masses, the W-boson mass in particular, very well; despite utilising a random sampling.

To investigate what drives the distortion of the distributions, several variables related to the angles between the reconstructed objects were examined to identify the cause of the strong deviations and the shape produced by the MMT regression. While the underlying structure remains largely elusive, the recovery of the shape by Gaussian sampling indicates that a major factor is that the reconstructed variables are often highly non-linear with respect to the neutrinos' momenta. Therefore, their computation does not commute with taking the mean of the variable over the possible neutrino momenta configurations, i.e.

$$\langle m(\ell^- \ell^+ b\bar{b}\nu\bar{\nu})\rangle_{\nu\bar{\nu}} \neq m(\ell^- \ell^+ b\bar{b}\langle\nu\bar{\nu}\rangle)\,. \tag{7.5}$$

Note that in this notation, the particles denote their four-momenta. The same holds for the computation of the spin-sensitive variables. This can lead to significant biases in the distribution by collapsing the neutrino momenta to the mean. Especially if the distribution has inherent multimodality, which, given the quadratic form of the mass-constraint (see section 2.2.3), is expected, a collapse of the distribution to the mean can lead to significant deviations. Note that a possible multi-modality in the conditional probability cannot be properly recovered by the Gaussian sampling here, as it lacks the expressivity to model such a distribution.

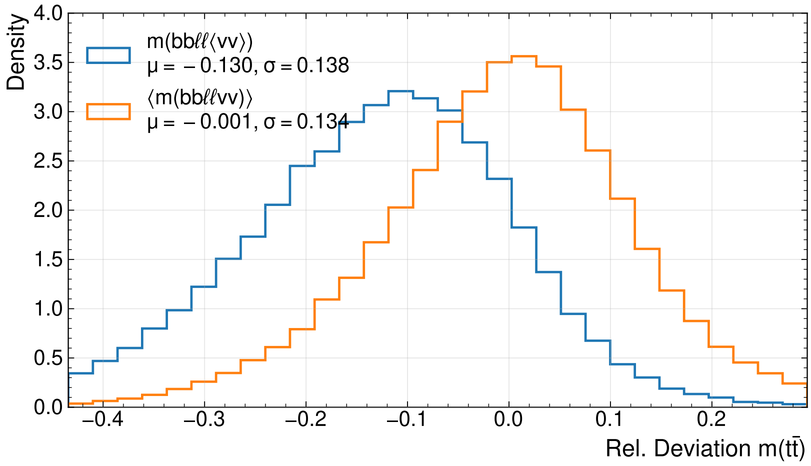


Figure 7.15.: Relative deviation between the reconstructed and true $m(t\bar{t})$ of the MMT trained with a Gaussian-Loss. The different curves show, whether the invariant mass was computed using the mean momentum or by propagating the neutrino probabilities and averaging of the obtained masses.

Plots showing the impact of propagating the posterior for the case of the Gaussian conditional probability density for the neutrino momenta are shown in the appendix under section B.2 for the variables discussed here. Due to the inaccessibility of the probability density of ν^2-Flow, a comparison of the methods' probabilities was not possible. It was found that by propagating the obtained conditional probability distribution to the reconstructed variables and averaging over the variable's distribution, one can obtain much lower biases and improved resolution. This can be seen in the figure 7.15. While random sampling reduces bias at the expense of resolution, this was not observed in this reconstruction strategy. Instead, both metrics improve when the probability distribution is propagated to the reconstructed variable.
Similar plots for the spin-sensitive variables can be found in figures B.3 and B.5 in the appendix. While this provides a promising approach to enhancing neutrino regression performance, it was not investigated further in this work due to time constraints.
In conclusion, the methods presented in this thesis can provide significant improvements in the reconstruction of variables critical to analyses. Generally, the evaluation depends very strongly on the variable and metrics one employs. Therefore, performance assessment is highly degenerate and warrants further investigation in future analyses.

8. Data-Driven Validation Studies

This chapter showcases the use of the developed tools in an analysis setting. For this, a setup similar to that used in the study of Quasi-bound-state effects in reference [2] is used to perform an Asimov fit. This demonstrates the tool's performance in a full-scale analysis and allows estimation of whether enhancements in measurement sensitivity are expected. Lastly, plots comparing the simulated distributions with experimental data collected by the Atlas Collaboration in 2018 are used to validate the agreement of model outputs on simulation and data.

8.1. Data and Simulation Samples

While the previous chapters used the nominal sample of top-quark pairs introduced in section 5.1 to set up and evaluate the ML tools, analysing event data requires a more comprehensive treatment of the contributing physics processes. Since the focus of this thesis lies on the machine-learning methods and their evaluation, an in-depth discussion of the samples used here is omitted. The samples used in this chapter mostly follow the reference [2]. Modelling the production of top-quark pairs follows the description in section 5.1, with a reweighting applied to match the cross-section calculation at next-to-next-to-leading order. This contribution is called pQCD, which is short for perturbative QCD.

Additionally, bound-state effects can modify the cross-section relative to the pQCD calculation. These are modelled using non-relativistic QCD (NRQCD). For this analysis, these contributions are modelled as described in reference [43]. The implementation details for event generation and the samples used here can be found in [2].

Finally, other Standard Model processes with similar experimental signatures have to be considered. These processes contribute to the event yield in the signal region without stemming from the process of interest – here $t\bar{t}$-production. These are referred to as backgrounds below. As before, this follows closely the description in [2]. The inclusion of backgrounds provides a realistic benchmark for testing different reconstruction methods in the analysis and for showing agreement of the tools between simulation and data. For simplicity, only the two leading background processes are considered in the fit; other, subdominant processes are noted below for completeness. One of the two considered processes is the production of a top quark in association with a W boson (tW), which was found to contribute about 4% of the total event yield in the signal regions. The second is the so-called Z+jets process, which primarily comprises leptonically decaying Z bosons from Drell-Yan production, with additional hadronic content in the event. Modelling this background introduces significant uncertainties; thus, its contribution is constrained and adjusted to the data in a dedicated control region. Two further contributions are neglected

in this analysis. Related background contributions from other processes involving the production of a single or pair of top quarks in conjunction with other particles were found to make up only 0.5% of the yield in the signal region [2] and are therefore not considered. Additionally, so-called fakes, where at least one fake or non-prompt lepton fulfils the lepton identification criteria described earlier, stem primarily from $t\bar{t}$ production with hadronic decays and were found to contribute around 1.5% of the total event yield [2]. Despite being larger than the other neglected contribution, fakes were likewise not considered for the analysis setup used in this work.
Finally, for the data-driven validation of the output distributions, data collected by the ATLAS Collaboration in 2018 is used. The amount of recorded collisions during this period is equivalent to a luminosity of $\mathcal{L} \approx 58\ \mathrm{fb}^{-1}$.

8.2. Analysis Setup

Table 8.1.: Summary of event selection criteria for the SRs and CRs in the analysis. The letter ℓ refers to e and μ.

Asimov-Fit		**Data-Validation**	
SRs	**CR-Z**	**CR-Z**	**CR-$\ell\ell$**
$\ell^{\pm}\ell'^{\mp}$ with $p_{\mathrm{T}}(\ell) \geq 10$ GeV ≥ 1 trigger-matched lepton with $p_{\mathrm{T}} \geq 28$ GeV ≥ 2 jets with $p_{\mathrm{T}} \geq 25$ GeV ≥ 1 b-tagged jet (85% efficiency WP) $m_{\ell\ell} \geq 15$ GeV for ee and $\mu\mu$ events $E_{\mathrm{T}}^{\mathrm{miss}} \geq 60$ GeV for ee and $\mu\mu$ events			
$m_{t\bar{t}} \leq 500$ GeV		$m_{t\bar{t}} > 500$ GeV	
$\lvert m_{\ell\ell} - m_Z \rvert > 10$ GeV	$\lvert m_{\ell\ell} - m_Z \rvert \leq 10$ GeV		$\lvert m_{\ell\ell} - m_Z \rvert \leq 10$ GeV

The setup of the analysis follows that developed in reference [2]; however, minor simplifications are made. The object definitions and reconstruction methods are the same as described in section 5.2. For the jet assignment and neutrino regression, three different setups are tested: ν-Weighter, ν^2-Flows, and MultiModalTransformer. Note that the two baseline reconstruction methods are used in conjunction with the χ^2-method for jet-assignment. Further, events where the ν-Weighter yields no solution are excluded for regions using the method. While previous analyses used fallback methods for such events, this would skew the comparison of the different methods conducted here.
To extract the parameter of interest, i.e. the cross-section of the $t\bar{t}_{\mathrm{NRQCD}}$-contribution, events are required to pass selection criteria and are then further categorised into regions. An overview of the selection criteria for the analysis regions is shown in table 8.1. The different regions can be categorised as signal and control regions, abbreviated to SR and CR, respectively. All share the same preselection, which puts the focus on events that contain the particles demanded by the final-state of dileptonic $t\bar{t}$-production, i.e. two

opposite charged leptons and at least two jets. Further requirements for these observed particles were found to reduce the contribution of background events. Events passing these criteria are then further divided. For the data-driven validation, four control regions are used, all of which require a cut on the invariant mass $m(t\bar{t}) > 500$ GeV. The selection of events with high invariant mass $m(t\bar{t})$ ensures that the regions used for data validation are disjoint from the signal regions, which are sensitive to the quasi-bound state effects. This is part of the blinding protocol and allows for comparison between data and simulation without biasing future analyses in the sensitive regions. The CR-Z is used to normalise the Z+jets contribution. For this region, the invariant mass of the two leptons is required to be in a window of 10 GeV around the nominal Z-boson mass. This enhances the contribution of events with leptons from resonant Drell-Yan production. The regions used for data and simulation validation are exactly disjoint with respect to the CR-Z; they use the same event selection but require events to be outside the dilepton mass window. For these events, three different control regions are defined based on the possible lepton flavour combinations, i.e. ee, $\mu\mu$, and $e\mu$.
A slightly different set of regions is defined for the Asimov-fit. While they share the pre-selection, the SRs and CR-Z both require the events to have an invariant mass $m(t\bar{t}) \leq 500$ GeV. This aims to enhance the contribution of non-relativistic QCD effects, which dominate near the production threshold. For the signal region, events are again required to have a dilepton mass outside the 10 GeV windows around the Z-boson mass. Events within this dilepton mass window constitute the CR-Z, which is used to extract the normalisation of the Z+jets background.
While a full analysis requires an in-depth investigation of systematic uncertainties from various sources, only statistical uncertainties are considered here. Due to the scope of this thesis statistical uncertainties are sufficient to evaluate the methods with respect to their sensitivity and validate the method on event data.
All fits are performed as binned profile-likelihood fits, where specific binning variables are used for each region. The fit-parameters are so-called normalisation factors, which measure the quotient of the measured cross-section and the prediction

$$\mu_X = \frac{\sigma_{\text{exp.}}(X)}{\sigma_{\text{pred.}}(X)}, \tag{8.1}$$

where $\sigma_{\text{exp./pred.}}(X)$ denote the measured and predicted cross-section of a process X, respectively.

8.3. Asimov Fit

For the Asimov-fit, only the SRs and the CR-Z are used. Due to the reduced fit-setup applied here, the fit parameters are $\mu(t\bar{t}_{\text{NRQCD}})$ (also *signal strength*), $\mu(t\bar{t}_{\text{pQCD}})$, and $\mu(Z\text{+jets})$. The latter is fitted in the CR-Z region, with events binned by dilepton mass. Normalisation factors for the $t\bar{t}$-production processes are adjusted in 9 signal regions, which require the event selection described in the previous section and are further categorised in SRs defined in figure 8.1 according to their values of c_{hel} and c_{han}. Within these regions, the events are binned with respect to the reconstructed $m(t\bar{t})$ with a width of

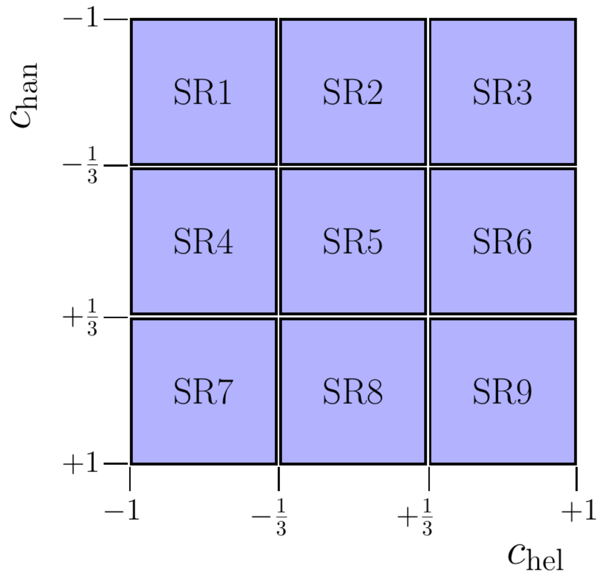


Figure 8.1.: Categorisation of the signal regions used in the fit. The events are divided based on their reconstructed values of the spin-sensitive variables c_{hel} and c_{han}. The Figure is adapted from reference [2].

Table 8.2.: Fit results for different reconstruction methods. The norm factors and their corresponding errors are shown. The significance is the number of standard deviations the extracted norm factor of $t\bar{t}_{\mathrm{NRQCD}}$ differs from zero.

Method	$\mu(Z+\text{jets})$	$\mu(t\bar{t}_{\mathrm{pQCD}})$	$\mu(t\bar{t}_{\mathrm{NRQCD}})$	Significance
ν-Weighter	1 ± 0.0512	1 ± 0.00659	1 ± 0.596	1.689σ
ν^2-Flows	1 ± 0.0253	1 ± 0.00345	1 ± 0.321	3.131σ
MultiModalTransformer	1 ± 0.0196	1 ± 0.00241	1 ± 0.193	5.222σ

50 GeV. Defining the signal regions with respect to the spin-sensitive variables allows for a more sophisticated test of the spin structure of the cross-section enhancement at the threshold. As it was outlined in section 2.2.4, different spin states show distinctive distributions of the markers c_{hel} and c_{han}, making the fit sensitive to not only the enhancement of the cross-section but also the internal structure of the decaying $t\bar{t}$-system. The Asimov data used for fitting is obtained by summing all samples with normalisation factors set to 1. The results for the Asimov-fit for the reconstruction methods investigated here are shown in table 8.2. Note that a computation of the expected significance demands a more comprehensive treatment of background and systematic uncertainties. As these contributions are largely independent of the reconstruction method used here, it still provides a meaningful comparison of the methods themselves. The results show substantial differences in the uncertainties of the norm factors and the extracted significance of the $t\bar{t}_{\mathrm{NRQCD}}$ process. Furthermore, the previously observed pattern in the performance of the different methods carries over to the significance of the analysis: the ν-Weighter has the largest error and, in turn, the lowest significance. While ν^2-Flows provides an improvement close

to a factor of 2, the use of the MMT enhances the significance by around 66% with respect to ν^2-Flows. Compared to the baseline of the ν-Weighter, this results in a 3-fold increase. Thus, it can be concluded that the sensitivity of an analysis can be increased by using the MultiModalTransformer. While a thorough investigation of the effect leading to the increased sensitivity would exceed the scope of this thesis, a few points should be made about the underlying properties of the reconstruction that influence this effect. One major aspect deteriorating the performance of the ν-Weighter is its limited efficiency; the reduced number of events in the signal region leads to larger statistical uncertainties. While the two ML-based methods share perfect efficiency, the previous chapter 7 found that the MMT provides considerably higher resolution. As the resolution quantifies the spread of the reconstructed variable around its true value, a method with higher resolution yields less diffuse peaks in the distributions. Considering that the quasi-bound state effects manifest as a narrow peak in the invariant mass distribution, as shown in figure 2.4, the sensitivity to such a signal gets enhanced by a reconstruction with a higher resolution. At reco-level this can be seen in figure A.24 in the appendix, which shows the simulated reconstructed distributions for the methods used here. Despite the apparent advantages of the MMT for reconstruction, it should be noted that a major problem of this method is the previously discussed bias introduced into many distributions. While this might be less relevant for a fit, it can make resulting distributions unphysical. One example is that the bulk of the invariant mass of the $t\bar{t}$ system lies below the MultiModalTransformer's production threshold.

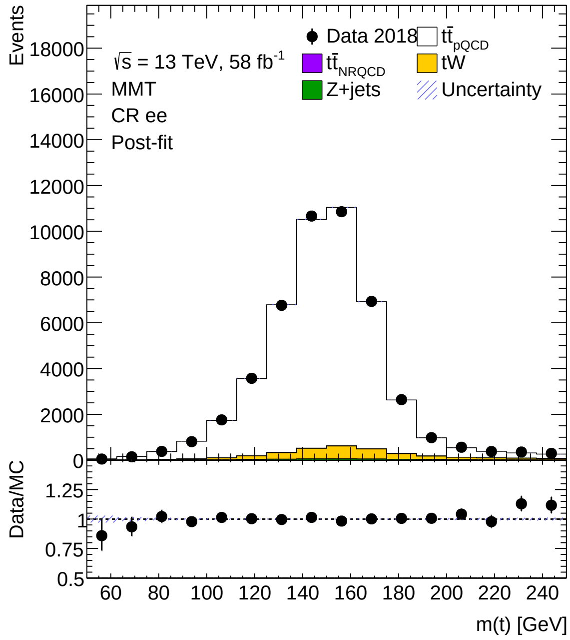
8.4. Validation Using ATLAS 2018 Data

To verify the distributions produced with the tool presented in this thesis, data and simulations are compared in the control regions introduced earlier. As before, the CR-Z region is used to normalise the Z+jets background. In the CR-$\ell\ell$, the normalisation of the top-quark production is fitted. Note that only $\mu(t\bar{t}\text{pQCD})$ is adjusted during the fit, as the non-relativistic production is expected to be negligible in the high-mass regions. For the likelihood fit, the events are binned by the reconstructed mass of the top quark. The result of the fit is best expressed by the optimal norm-factors, which are found to be

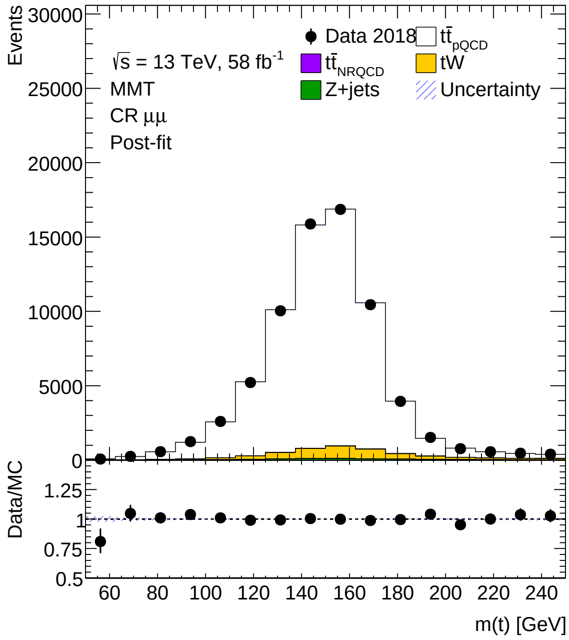
$$\mu(t\bar{t}_{\text{pQCD}}) = 0.966^{+0.002}_{-0.002}, \qquad \mu(Z\text{+jets}) = 1.08^{+0.03}_{-0.03}. \tag{8.2}$$

For both processes, the normalisations are close to the nominal value of one. While the normalisation for top-quark pair production deviates from unity beyond its uncertainties, this can be attributed to the simplified modelling of the background and the absence of systematic uncertainties. This is further supported by the observation that the norm factors exhibit similar departures from their nominal values across the baseline reconstruction methods. Thus, this deviation does not originate from the reconstruction of the MultiModalTransformer. The corresponding fit results from the baseline methods are shown in table A.5 in the appendix. figure 8.2 summarises the distributions in the different regions with fit-adjusted normalisations (Post-fit). Generally, the data seems to be in superb agreement with the simulation. The distribution matches very closely in all bins with sufficient events. Moreover, the distributions are very similar for the different leptons. Notably, all distributions show a peak for the reconstructed top-quark mass significantly below the nominal value. This is consistent with the biases discussed in the chapters before. Other variables, which rely on reconstruction and neutrino assignment, were investigated, such as $p_T(t/W), E(t/W), \eta(t/W)$. For these variables, similar agreement between the data and the simulation was observed. Their distributions can be found in figures A.27 and A.28.

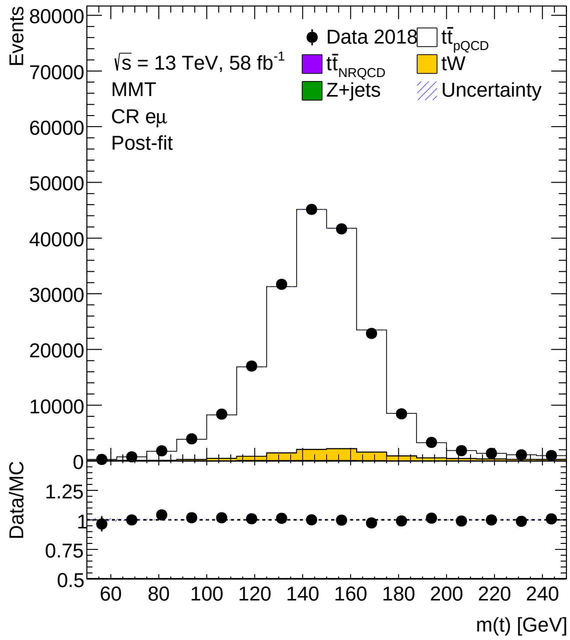
In summary, this chapter has demonstrated the practical relevance of the reconstruction methods studied throughout this thesis within a realistic analysis setting. The Asimov fit showed that the choice of jet-assignment and neutrino-regression method has a direct and substantial impact on physics sensitivity: moving from the ν-Weighter to the MultiModalTransformer nearly triples the expected significance of the $t\bar{t}_{\mathrm{NRQCD}}$ signal strength, a gain that can be traced back to the combination of full reconstruction efficiency and improved resolution established in chapter 7. At the same time, the validation against 2018 ATLAS data confirmed that this improved performance is not an artefact of simulation: the MultiModalTransformer reproduces the same mass, momentum, and angular distributions on data as on simulated events, including the characteristic downward bias in the reconstructed top-quark mass identified earlier in this work. Taken together, these results indicate that the sensitivity gains offered by the MultiModalTransformer are both substantial and reliable, supporting its use as a competitive reconstruction method for future analyses of threshold effects in top-quark pair production – provided that the biases discussed here are properly accounted for in such applications.



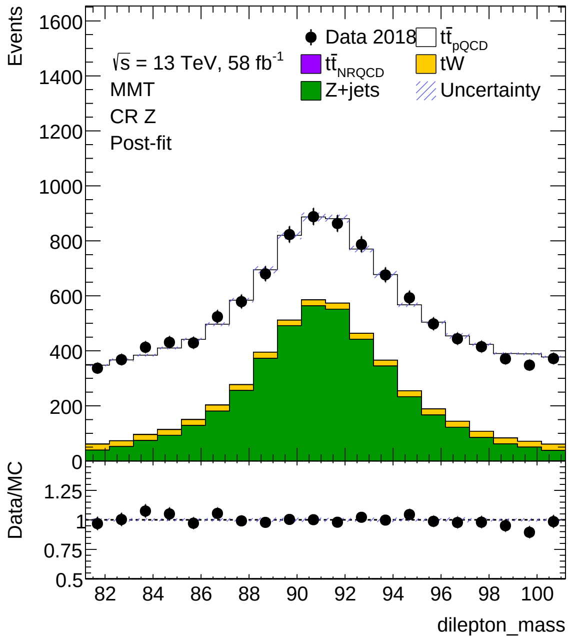
(a) CR *ee*



(b) CR $\mu\mu$



(c) CR $e\mu$



(d) CR Z

Figure 8.2.: Plots showing the comparison between data and simulation of top-quark pair production and its leading backgrounds. The different control regions used in the fits are shown as stacked histograms of the Post-fit distributions. Note that the reconstruction of the $t\bar{t}$-system is performed using the MultiModalTransformer.

9. Conclusion

9.1. Summary

Dileptonic decays of top quarks enable the exploration of numerous open questions and facilitate precise measurements. Although this channel exhibits a low branching ratio, it offers a particularly clean event environment. Recent studies of quantum and quasi-bound-state effects near the production threshold have motivated further investigation of this region, where dileptonic decays offer unique precision for probing spin structures in the underlying physical processes. Despite the clean environment provided by this channel, several challenges persist. The presence of two neutrinos from the leptonic decay of the W bosons results in six unknown momentum components, while the selection and assignment of b-jets to their parent top quarks introduces further ambiguities. Neutrino regression has been previously investigated, but neutrino assignment remains comparatively understudied, even though both tasks are inherently interconnected. Since accurate measurement of variables related to the internal structure requires full reconstruction of the top-quark four-momenta, reconstruction quality is critical to the sensitivity of top-quark pair studies.

To address the jet-assignment problem, a transformer-based machine-learning model was developed and trained. Various architectures and layouts were evaluated, all demonstrating similar performance despite incorporating different kinematic symmetries. The resulting model substantially improved jet-assignment efficiency, which in turn reduced reconstruction errors and improved the resolution of key variables, thereby increasing the sensitivity of subsequent measurements. Beyond these overall improvements, a detailed analysis identified specific regions of phase space that benefit most from machine-learning-based reconstruction. The region near the production threshold proved particularly challenging: no significant improvement in accuracy over baseline methods was observed there, and while certain kinematic features contributing to this difficulty were identified, some effects remain unexplained.

To further enhance event reconstruction and address the interdependence between jet assignment and neutrino regression, a model capable of performing both tasks was developed, extending the previous architecture with an additional output head for neutrino momenta. Although resolving this interdependency was an initial objective, the combined approach yielded only limited improvements: comparisons between single-task and multi-task models revealed no enhancement in jet-assignment accuracy and only minor gains in regression. Some cross-talk between the tasks was nonetheless observed, suggesting that existing tools for neutrino regression – whether analytical or machine-learning-based – can still provide valuable information for jet assignment.

The developed methods were also found to reduce errors in neutrino regression and improve the resolution of derived variables, such as spin-sensitive markers and the invariant mass of the $t\bar{t}$ system. Despite these improvements, the model exhibited significant biases in the reconstructed variables, distorting their distributions. The primary cause of this effect is likely the probabilistic nature of the neutrino momenta: since the neutrinos are inherently underdetermined, they are best represented by a conditional probability density, and training a regression model collapses the output to the mean of this distribution, which can introduce bias when computing nonlinear functions. Some corrective measures were identified and discussed; in particular, this issue can be mitigated by modelling and propagating the conditional density from the neutrino momenta to the reconstructed variables and averaging over their distributions, providing a more comprehensive representation of the kinematics.

The performance of the developed methods was further demonstrated by reproducing the analysis of bound-state effects at the production threshold in a simplified setting. An Asimov fit, designed to assess the potential analytical impact, indicated that uncertainties associated with the signal strength are significantly reduced by methods with enhanced resolution, such as ν^2-Flows and the MultiModalTransformer – the latter achieving a threefold increase in the significance of the signal strength. In practical applications, however, this enhancement will be constrained by systematic uncertainties, and the improved reconstruction provided by the MultiModalTransformer presents a notable limitation of its own: reconstructed variables of the top quarks, such as the invariant mass of the $t\bar{t}$ system and the spin-sensitive c_{hel} and c_{han}, exhibit significant biases and distortions in their shapes. It has been demonstrated that this does not affect results when both data and simulation are similarly distorted, but the issue can be critical for analyses that extract information directly from the distributions – for example, the measurement of quantum effects in reference [1], which utilised the slope of the c_{hel} distribution as a marker for entanglement. Notably, the distributions of reconstructed masses indicate that this machine-learning-based method yields conceptually distinct results compared with other approaches, more frequently violating physical constraints such as the production threshold or the masses of intermediate particles. Additional consideration is therefore required for such analyses, and it will be necessary to determine which reconstruction method yields the most reliable results for a given analysis and to adapt the approach accordingly.

For reconstructed variables to be useful in an analysis, it is essential that they are accurately modelled by simulation. To assess this for the MultiModalTransformer, distributions of reconstructed variables, such as the top-quark mass, were compared between simulation and data collected by Atlas. A very good agreement between the distributions was generally found, demonstrating that the tool is applicable to both simulation and event data and supporting its robustness for use in future analyses.

Combined with the thorough evaluation of the method's performance, these findings provide strong motivation to adopt the approach developed here, or similar methods, in future research. Integrating the two reconstruction tasks into a single model not only addresses their interdependence but also reduces computational demands – an increasingly important property as the transition to the HL-LHC, which will increase event numbers by up to an order of magnitude, introduces further challenges related to analysis computation time.

9.2. Outlook

This thesis has comprehensively examined the strengths and weaknesses of machine-learning-based reconstruction methods for the dilepton channel, and a key finding is the inherent interdependence of the jet-assignment and neutrino-regression tasks. Based on this and the other results presented, several new research directions emerge.

The analysis presented here demonstrates that the two reconstruction tasks are inherently linked. Therefore, it is strongly recommended that tools such as ν^2-Flows, which perform neutrino regression, be extended to include jet assignment. This recommendation is based on the observation that a transformer trained for neutrino regression develops an internal understanding of the assignment structure. Consequently, a tool should generally provide both functionalities, which would also enhance consistency between jet assignment and neutrino regression – consistency that may be compromised when using separate tools.

An important open question concerns the propagation of the neutrino posterior to derived event variables. It was demonstrated that averaging over the posterior of event variables, rather than computing event variables from averaged momenta, can reduce biases and enhance resolution. Extending this approach to more sophisticated models of posterior distributions, such as ν^2-Flows, is expected to significantly improve reconstruction performance. Beyond point estimates, incorporating the full posterior of an event variable into a fit or an uncertainty estimation may likewise be beneficial, providing a more comprehensive description of the underdetermined neutrino kinematics than a single averaged value can offer.

A related direction is to incorporate the probability distribution of reconstructed variables directly into the model's loss function. By employing automatic differentiation to compute the Jacobian, the probability density of the neutrinos can be transformed into the probability density of the reconstructed variables. This would enable optimisation of the model with respect to the variables of primary physical interest, such as $m(t\bar{t})$, c_{hel}, or c_{han}, rather than the neutrino momenta alone; direct regression of these variables could also be explored as a simpler alternative.

In scenarios where interpretability is less critical than the discriminative power of a variable, training a dedicated discriminator may be advantageous instead. For example, in the search for contributions from non-relativistic bound-state effects, a discriminative neural network could be trained to distinguish events originating from the pQCD and NRQCD states, thereby providing one or more sensitive variables tailored specifically to this signal.

A further open question concerns the biases identified in the reconstructed variables produced by the MultiModalTransformer. While this thesis identified their likely origin and demonstrated a mitigation strategy based on posterior averaging, a systematic study of how these biases propagate into and interact with experimental systematic uncertainties has not been performed. Since this thesis considered only statistical uncertainties, understanding this interplay and developing correction or unfolding procedures where necessary will be an important step before the method can be applied to precision measurements that rely on absolute distribution shapes rather than relative comparisons between data and simulation.

As the tool has demonstrated improvements in analyses involving dileptonic decays of top-quark pairs, it is anticipated to be utilised in future studies. The most prominent application remains the investigation of bound-state effects at the $t\bar{t}$ production threshold, where a significant reduction in uncertainty is expected for cross-section measurements using the MultiModalTransformer. To support its application in measurements by ATLAS, the tool has been integrated into the central software for top-related analyses: TopCPToolkit [78]. Combined with the increased statistics provided by Run 3 in the near future, substantial progress in understanding the dynamics at the top-quark pair-production threshold is anticipated.

A. Additional Plots

A.1. Data Feature Plots

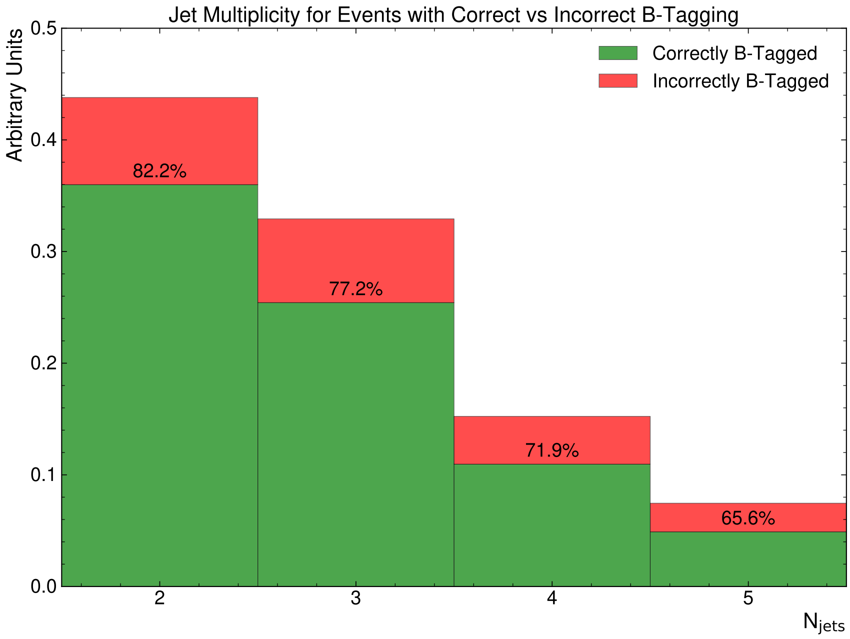


Figure A.1.: This plot shows the b-tagging performance for different numbers of jets in the event. For each jet multiplicity, the number of events, with only jets from top-decays being b-tagged and the number of events with additional or missing b-tagged jets is stacked. The percentage shows the fraction of events with only jets from top-decay being b-tagged.

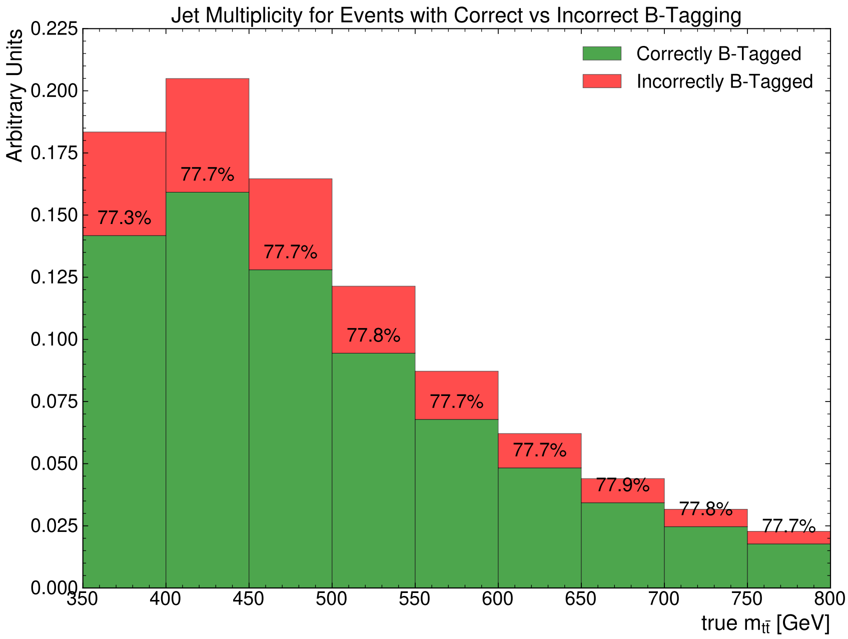


Figure A.2.: This plot shows the b-tagging performance for different bins of the invariant mass $m(t\bar{t})$ at parton level.

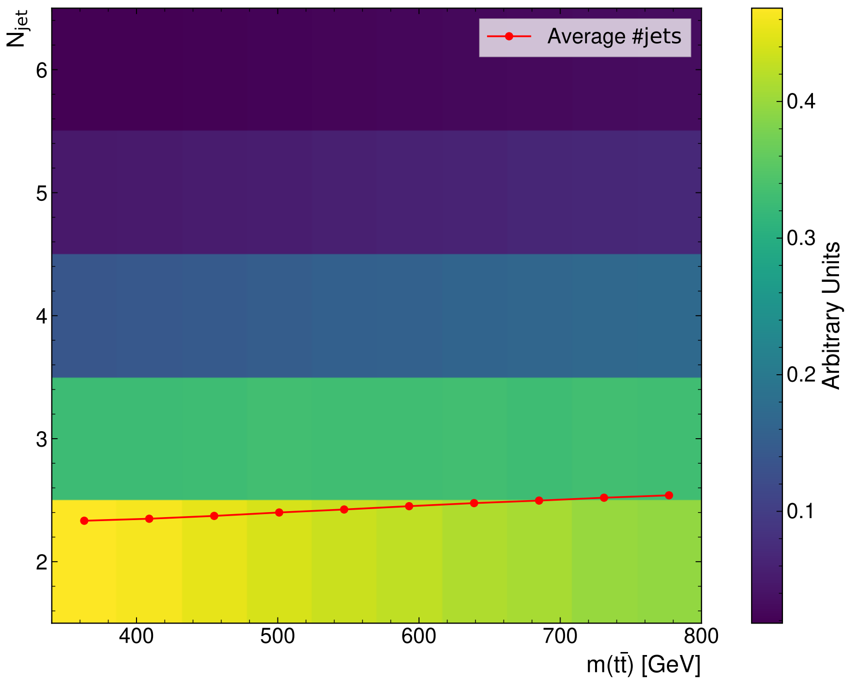


Figure A.3.: This plot shows the b-tagging performance for different bins of the invariant mass $m(t\bar{t})$ at parton level. For each bin, the number of events, with only jets from top-decays being b-tagged and the number of events with additional or missing b-tagged jets is stacked. The percentage shows the fraction of events with only jets from top-decay being b-tagged.

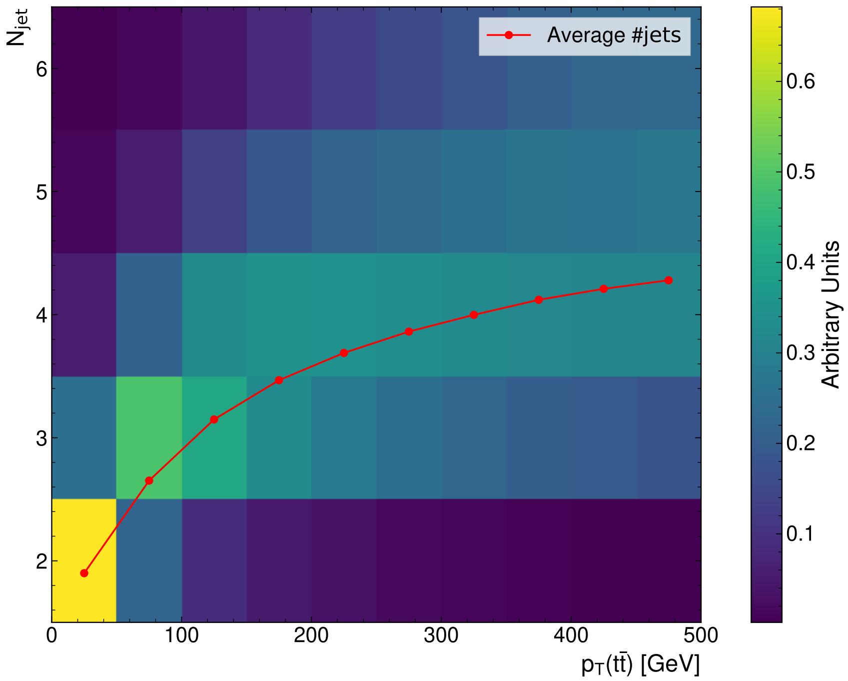


Figure A.4.: This plot shows the distribution of events binned in the parton-level $p_T(t\bar{t})$ and number of jets per event.

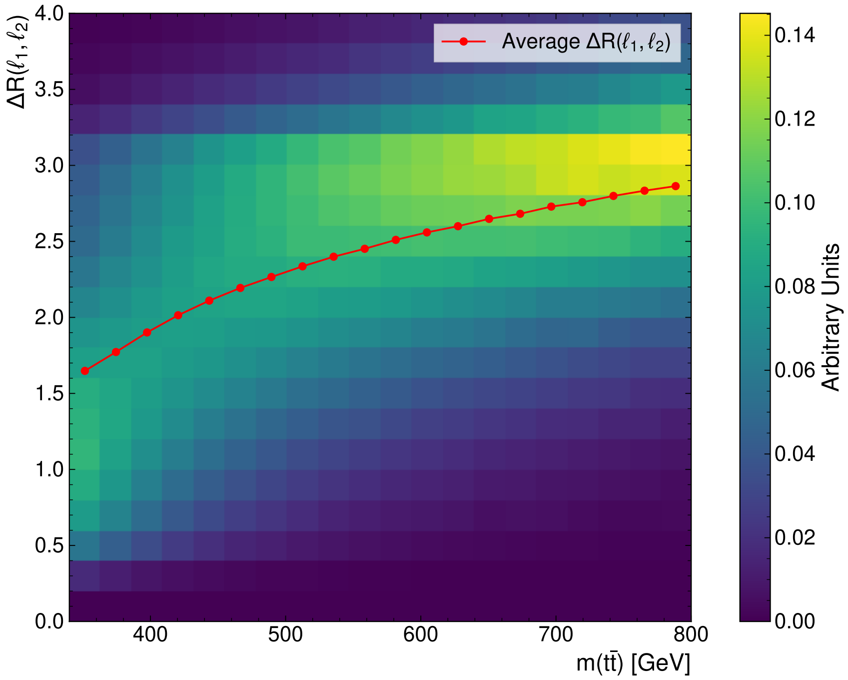


Figure A.5.: This plot shows the distribution of events binned in the parton-level $m(t\bar{t})$ and the angular distance $\Delta R(\ell_1, \ell_2)$.

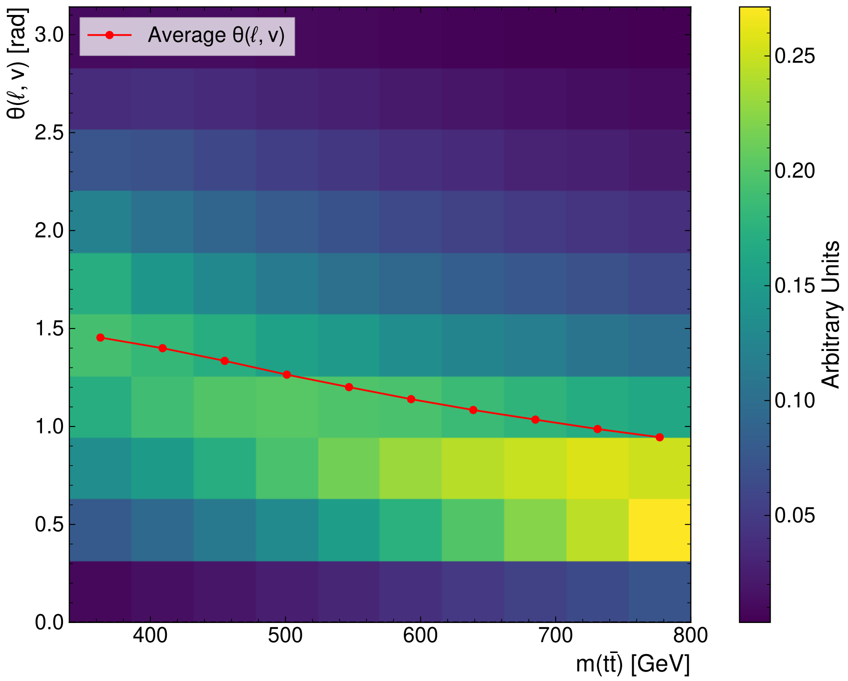
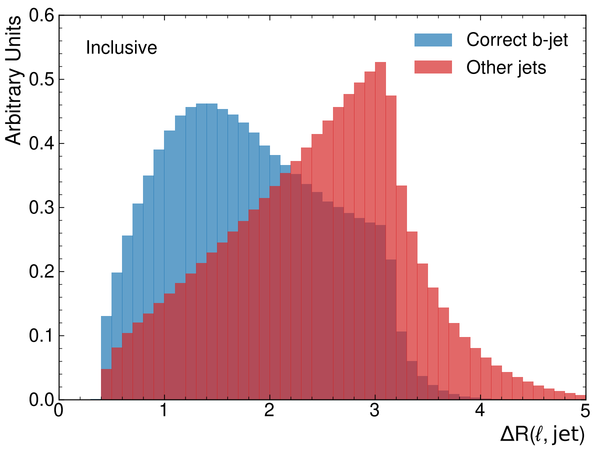
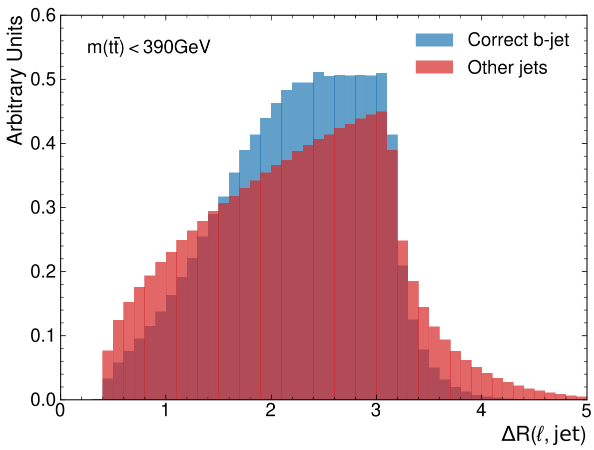


Figure A.6.: This plot shows the distribution of events binned in the parton-level $m(t\bar{t})$ and the angle $\theta(\nu, \ell)$ between the lepton and neutrino at parton level.

Threshold Topology

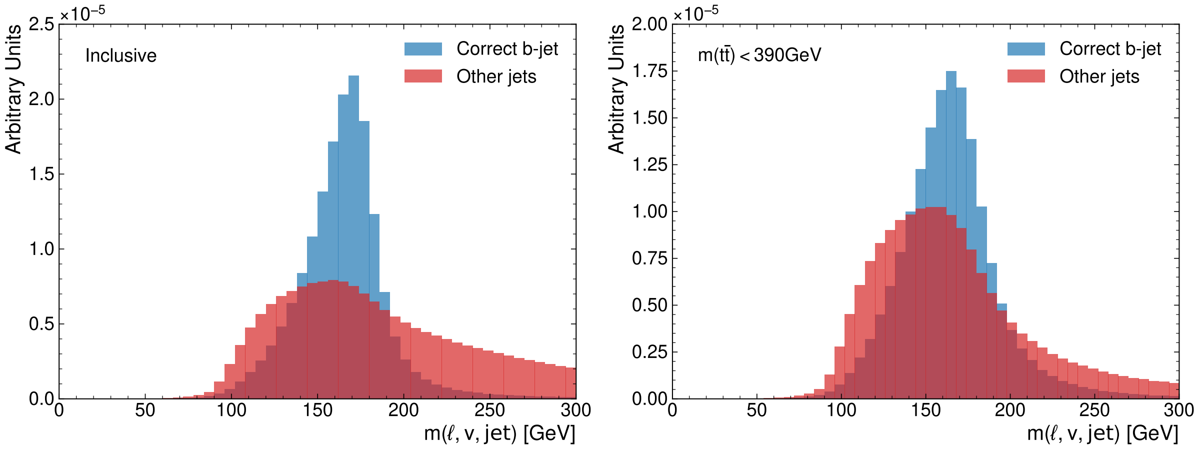


(a) Inclusive



(b) Near-Threshold

Figure A.7.: Distribution of $\Delta R(\ell, j)$ for correct and incorrect pairing of jets and leptons. Evaluated on the full sample in (a) and near-threshold in (b).



(a) Inclusive **(b)** Near-Threshold

Figure A.8.: Distribution of $m(\ell, \nu, j)$ for correct and incorrect pairing of jets and leptons. Evaluated on the full sample in (a) and near-threshold in (b). Note that the regressed neutrino momenta from nu^2-Flows were used.

A.2. Hyperparameter Optimisation Results

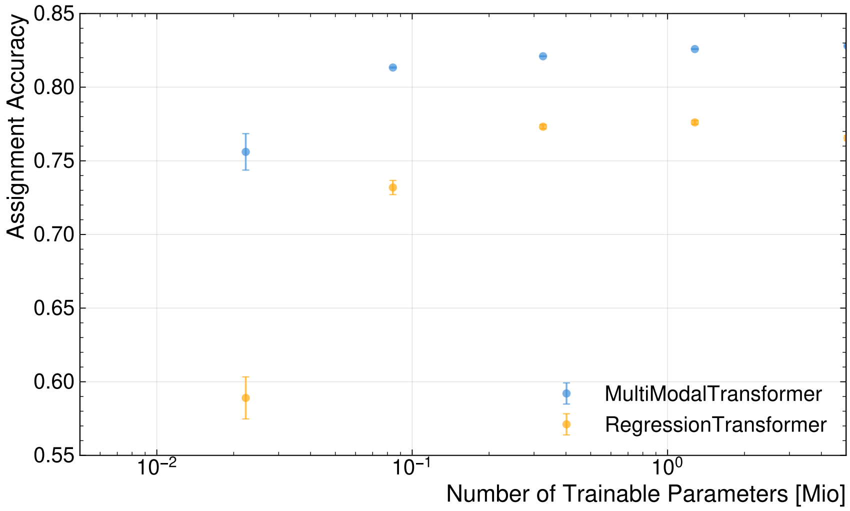
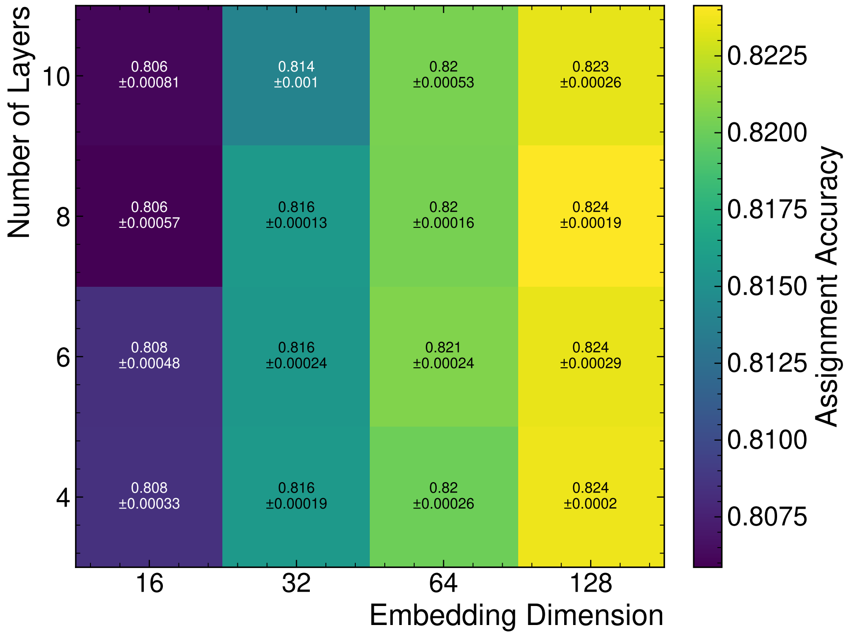
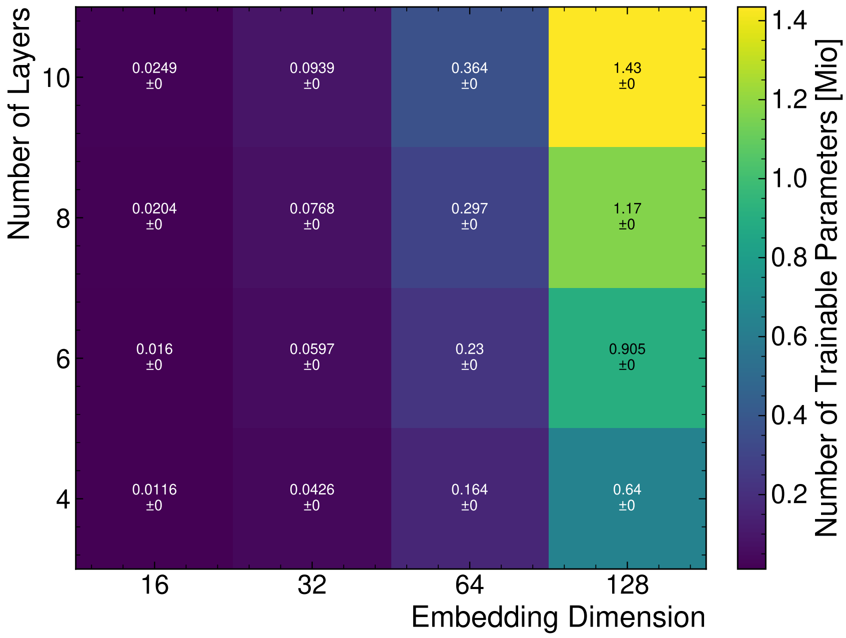


Figure A.9.: Assignment accuracy compared to the true values for the MultiModal-Transformer and the RegressionTransformer as a function of the number of trainable parameters.

A.2.1. FeatureConcatTransformer



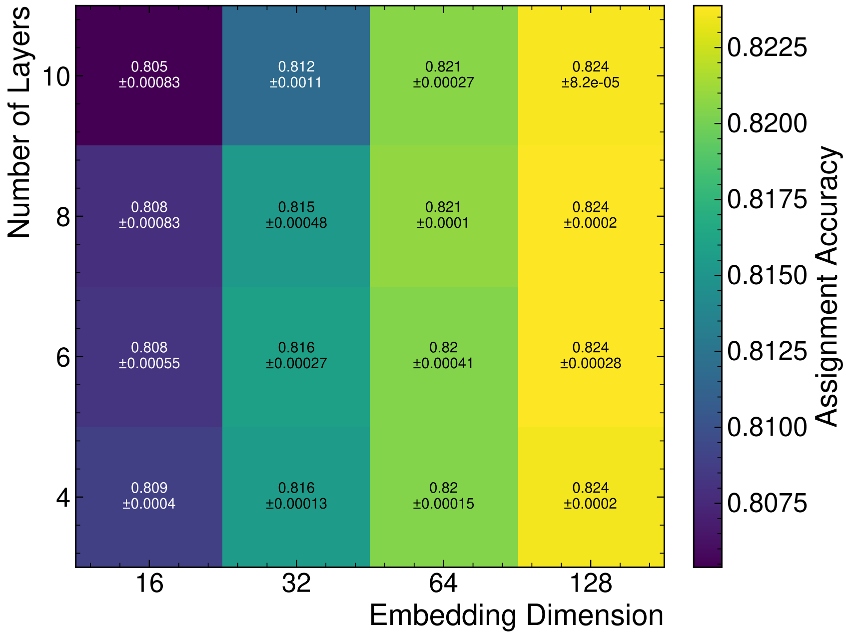
(a) Assignment Accuracy



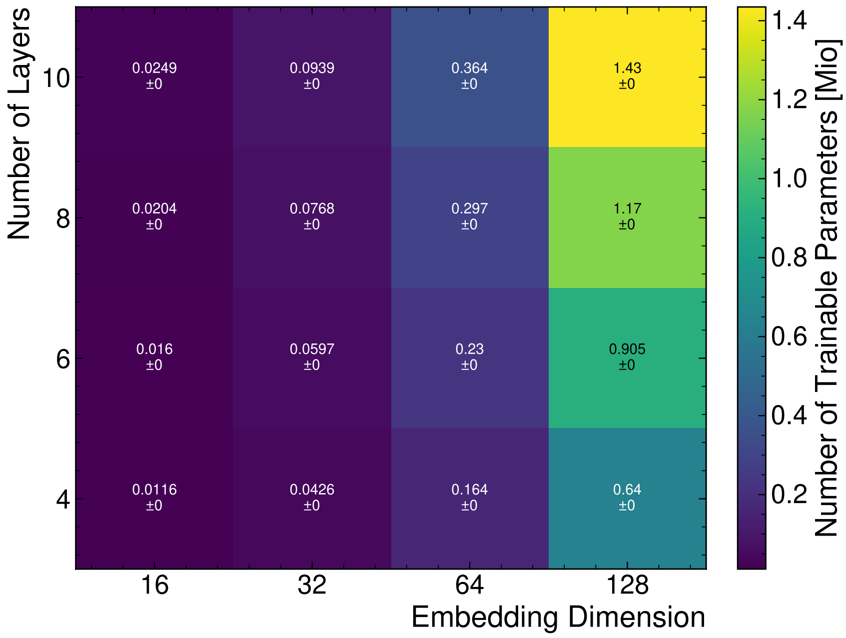
(b) Number of Trainable Parameters

Figure A.10.: The Figures (a) and (b) show the assignment accuracy and the number of trainable parameters of the FeatureConcatTransformer architecture as a function of the embedding dimension and the number of transformer layers.

A.2.2. CrossAttentionTransformer



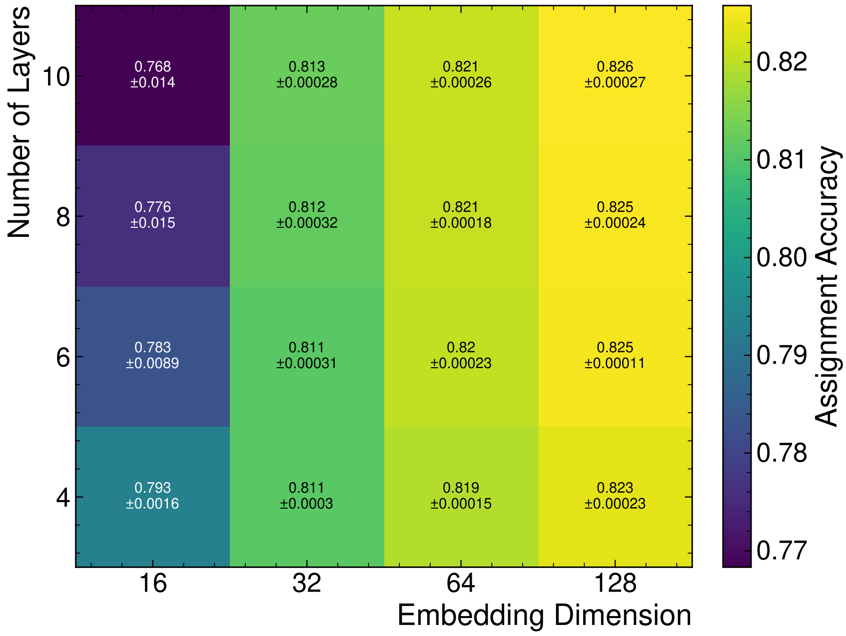
(a) Assignment Accuracy



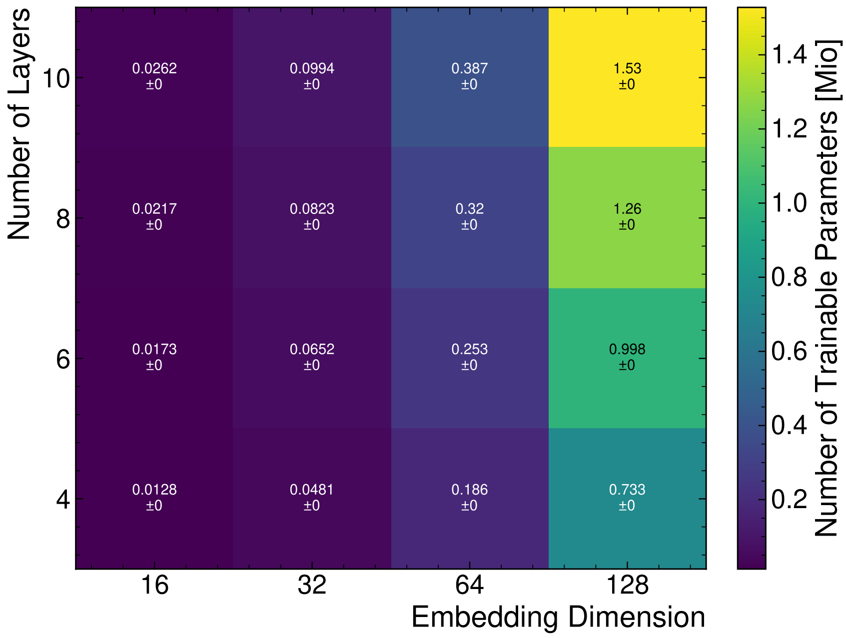
(b) Number of Trainable Parameters

Figure A.11.: The Figures (a) and (b) show the assignment accuracy and the number of trainable parameters of the CrossAttentionTransformer architecture as a function of the embedding dimension and the number of transformer layers.

A.2.3. CompactTransformer



(a) Assignment Accuracy



(b) Number of Trainable Parameters

Figure A.12.: The Figures (a) and (b) show the assignment accuracy and the number of trainable parameters of the CompactTransformer architecture as a function of the embedding dimension and the number of transformer layers.

A.3. Performance Plots

A.3.1. Assignment Performance

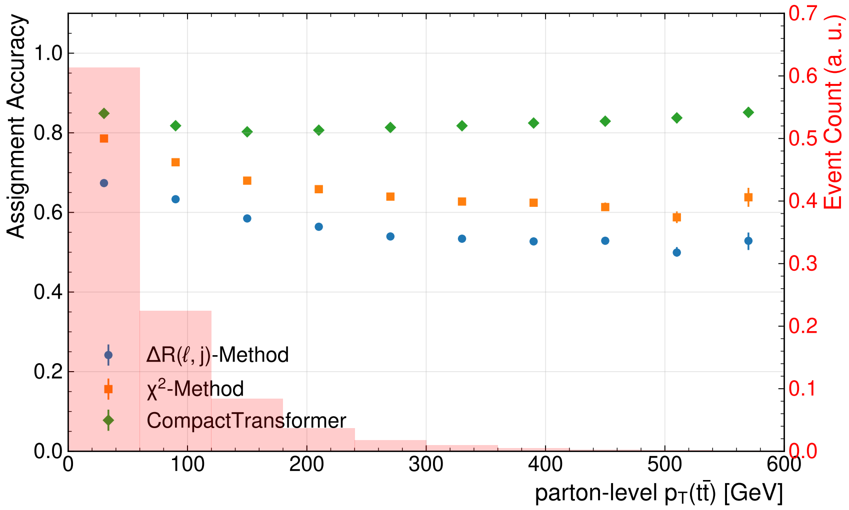


Figure A.13.: Assignment accuracy binned in the transverse momentum of top-quark pair system. The background histograms show of the distribution of events as a function of the number of jets. The uncertainties represent the statistical variations in the evaluation dataset.

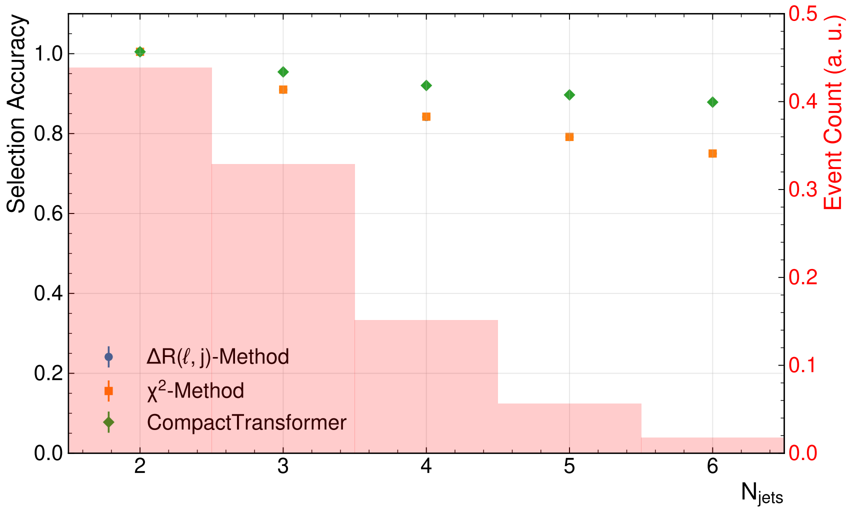


Figure A.14.: Selection accuracy as a function of the number of jets in the event for different reconstruction methods.

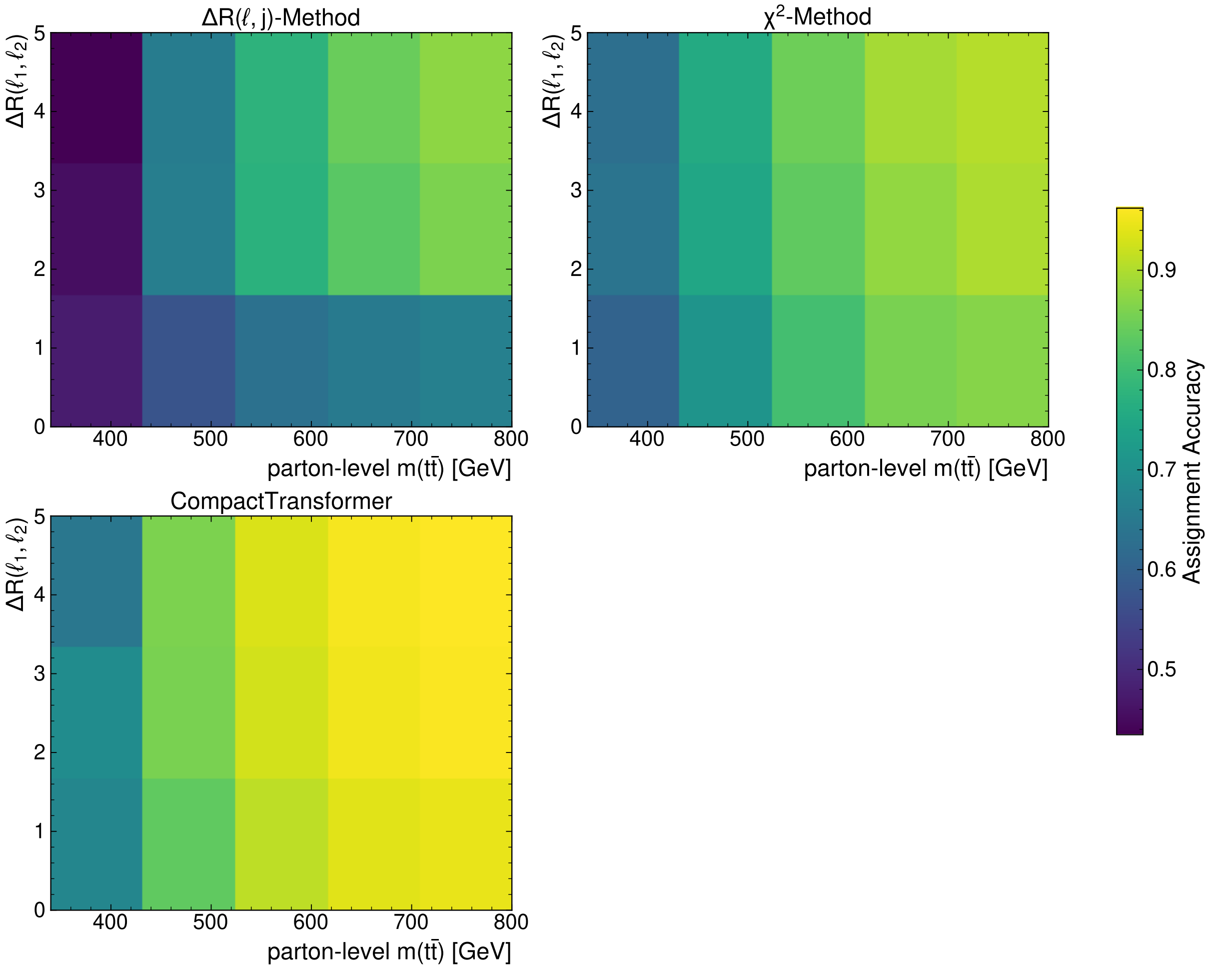


Figure A.15.: Assignment accuracy binned with respect to parton-level $m(t\bar{t})$ and $\Delta R(\ell_1, \ell_2)$.

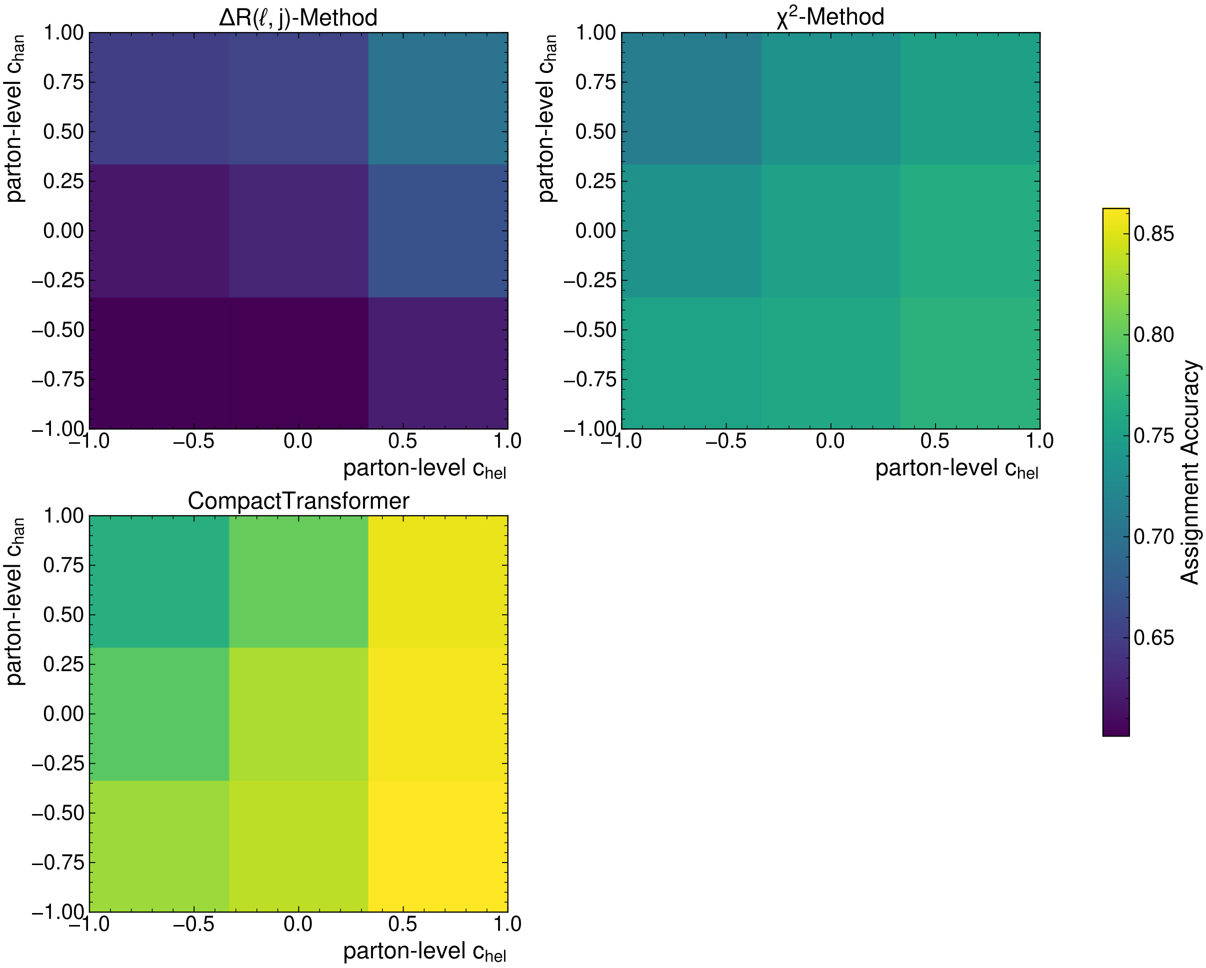


Figure A.16.: Assignment accuracy binned with respect to parton-level c_{hel} and c_{han}.

A.3.2. Neutrino Regression

Table A.1.: Summary of the mean absolute error (MAE) of the different neutrino regression methods for the cartesian coordinates of the neutrinos in the lab frame. For the MultiModalTransformer, the bracket shows whether the mean of the Gaussian or a random sampling was used. The uncertainties represent the statistical variations in the dataset.

Method	MAE (p_x) [GeV]	MAE (p_y) [GeV]	MAE (p_z) [GeV]
ν^2-Flows	$33.6^{+0.0}_{-0.0}$	$33.6^{+0.0}_{-0.0}$	88^{+0}_{-0}
ν-Weighter	$51.1^{+0.0}_{-0.1}$	$51.1^{+0.0}_{-0.0}$	124^{+0}_{-0}
MMT (Mean)	$23.4^{+0.0}_{-0.0}$	$23.4^{+0.0}_{-0.0}$	$63.1^{+0.1}_{-0.0}$
MMT (Gaussian Sampling)	$33.3^{+0.0}_{-0.0}$	$33.4^{+0.0}_{-0.0}$	90^{+0}_{-0}

Table A.2.: Summary of the mean absolute error of the neutrinos' p_T, η, ϕ, and E, evaluated for the different methods. For the MultiModalTransformer, the bracket shows whether the mean of the Gaussian or a random sampling was used. Note that the computation of the error in ϕ was performed periodically.

Method	MAE (p_T) [GeV]	MAE (η)	MAE (ϕ) [GeV]	MAE (E) [GeV]
ν^2-Flows	$30.0^{+0.0}_{-0.0}$	$1.003^{+0.001}_{-0.001}$	$0.964^{+0.001}_{-0.001}$	75^{+0}_{-0}
ν-Weighter	$51.3^{+0.1}_{-0.1}$	$1.068^{+0.000}_{-0.001}$	$1.081^{+0.001}_{-0.001}$	114^{+0}_{-0}
MMT (Mean)	$23.5^{+0.0}_{-0.0}$	$0.910^{+0.001}_{-0.001}$	$0.764^{+0.001}_{-0.001}$	$54.3^{+0.0}_{-0.0}$
MMT (Gaussian Sampling)	$28.5^{+0.0}_{-0.0}$	$1.095^{+0.000}_{-0.001}$	$0.993^{+0.001}_{-0.001}$	$69.4^{+0.1}_{-0.1}$

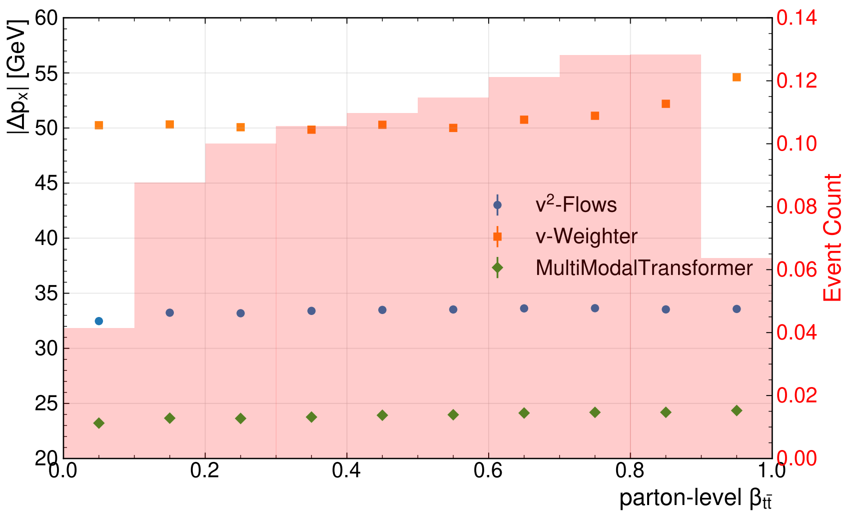


Figure A.17.: Mean absolute error of the neutrinos' p_x component binned with respect to the boost parameters.

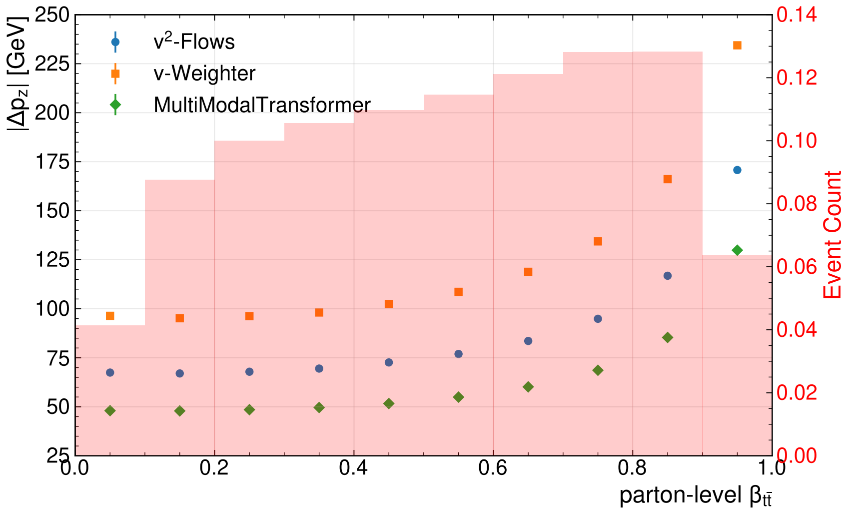


Figure A.18.: Mean absolute error of the neutrinos' p_z component binned with respect to the boost parameters.

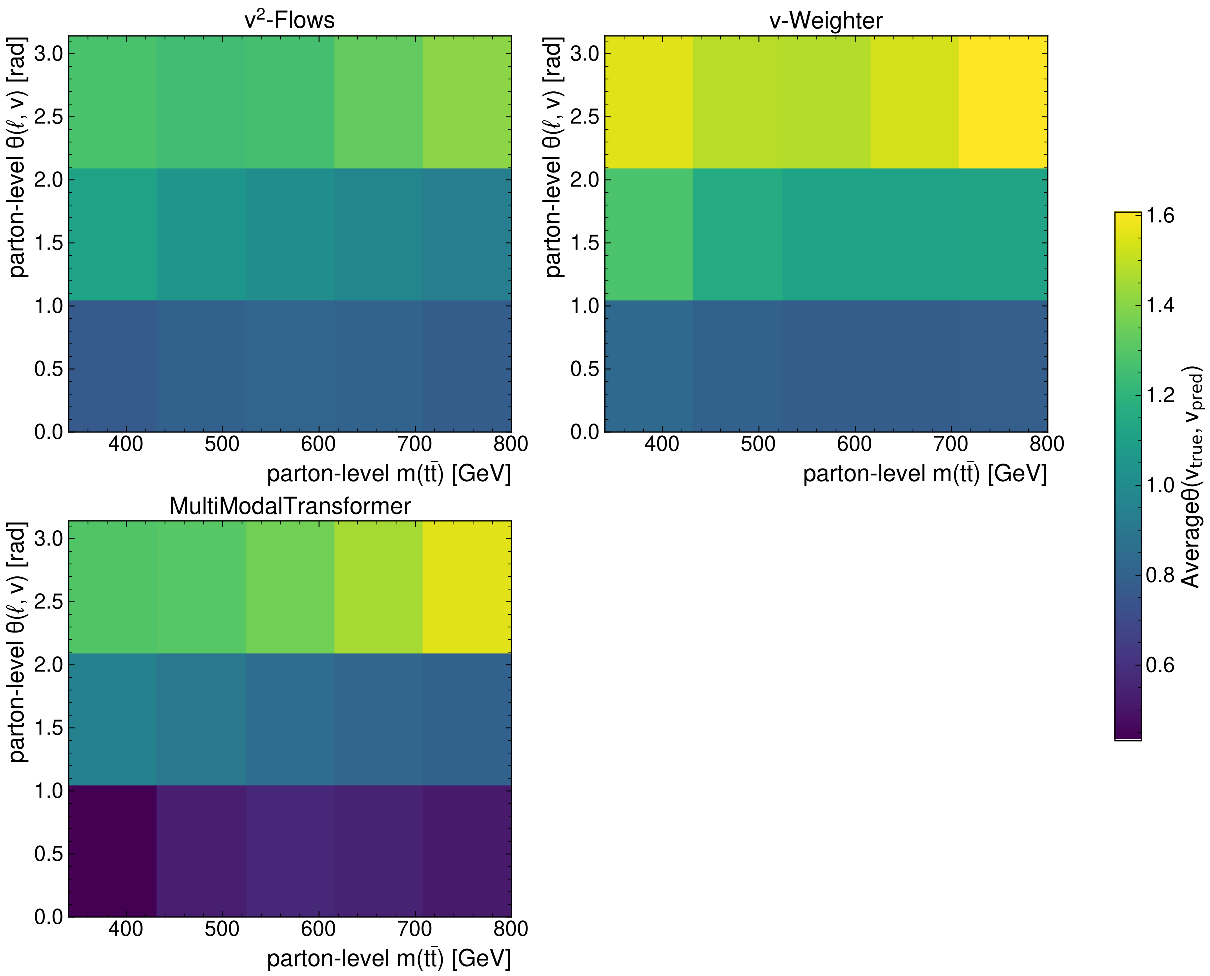


Figure A.19.: Mean angle between true and predicted neutrino binned in the parton-level $m(t\bar{t})$ and parton-level angle between lepton and neutrino.

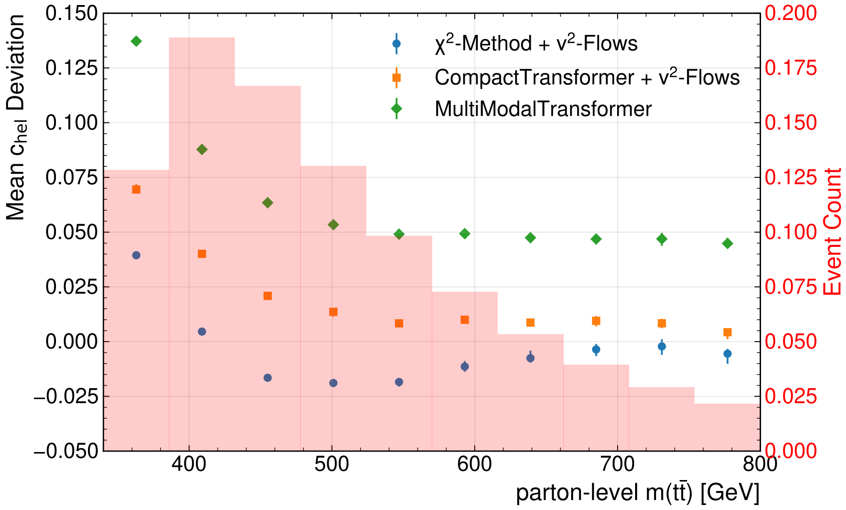
A.3.3. Variable Reconstruction

Method	Mean Rel. Deviation $m(t\bar{t})$	Rel. Resolution $m(t\bar{t})$
$\Delta R(\ell, j)$-Method + ν^2-Flows	$0.0025^{+0.0002}_{-0.0003}$	$0.2887^{+0.0016}_{-0.0010}$
χ^2-Method + ν^2-Flows	$0.0023^{+0.0005}_{-0.0005}$	$0.2888^{+0.0006}_{-0.0011}$
CompactTransformer + ν^2-Flows	$-0.0203^{+0.0004}_{-0.0003}$	$0.2421^{+0.0012}_{-0.0011}$
True Assignment + ν^2-Flows	$-0.0215^{+0.0004}_{-0.0003}$	$0.2400^{+0.0009}_{-0.0009}$

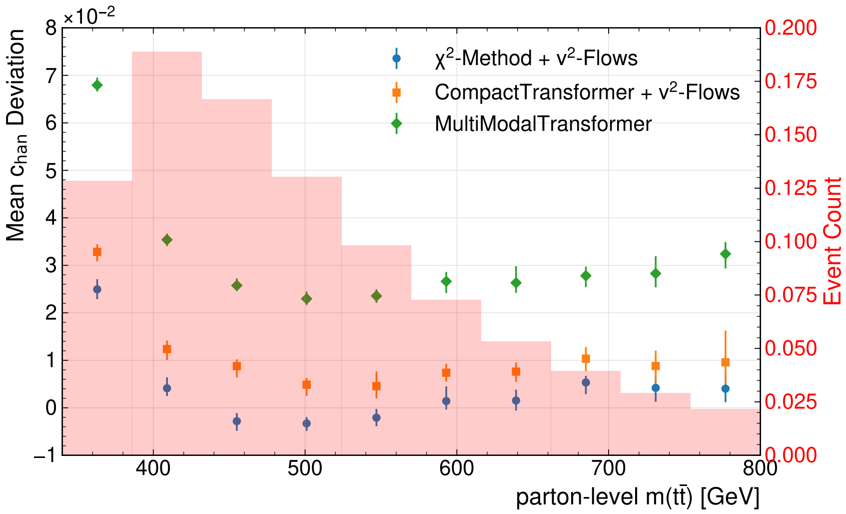
Table A.3.: Summary table showing the mean relative deviation and relative resolution for the invariant mass of the $t\bar{t}$-system. Compared are the baseline-methods for assignment, the CompactTransformer, and the true assignment, all in conjunction with ν^2-Flows.

Method	Mean Rel. Deviation $m(t\bar{t})$	Rel. Resolution $m(t\bar{t})$
CompactTransformer + ν^2-Flows	$-0.0203^{+0.0003}_{-0.0004}$	$0.2419^{+0.0010}_{-0.0012}$
MMT (Cartesian)	$-0.13347^{+0.00009}_{-0.00016}$	$0.1428^{+0.0002}_{-0.0003}$
MMT (PtEtaPhi)	$-0.12957^{+0.00008}_{-0.00011}$	$0.1279^{+0.0002}_{-0.0003}$
MMT (MagDir)	$-0.07700^{+0.00011}_{-0.00019}$	$0.13210^{+0.00016}_{-0.00023}$

Table A.4.: Summary table showing the mean relative deviation and relative resolution for the invariant mass of the $t\bar{t}$-system. Compared are MMT models trained with different losses as well as the CPT interfaced with ν^2-Flows



(a) Mean Deviation of c_{hel}



(b) Mean Deviation of c_{han}

Figure A.20.: Mean Deviation of c_{hel} (a) and c_{han} (b) computed in bins of the parton-level invariant mass of the $t\bar{t}$-system. The different setups for assignment and regression are compared. The displayed error bars represent the statistical uncertainties of the dataset obtained using bootstrapping.

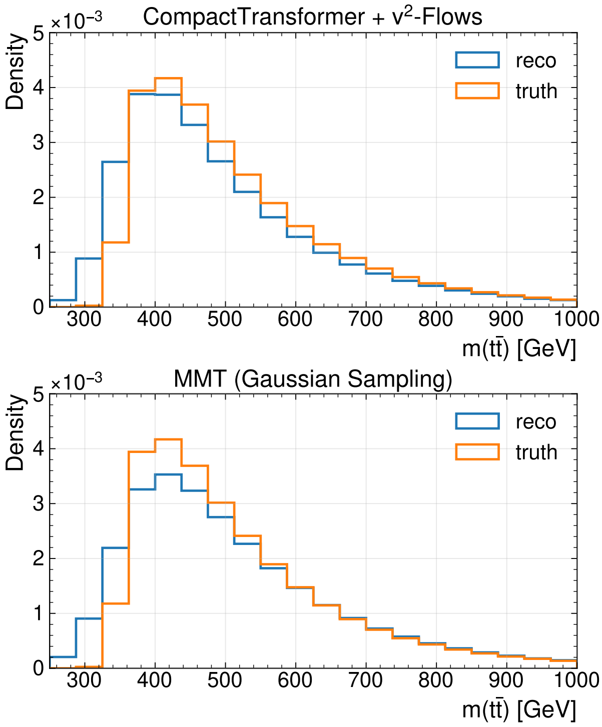


Figure A.21.: Comparison of $m(t\bar{t})$ distributions at reco- and truth-level for the different regression settings of the MultiModalTransformer trained with the Gaussian-loss.

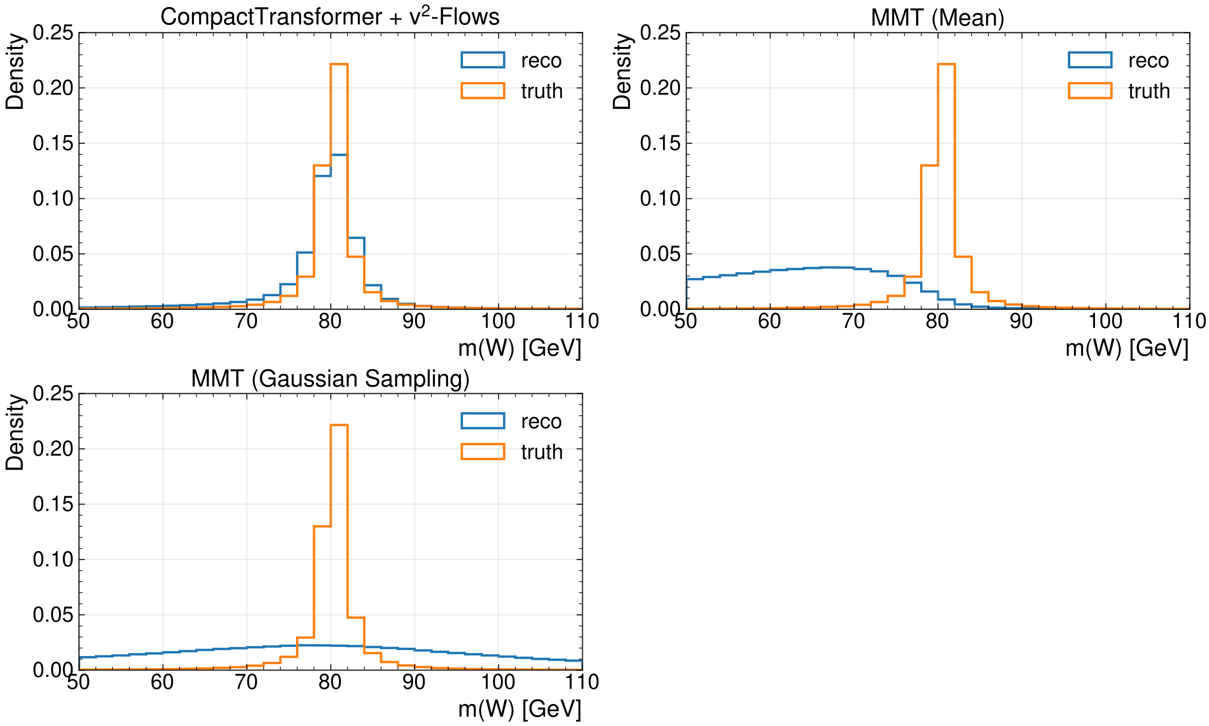


Figure A.22.: Comparison of $m(W)$ distributions at reco- and truth-level for the different regression settings of the MultiModalTransformer trained with the Gaussian-loss.

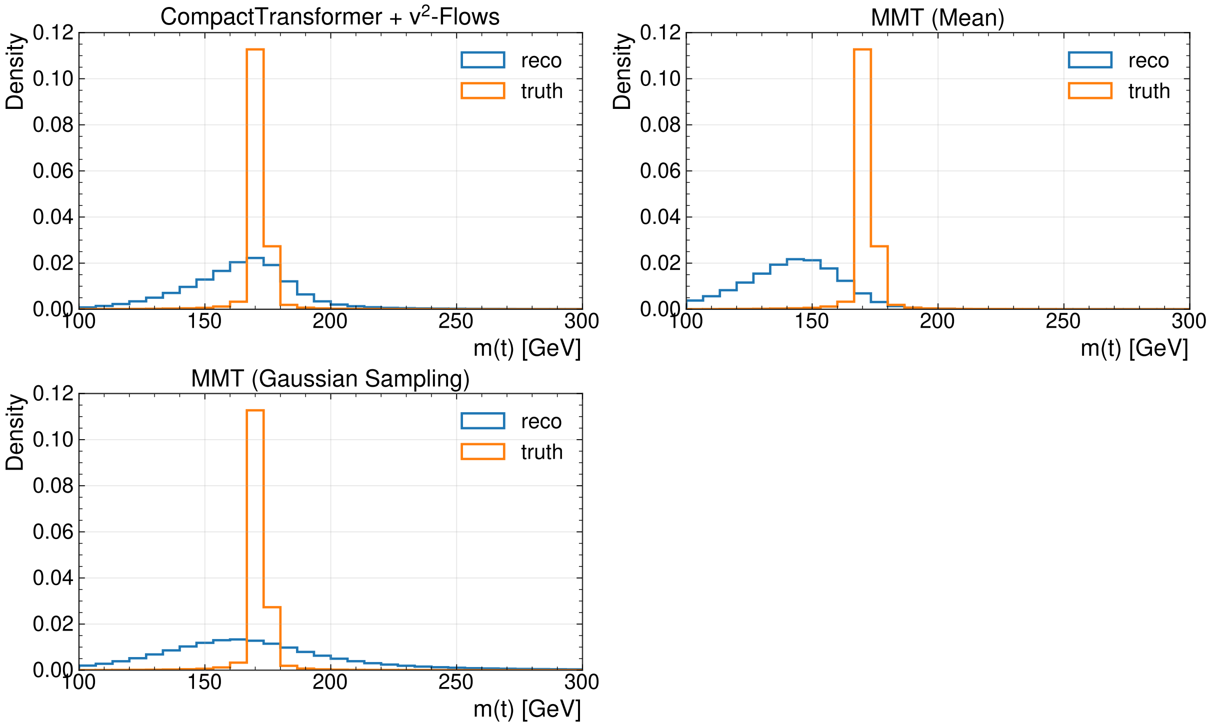
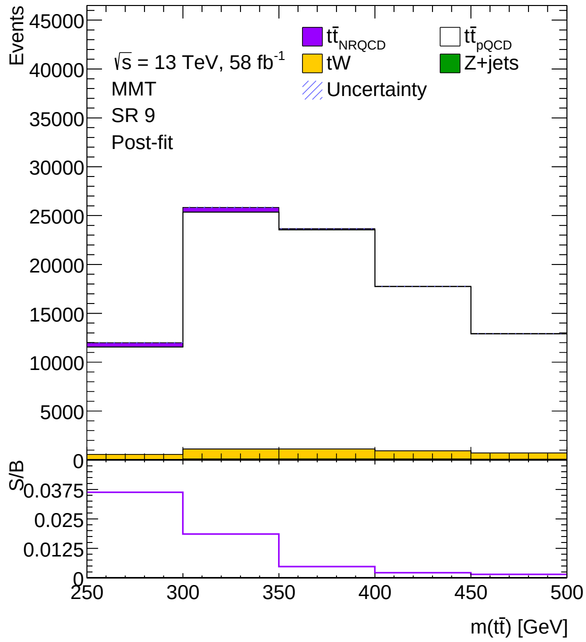


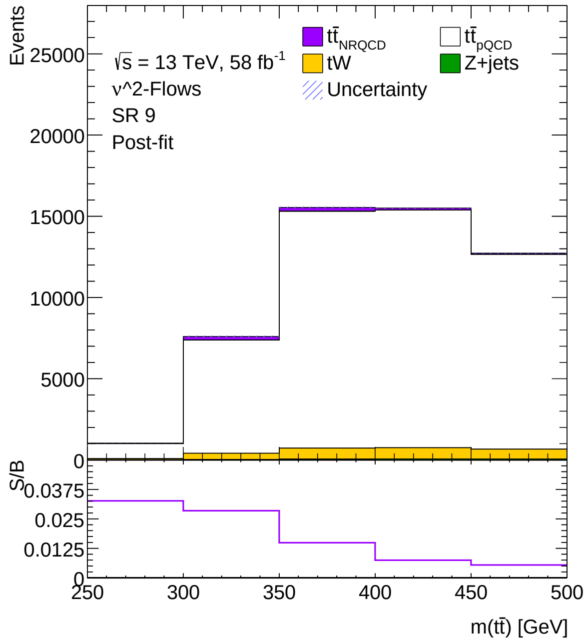
Figure A.23.: Comparison of $m(t)$ distributions at reco- and truth-level for the different regression settings of the MultiModalTransformer trained with the Gaussian-loss.

A.4. Fit Results

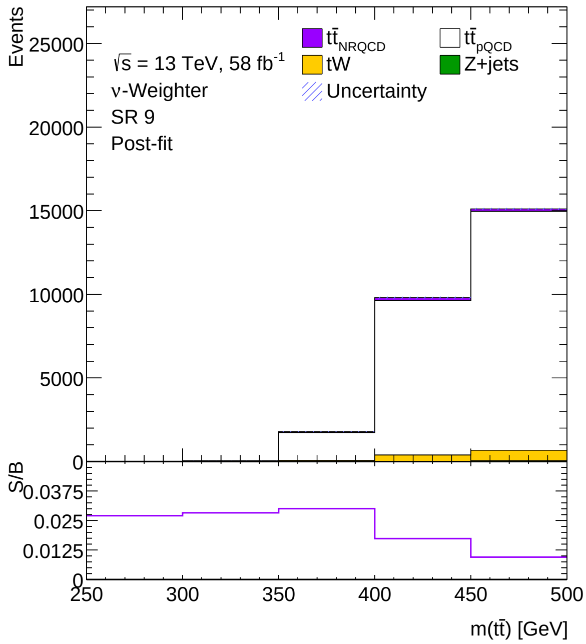
A.4.1. Asimov-Fit



(a) MultiModalTransformer



(b) ν^2-Flows



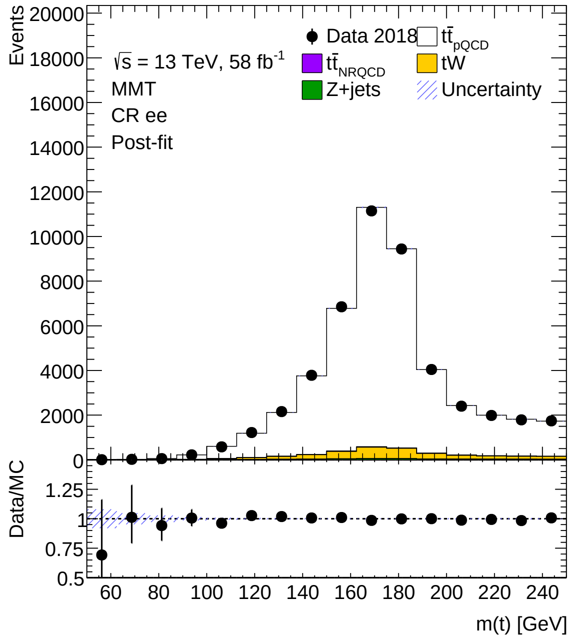
(c) ν-Weighter

Figure A.24.: Plots of SR 9 for the different reconstruction methods used. The upper panels show the stacked event yield per bin from different components. The lower panels show the ratio of signal over background in the corresponding bins, where signal refers to the contribution from non-relativistic QCD and the background correspondingly all other contributions.

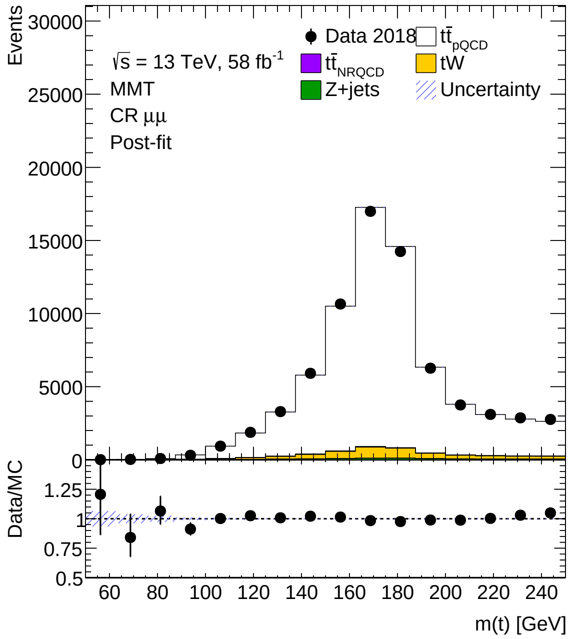
A.4.2. Data-Driven Validation

Table A.5.: Fit results for the baseline-methods obtained by fitting the simulated processes for dileptonic decay $t\bar{t}$-production with data collected by ATLAS in 2018.

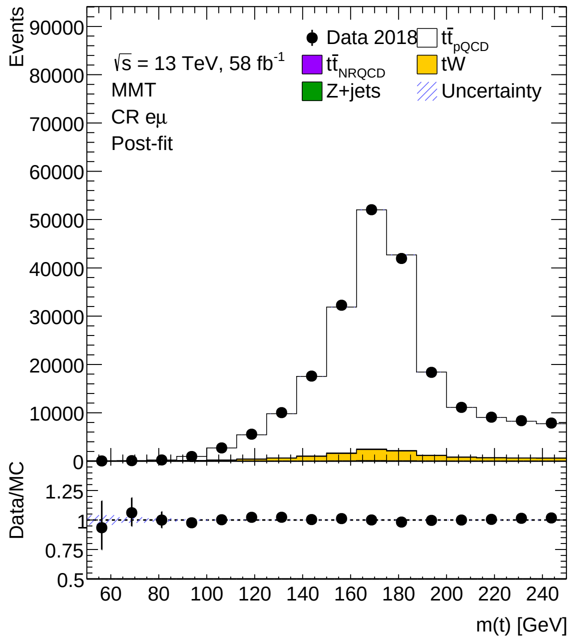
Method	$\mu(t\bar{t}_{\mathrm{pQCD}})$	$\mu(Z\text{+jets})$
ν-Weighter	$0.963^{+0.0017}_{-0.0017}$	$1.01^{+0.02}_{-0.02}$
ν^2-Flows	$0.958^{+0.0018}_{-0.0018}$	$1.073^{+0.02}_{-0.02}$



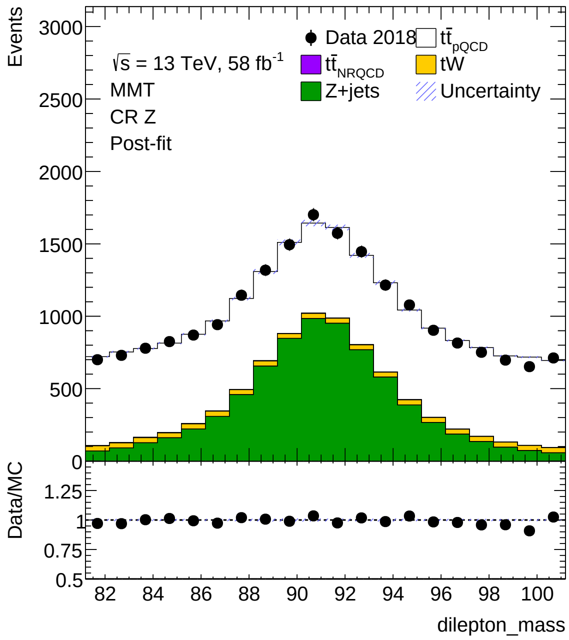
(a) CR ee



(b) CR $\mu\mu$

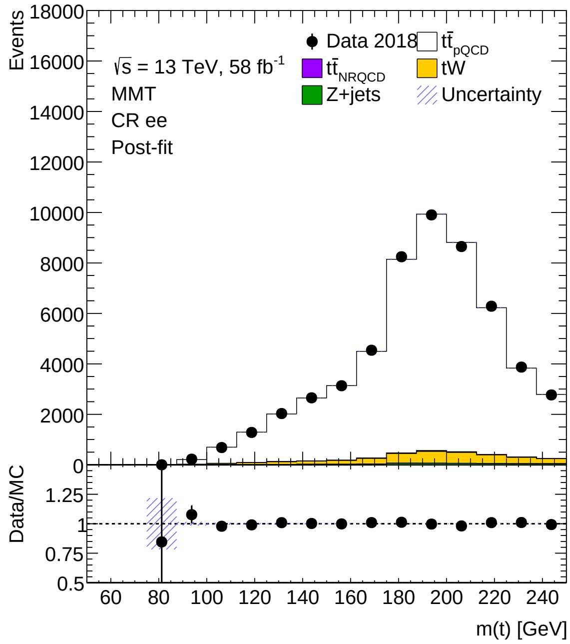


(c) CR $e\mu$

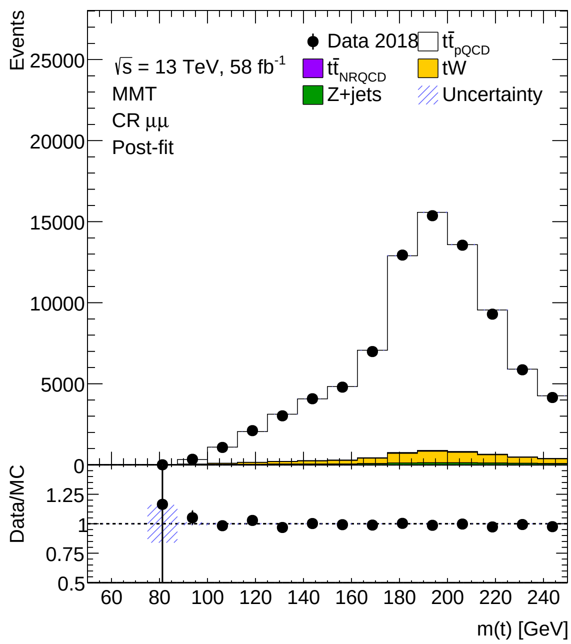


(d) CR Z

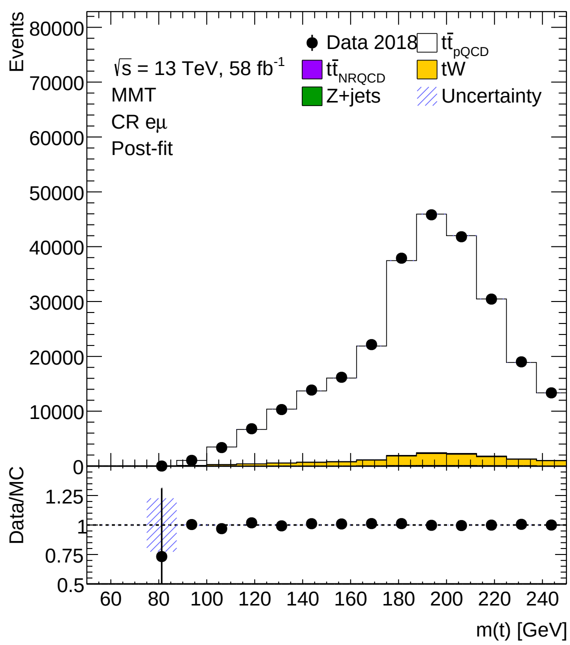
Figure A.25.: Plots showing the comparison between data and simulation of top-quark pair production and its leading backgrounds. The different control regions used in the fits are shown as stacked histogrmas of the different distributions, where the normalisations for $t\bar{t}_{\text{pQCD}}$ and Z+jets are extracted from the fit as summarised in equation (8.2). Note that the reconstruction of the $t\bar{t}$-system is performed using ν^2-Flows.



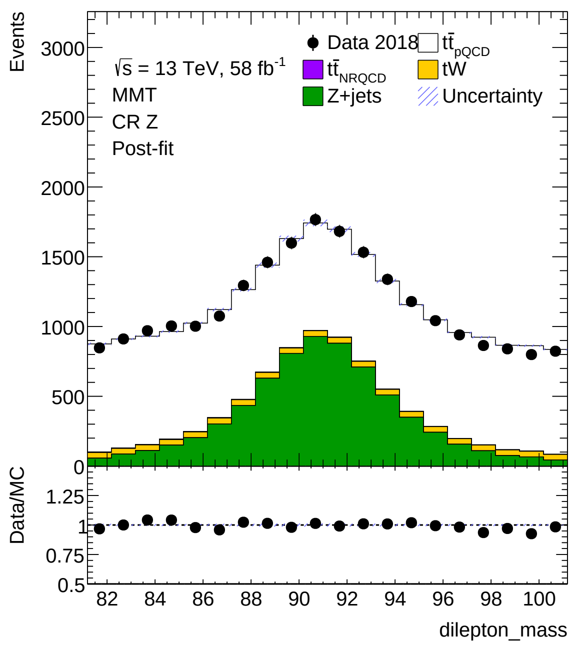
(a) CR ee



(b) CR $\mu\mu$

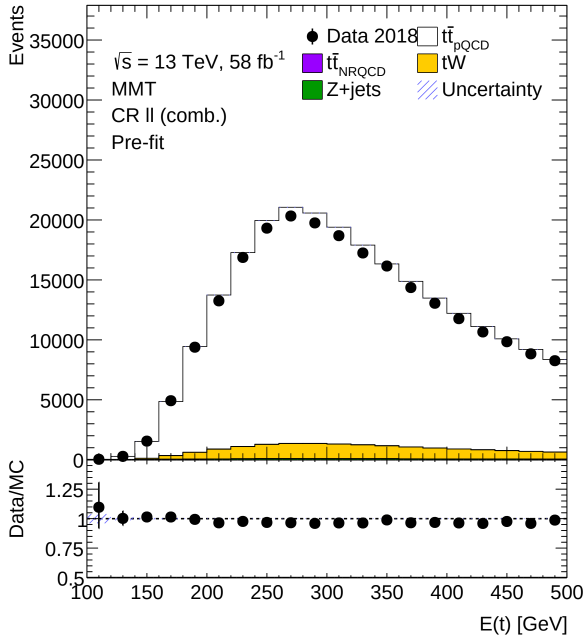


(c) CR $e\mu$

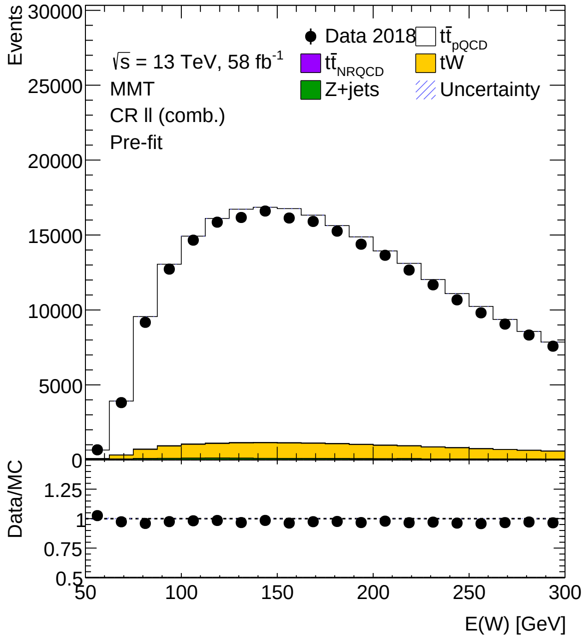


(d) CR Z

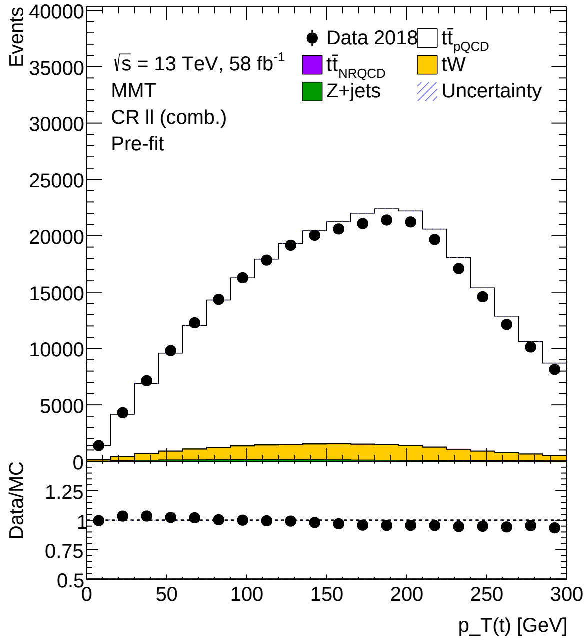
Figure A.26.: Plots showing the comparison between data and simulation of top-quark pair production and its leading backgrounds. The different control regions used in the fits are shown as stacked histogrmas of the different distributions, where the normalisations for $t\bar{t}_{\mathrm{pQCD}}$ and Z+jets are extracted from the fit as summarised in equation (8.2). Note that the reconstruction of the $t\bar{t}$-system is performed using ν-Weighter.



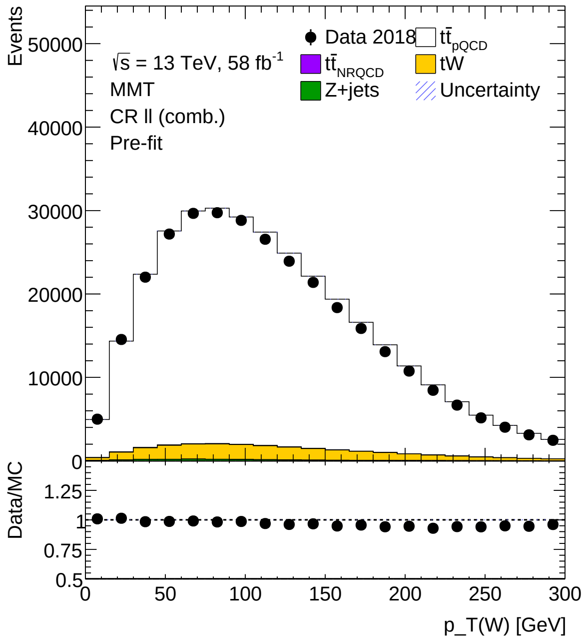
(a) $E(t)$



(b) $E(W)$

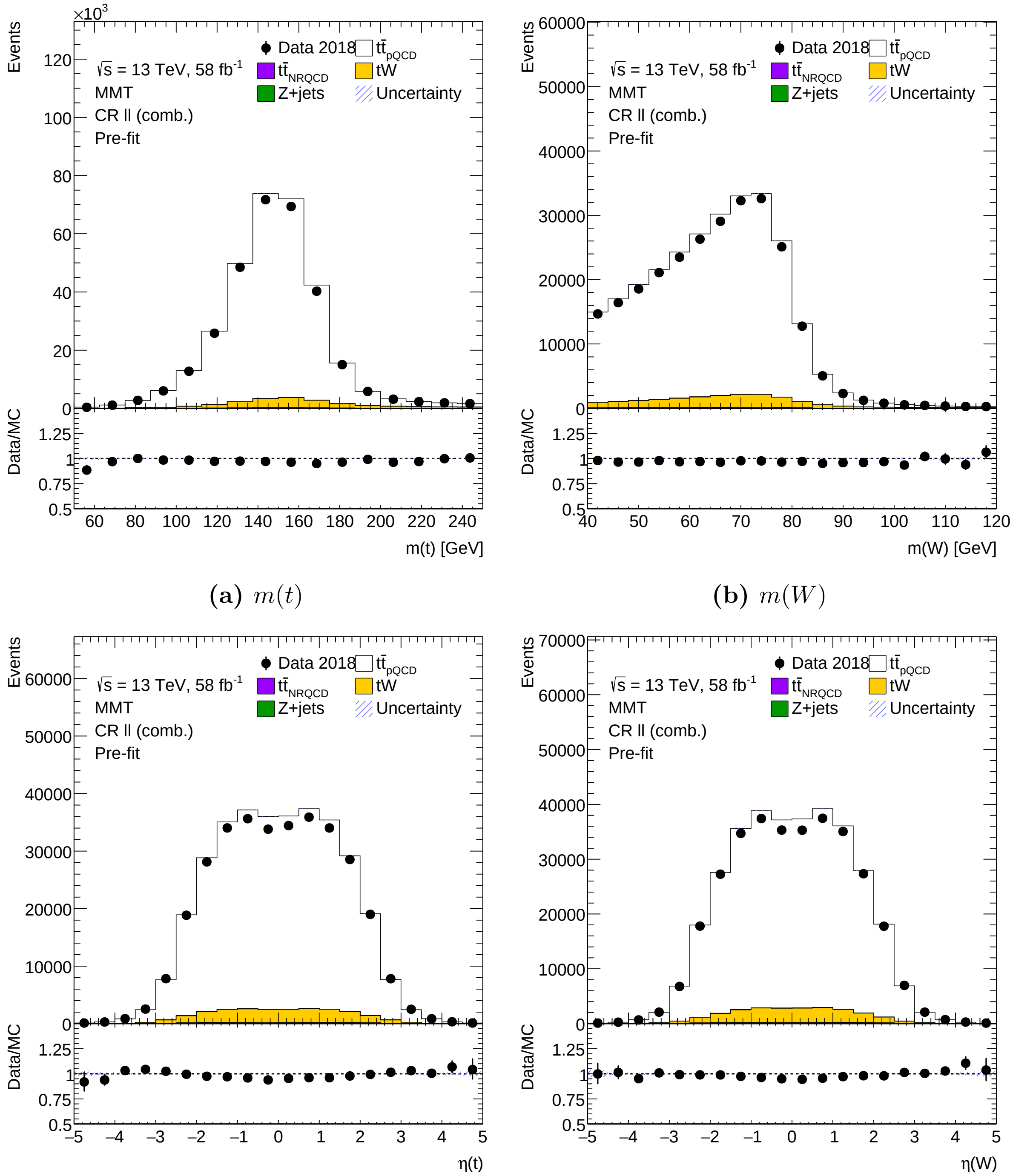


(c) $p_T(t)$



(d) $p_T(W)$

Figure A.27.: Plots showing the comparison between data and simulation for different variables reconstructed with the MultiModalTransformer. The simulations used the data-driven adjustments of the normalisations. Note, that the region here is a combination of all CR-$\ell\ell$ combining all leptons flavours.



(a) $m(t)$ **(b)** $m(W)$

(c) $\eta(t)$ **(d)** $\eta(W)$

Figure A.28.: Plots showing the comparison between data and simulation for different variables reconstructed with the MultiModalTransformer. The simulations used the data-driven adjustments of the normalisations. Note, that the region here is a combination of all CR-$\ell\ell$ combining all leptons flavours.

B. Supplementary Material

B.1. Mathematical Foundation of Machine Learning

B.1.1. Mean Squared Error Loss and Conditional Expectation

When performing a minimisation of the mean squared error (MSE) loss function in a supervised learning setting, one can show that the optimal predictor corresponds to the conditional expectation of the target variable given the input features. The MSE loss function can be written as:

$$L_{\text{MSE}}(\mathbf{y}, f_\theta(\mathbf{x})) = \frac{1}{N}\sum_{i=1}^{N}(y_i - f_\theta(\mathbf{x}_i))^2 \tag{B.1}$$

Using a simple calculus argument, one can show that the function f_θ that minimises this loss function is given by:

$$\begin{aligned} \frac{d}{d\theta}L_{\text{MSE}}(\mathbf{y}, f_\theta(\mathbf{x})|\mathbf{x}) &= \frac{d}{df}\int (y - f_\theta(x))^2 p(y|x)dy \cdot \frac{\partial f}{\partial \theta} && \text{(B.2)} \\ &= \int 2(f_\theta(x) - y)p(y|x)dy \cdot \frac{\partial f}{\partial \theta} \overset{!}{=} 0 && \text{(B.3)} \\ \Rightarrow f_\theta(x) &= \int y p(y|x)dy = \mathbb{E}[y|x] && \text{(B.4)} \end{aligned}$$

B.1.2. Equivalence of Maximum Likelihood and KL-Divergence Minimisation

Let $p_{\text{data}}(x)$ denote the true data distribution and $p_\theta(x)$ a parametric model. The KL-divergence between them is defined as

$$D_{\text{KL}}(p_{\text{data}}\|p_\theta) = \mathbb{E}_{x\sim p_{\text{data}}}\left[\log\frac{p_{\text{data}}(x)}{p_\theta(x)}\right] = \mathbb{E}_{x\sim p_{\text{data}}}\left[\log p_{\text{data}}(x)\right] - \mathbb{E}_{x\sim p_{\text{data}}}\left[\log p_\theta(x)\right]. \tag{B.5}$$

Since the first term, $-H(p_{\text{data}})$, is the negative entropy of the data distribution and does not depend on θ, minimising the KL-divergence over θ is equivalent to

$$\hat{\theta} = \arg\min_\theta D_{\text{KL}}(p_{\text{data}}\|p_\theta) = \arg\max_\theta \mathbb{E}_{x\sim p_{\text{data}}}\left[\log p_\theta(x)\right]. \tag{B.6}$$

Approximating the expectation over p_{data} by the empirical mean over a dataset $\{x_i\}_{i=1}^N$ of N independent observations gives

$$\hat{\theta} = \arg\max_\theta \frac{1}{N}\sum_{i=1}^N \log p_\theta(x_i), = \arg\min_\theta \frac{1}{N}\sum_{i=1}^N -\log p_\theta(x_i) \tag{B.7}$$

which is precisely the maximum likelihood estimator. Note that this simply generalises to the case of a conditional probability.

B.1.3. Gaussian-Loss as Minimisation of the KL-divergence

Assume the conditional parametric model to have the form of a (multidimensional) Gaussian distribution

$$p_\theta(\vec{x}|y) = \prod_i \frac{1}{\sqrt{2\pi\sigma_{i,\theta}^2(y)}} \exp\left(-\frac{(x_i - \mu_{i,\theta}^2(y))^2}{2\sigma_{i,\theta}^2(y)}\right) . \tag{B.8}$$

For this, one can now consider the negative log-likelihood (NLL) of a dataset of $\{x_j\}_{j=1}^N$ of N independent observations:

$$\text{NLL}(p_\theta) = \frac{1}{N}\sum_{j=1}^N -\log(p_\theta(\vec{x}|y)) \tag{B.9}$$

$$= \frac{1}{N}\sum_{j=1}^N\sum_{i=1}^N \left(\frac{1}{2}\log(2\pi) + \frac{1}{2}\log(\sigma_{i,\theta}^2(y)) + \frac{(x_i - \mu_{i,\theta}^2(y))^2}{2\sigma_{i,\theta}^2(y)}\right) \tag{B.10}$$

The constant term $\log(2\pi)$ can be dropped for the minimisation, which results in

$$\hat{\theta} = \arg\min_\theta \text{NLL}(p_\theta) \tag{B.11}$$

$$= \arg\min_\theta \frac{1}{N}\sum_{j=1}^N\sum_{i=1}^N \left(\log(\sigma_{i,\theta}^2(y)) + \frac{(x_i - \mu_{i,\theta}^2(y))^2}{\sigma_{i,\theta}^2(y)}\right) \tag{B.12}$$

B.2. Neutrino Momenta Posterior Distributions

B.2.1. Invariant Mass of Top-Quark Pair

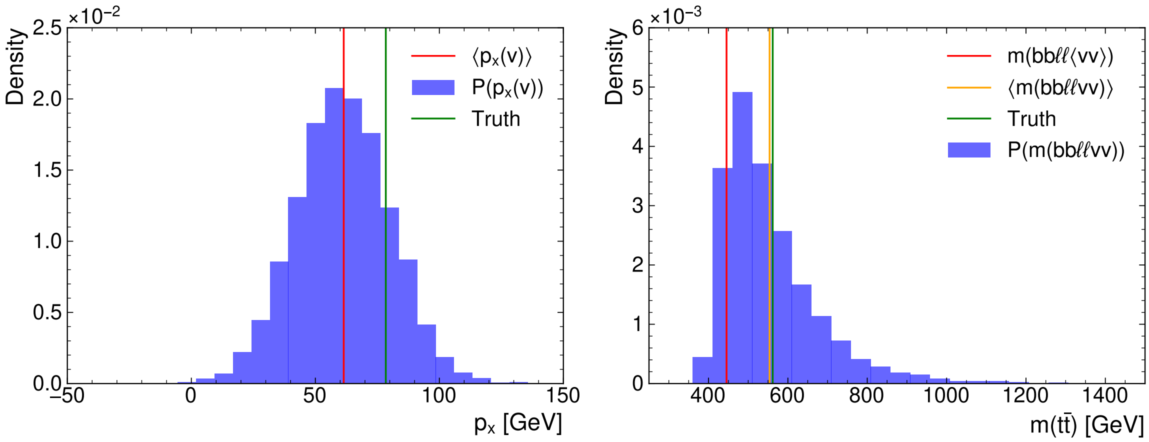


Figure B.1.: Plots showing the probability distribution of the neutrino momentum component p_x and the reconstructed invariant mass $m(t\bar{t})$ based on the mean and standard deviation predicted by the MMT-model trained with a Gaussian-loss for a random event.

B.2.2. Spin-Sensitive Variable c_{hel}

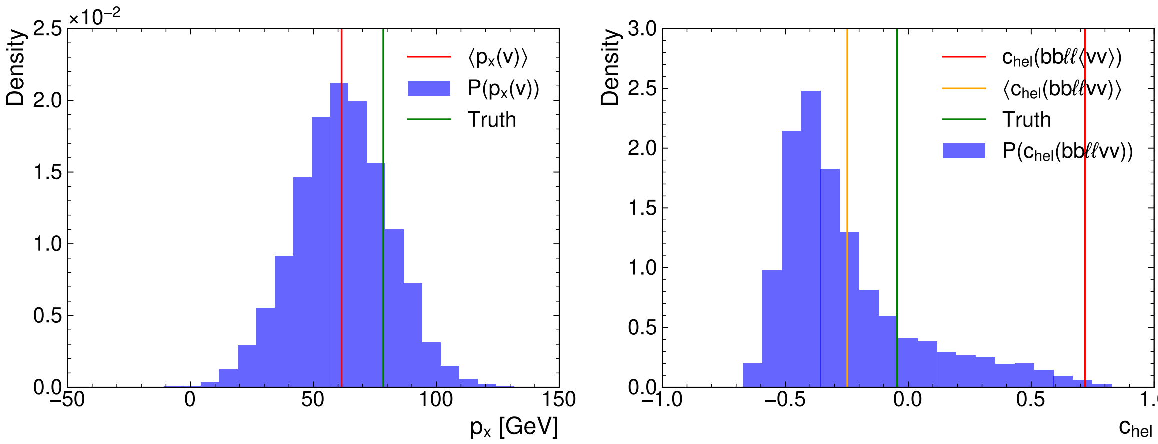


Figure B.2.: Plots showing the probability distribution of the neutrino momentum component p_x and the reconstructed c_{hel} based on the mean and standard deviation predicted by the MMT-model trained with a Gaussian-loss for a random event. Indicated are the mean and true values.

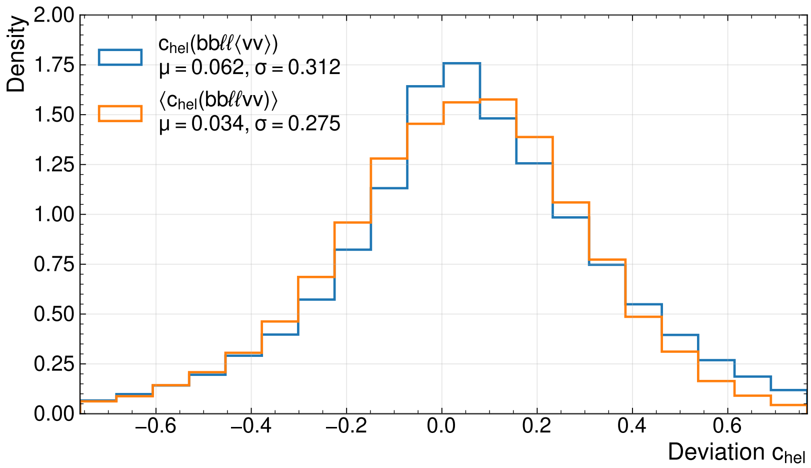


Figure B.3.: Deviation between the reconstructed and true c_{hel}. The different curves show, whether the invariant mass was computed using the mean momentum or by propagating the neutrino probabilities and averaging of the obtained spin-variable.

B.2.3. Spin-Sensitive Variable c_{han}

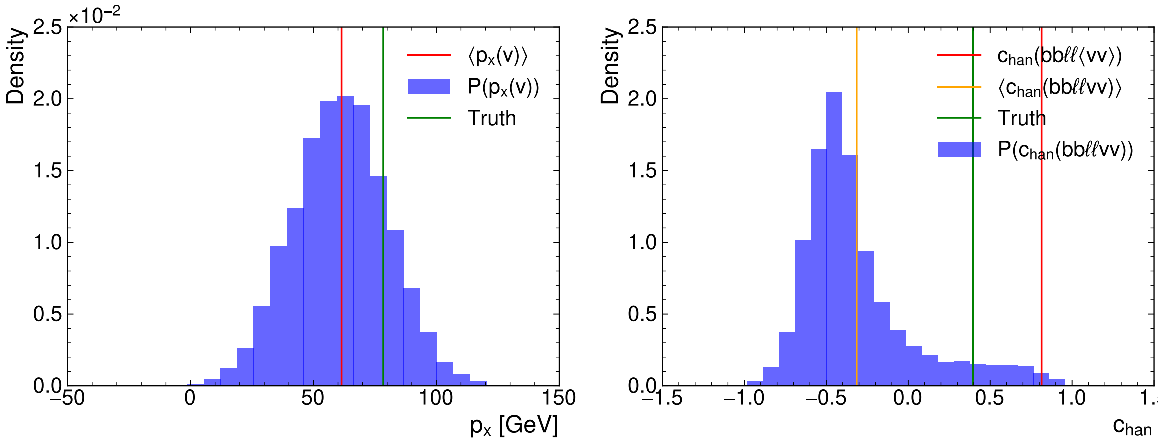


Figure B.4.: Plots showing the probability distribution of the neutrino momentum component p_x and the reconstructed c_{han} based on the mean and standard deviation predicted by the MMT-model trained with a Gaussian-loss for a random event. Indicated are the mean and true values.

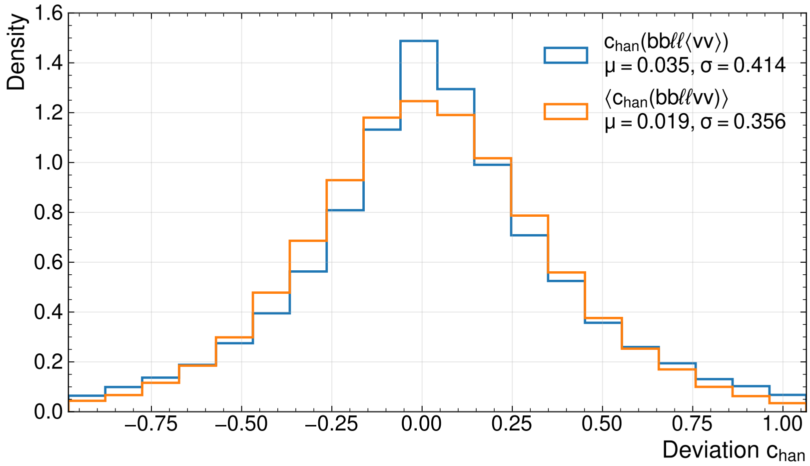


Figure B.5.: Deviation between the reconstructed and true c_{han}. The different curves show, whether the invariant mass was computed using the mean momentum or by propagating the neutrino probabilities and averaging of the obtained spin-variable.

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Erklärung nach §12(9) der Prüfungsordnung für den Master-Studiengang Physik an der Universität Göttingen:

Hiermit erkläre ich, dass ich diese Abschlussarbeit selbständig verfasst habe, keine anderen als die angegebenen Quellen und Hilfsmittel benutzt habe und alle Stellen, die wörtlich oder sinngemäß aus veröffentlichten Schriften entnommen wurden, als solche kenntlich gemacht habe.

Darüberhinaus erkläre ich, dass diese Abschlussarbeit nicht, auch nicht auszugsweise, im Rahmen einer nichtbestandenen Prüfung an dieser oder einer anderen Hochschule eingereicht wurde.

Göttingen, den 2. Juli 2026

(Siemen Henning Aulich)